

# CHAPTER 4

## OVERVIEW OF VARIOUS VIBRATION TRANSDUCERS AND HOW TO PROPERLY SELECT THEM

### 4.0 INTRODUCTION

The purpose of **Chapter 4** is to introduce the reader to vibration transducers, the various types available, how they function, how to select them and to provide information on transducers currently available on the market today. Each of the following transducers will be covered:

- 1) **Accelerometers**
- 2) **Velocity Pickups**
- 3) **Non-contact Eddy Current Displacement Probes**
- 4) **Shaft Contact Displacement Probes (including Shaft Sticks and Shaft Riders)**

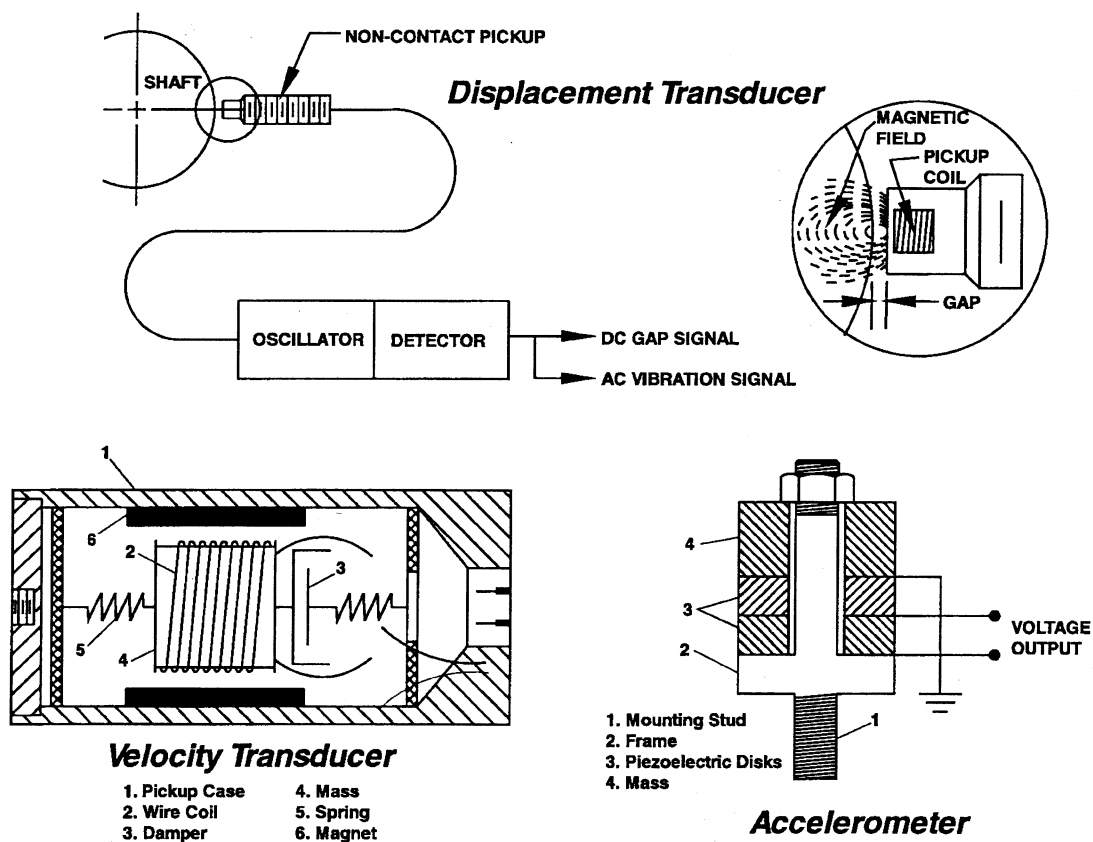
Figure 1 illustrates the three most common vibration transducers in use today, including the non-contact displacement probes, velocity pickups and accelerometers. Also shown are the components making up these transducers, each of which will be discussed in more detail under separate sections. Table I includes a general summary of some of the more important characteristics and general specifications for each of the various categories of accelerometers, velocity pickups and non-contact probes. In addition, it includes example models of transducers made by various manufacturers.

Depending on the measurement to be made, each of these transducers have their own optimum applications which will be discussed in this chapter. Also, particularly in the case of accelerometers, there is a wide variety of individual transducers. Finally, mounting techniques and their influence on the vibration measurement accuracy will likewise be discussed in this chapter.

As with **CHAPTER 3 - "OVERVIEW OF THE STRENGTH AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS"**, it is important to note that the following information is meant to reflect the general state of transducer design. However, the specific design characteristics for each individual transducer may vary from vendor to vendor. The reader should use this chapter as a guide for discussing specific needs with a vendor before purchasing a particular transducer or set of transducers.

### 4.1 TYPES OF VIBRATION TRANSDUCERS AND THEIR OPTIMUM APPLICATIONS

Typically, there are four types of vibration measurement transducers. They are known as accelerometers, velocity pickups, non-contact eddy current displacement probes and shaft contact displacement probes (including shaft sticks and shaft riders). Each type of transducer has applications which justify its use for monitoring machinery. These applications are important to understand so that a vibration analyst may best take advantage of them. For instance, when ordering large, high-speed machinery (i.e. turbine-generators or centrifugal air compressors), a



**FIGURE 1**  
**EXAMPLES OF VARIOUS TYPES OF VIBRATION TRANSDUCERS (REF. 5)**

**TABLE I  
GENERAL TRANSDUCER CHARACTERISTICS**

| TYPICAL SPECIFICATIONS                                                 | ACCELEROMETERS                                                                                                                                                           |                                                                                                                                                                          |                                                                           |                                                                                  |                                                                        | VELOCITY PICKUPS         |                                                                             | NONCONTACT DISPLACEMENT PROBES          |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------|-----------------------------------------|
|                                                                        | GENERAL PURPOSE                                                                                                                                                          | LOW FREQUENCY                                                                                                                                                            | HIGH FREQUENCY                                                            | PERMANENTLY MOUNTED                                                              | TRIAXIAL                                                               | SEISMIC                  | PIEZOELECTRIC                                                               |                                         |
| TYPICAL SENSITIVITY RANGE (mV/g Unless Noted)                          | 10-100                                                                                                                                                                   | 500 - 10,000                                                                                                                                                             | 0.4 - 20                                                                  | 10 - 100                                                                         | 10 - 100                                                               | (mV/in/s)<br>100 - 1,000 | (mV/in/s)<br>100 - 1,000                                                    | (mV/mil)<br>100 - 200                   |
| TYPICAL MEASUREMENT FREQUENCY RANGE (CPM)                              | 60-600,000                                                                                                                                                               | 6 - 60,000                                                                                                                                                               | 600 - 3,600,000                                                           | 180 - 600,000                                                                    | 120 - 600,000                                                          | 600 - 60,000             | 60 - 270,000                                                                | 0 - 120,000                             |
| TYPICAL STUD MOUNTED NATURAL FREQUENCIES (CPM)                         | 960,000 - 2,700,000                                                                                                                                                      | 390,000 - 2,100,000                                                                                                                                                      | 4,200,000 - 10,800,000                                                    | 1,080,000 - 1,800,000                                                            | 480,000 - 1,200,000                                                    | APPROXIMATELY 200,000    | APPROXIMATELY 200,000                                                       | N.A.                                    |
| TYPICAL WEIGHT RANGE (grams)                                           | 17-110                                                                                                                                                                   | 135 - 1,000                                                                                                                                                              | 1.2 - 15                                                                  | 60 - 180                                                                         | 17 - 130                                                               | 100 - 150                | 150 - 250                                                                   | 230 - 280                               |
| TYPICAL USABLE TEMPERATURE RANGE (°F)                                  | (-100°) - 250°                                                                                                                                                           | (-100°) - 250°                                                                                                                                                           | (-40°) - 350°                                                             | (-50°) - 250°                                                                    | (-100°) - 250°                                                         | (-50°) - 250°            | (-50°) - 250°                                                               | (-30°) - 350°                           |
| EXAMPLE TRANSDUCER MODEL NUMBERS AND MANUFACTURER (Alphabetical Order) | B & K 4370/81<br>B & K 4390<br>IRD 943<br>IRD 970<br>PCB 302A06<br>PCB 302A09<br>PCB 307A<br>PCB 308B<br>RION PV-44A<br>VIBRAMET. 1020<br>VIBRAMET. 1021<br>WILCOXON 793 | B & K 8318 (S)<br>PCB 326A01 (S)<br>PCB 326A02 (S)<br>PCB 326A03 (S)<br>PCB 393C (C)<br>VIBRAMET. 5100 (S)<br>WILCOXON 731 (C)<br>WILCOXON 793L (C)<br>WILCOXON 797L (S) | B & K 8309<br>PCB 303A02<br>RION PV 90B<br>VIBRAMET. 6032<br>WILCOXON 732 | B & K 8310/24<br>B & K 8317<br>IRD 941B<br>IRD 942<br>PCB 328B<br>VIBRAMET. 1024 | B & K 4321<br>PCB 306A<br>PCB 306A06<br>VIBRAMET. 3130<br>WILCOXON 733 | BENTLY. 47633<br>IRD 544 | BENTLY. 26862<br>IRD 560<br>WILCOXON 756V<br>WILCOXON 793V<br>WILCOXON 797V | BENTLY. 3300<br>BENTLY. 7200<br>IRD 403 |

**NOTES:**

1. For detailed information on particular transducers, see the APPENDIX which includes specifications for many of them.
2. By inclusion of specific transducers within this table, Technical Associates does not necessarily recommend them.

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plant may wish to include internally mounted non-contact displacement probes in the specifications to measure relative shaft vibration, since much of the energy will not transmit through the heavy casings to the exterior housing where an accelerometer or velocity pickup may be magnetically mounted by PMP personnel on a data collection route. The following sections describe the most common applications for the various transducer types in use today.

#### 4.11 ACCELEROMETERS

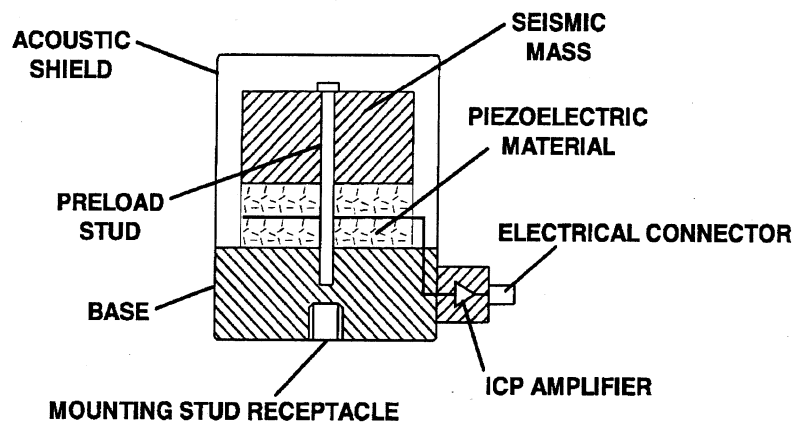
As the name implies, accelerometers are sensors which provide the direct measurement of acceleration (g). Typically, there are three primary performance characteristics which accelerometers possess affecting their performance. These include voltage sensitivity (mV/g), frequency response (Hz or CPM) and weight (oz. or grams). This section will take a brief look at each of these characteristics and will make some introductory remarks concerning their applications.

Figures 2, 3 and 4 illustrate the three most common types of piezoelectric accelerometers in use today (in simple terms, the piezoelectric effect of a crystal is its ability to accumulate electrical charges on its surface when the crystal is mechanically stressed). Figures 2 and 3 show one of the two major piezoelectric accelerometer types called the "compression mode accelerometer" (note that Figure 3 is an inverted compression type accelerometer). Figure 4 shows the other major type known as a shear mode accelerometer (differences of which will later be discussed). Each of the three accelerometers pictured in Figures 2 thru 4 have built-in electronics known as ICP circuits which are most often supplied with 4 to 20 milliamp (mA) current either from an external power supply or directly from the vibration instrument itself. The term "ICP" stands for "integrated circuit piezoelectric" type.

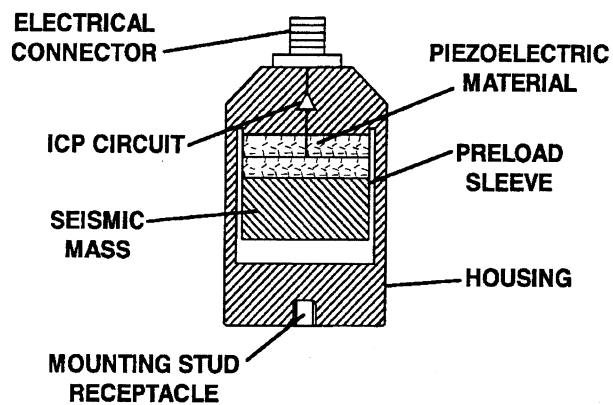
Typically, accelerometers are used with rolling element bearing supported shafts. This is because rolling element bearings transfer the majority of the shaft vibration into the machine housing and can be readily picked up by a transducer mounted on the external housing. However, they sometimes also work well with journal bearings because of advancements made in the sensitivity of certain accelerometers allowing them to pick up low amplitude vibration. However, be aware that readings from a non-contact eddy current displacement probe (measuring actual shaft vibration) may be quite a bit higher in amplitude than that taken on the casing itself (depending on the oil film characteristics as well as on the mass and distance from the shaft out to the bearing housings). Sometimes, the shaft vibration may be as much as 20 times higher on a journal bearing machine at a distance of 3 to 6 feet to the bearing surface meaning that potentially serious problems such as oil whirl or oil whip may be missed altogether if readings are not taken directly from the shaft itself due to the loss of signal through the oil film, as well as at each of the metal interfaces and thicknesses out to the machine housing.

Accelerometers are the most common PMP transducers in use today due to a number of reasons. They have relatively inexpensive costs when compared to velocity pickups and have much wider frequency range capabilities than either velocity pickups, non-contact probes, shaft sticks or shaft riders. Also they can be easily mounted onto machinery by stud, adhesive, magnet or (if absolutely necessary) extension probe. On the other hand, non-contact probes must be permanently mounted to give good, repeatable data required in predictive maintenance programs (and great effort has to be spent to prepare the target area at which the probe will be pointed, ensuring a very smooth surface which is free of cracks and other defects).

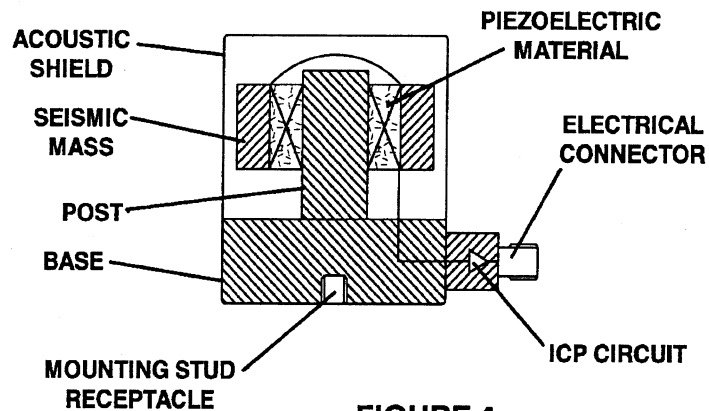
There are a large variety of accelerometers available which have a wide selection of frequency ranges. This is shown in Table I which will summarize much of the information within Chapter 4, not only providing general information about all the major transducers discussed, but also specific manufacturer and model numbers of each of the various types of accelerometers, velocity pickups and non-contact probes. Table I shows that the "General Purpose Accelerometer" has a relatively wide "Typical Frequency Range" of approximately 120 to 600,000 CPM (2 - 10,000 Hz).



**FIGURE 2**  
**COMPRESSION TYPE ACCELEROMETER (REF. 2)**



**FIGURE 3**  
**INVERTED COMPRESSION TYPE ACCELEROMETER (REF. 2)**



**FIGURE 4**  
**SHEAR TYPE ACCELEROMETER (REF. 2)**

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However, it also shows that in order to make very low frequency or high frequency measurements, it will likely be necessary to acquire other specially designed accelerometers. For example, the typical low frequency accelerometer might measure down to about 6 CPM, but up to only about 60,000 CPM (0.1 - 1,000 Hz); whereas a typical high frequency accelerometer might effectively measure from approximately 600 to as high as 3,600,000 CPM (10 - 60,000 Hz).

In most cases, if desiring to take low frequency measurements, it will be necessary to choose one of the low frequency accelerometers listed in Table I which typically are much higher weight and normally have voltage sensitivities (mV/g) much higher than the general purpose accelerometers as shown in the table. The reason for this is that even large vibration displacement (mils) produces very small acceleration levels (g) when at frequencies of less than 60 CPM (1 Hz), making it of great importance to have high voltage sensitivity to bring the vibration signal above the noise. For example, an excessive displacement of 100 mils (Pk-Pk) (.100 inch) at a frequency of only 60 CPM (1 Hz) corresponds to an acceleration of only .005 g Pk at this frequency. On the other hand, if desiring to make measurements on machines generating very high frequencies (typically above 600,000 CPM or 10,000 Hz), it will likely be necessary to acquire one of the high frequency accelerometers listed in Table I. Typically, these high frequency accelerometers are much smaller accelerometers which likewise have smaller voltage sensitivity, usually on the order of only about 10 mV/g.

One of the common misconceptions is that the higher the voltage sensitivity (mV/g), the better the accelerometer. This is not the case in all measurement situations. For example, one of the problems with some of the seismic accelerometers used in low frequency measurements is that they are very sensitive to temperature changes; and, if dropped, they can fail due to the tremendous voltage response which oversaturates their built-in electronics (a general purpose or high frequency accelerometer can generally be dropped with no damage to the transducer).

Note that Table I also lists some accelerometers specially designed for permanently mounting as well as those designed to make triaxial measurements (simultaneous measurements in the horizontal, vertical and axial directions). The permanently mounted transducers are designed so that they can withstand harsh environments and sometimes can be placed under water or within other fluids such as lubricating oil or cutting oils. On the other hand, triaxial accelerometers give the analyst the advantage of simultaneously making measurements in all three directions, thereby increasing his measurement speed on a PMP route in the field.

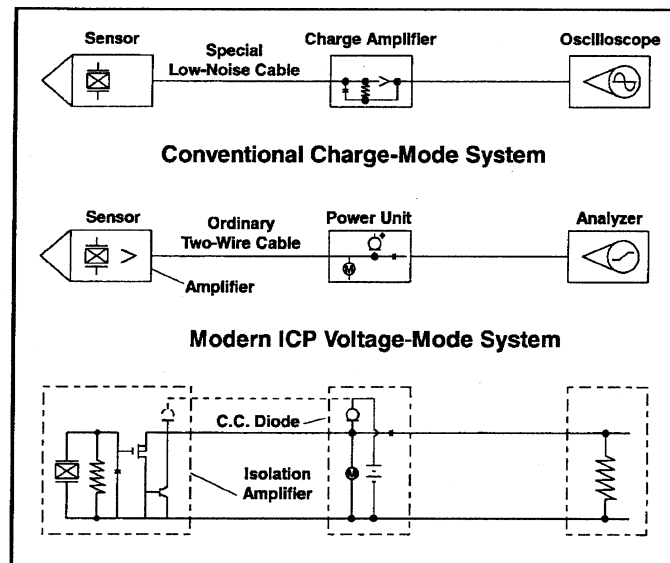
Returning to Figures 2 thru 4, note that these illustrate the three most commonly used accelerometer designs today. Note that there are two types of compression designs (upright and inverted positioning of the seismic mass) in which the piezoelectric crystal is sandwiched between a seismic mass and a base or housing. In this case, the amount of electrical output generated by the crystal is proportional to the amount of acceleration experienced by the seismic mass which is proportional to the amount of force applied to it (as per Newton's law of motion,  $F = ma$ ). The compression designs in Figures 2 and 3 are probably the most widely utilized designs today, mainly due to their simplicity and lower cost, along with their wide frequency range. However, one of the disadvantages with compression mode accelerometers is that they are often subject to problems due to thermal transients and base strain sensitivity which can oversaturate their electronics and require much time for the transducer to "settle".

In those situations where there are large differences in temperature on which the accelerometer will be placed (or if either the temperature of the surface on which the accelerometer will be placed is continually changing or if high pressure air is blowing on the accelerometer), the shear type accelerometer pictured in Figure 4 can help overcome these settling problems. These shear mode designs tend to isolate the crystal element from the base and housing by sandwiching the element between the seismic mass and a center post. When accelerated, the seismic mass will apply a shear force to the crystal element which will emit a voltage proportional to the acceleration (and therefore to the force) applied to it. The advantages of shear mode

accelerometers include a stable output signal in the presence of thermal transients and/or high pressure air blow-by, especially when measuring at low frequencies. Also these are typically smaller in size and mass. Disadvantages of these units typically include a higher cost due to the added components required to make up the shear configuration.

Later on, when an analyst begins taking low frequency measurements (typically below 60 CPM or 1 Hz), this difference between shear mode and compression mode accelerometers will become important. In this case, it may require a compression mode accelerometer 3 or 4 minutes for it to stabilize before any measurements can be taken whatsoever. However, in these cases, a shear mode accelerometer will not be so sensitive to such thermal transients, and will stabilize quickly, allowing the analyst to begin the measurement almost immediately.

One other important difference concerns the type of signal conditioning electronics serving the piezoelectric accelerometers. There are two types of such signal conditioning including (1) high impedance, charge mode or charge type which requires an external signal conditioner; and (2) low impedance, voltage mode, integrated circuit or "ICP" (integrated circuit piezoelectric) types which contain built-in signal conditioning electronics. Each of these is pictured in Figure 5. One of the biggest disadvantages of the charge mode accelerometer is that they require a separate charge amplifier between the analyzer and the sensor as shown in Figure 5. On the other hand, ICP current supply is now included within most all FFT analyzers (including both data collectors and spectrum analyzers) allowing an analyzer to directly power the ICP, voltage-mode accelerometer. Another disadvantage with the charge mode units is that special, low-noise cable of fixed length most often has to be used between the charge amplifier and sensor in the field whereas the ICP voltage-mode sensor requires only ordinary 2-wire coaxial cable and the ICP accelerometer can usually be powered at distances up to about 1000 feet without additional power sources. In the case of the charge mode unit, accelerometer sensitivity is usually defined in units of picocoulombs per g (pC/g), whereas the sensitivity of voltage-mode units is normally expressed directly in millivolts per g (mV/g). Advantages of the charge mode accelerometer include the ability to operate at high temperatures. However, its disadvantages of requiring the low-noise, fixed length cable, along with a clean and moisture free environment as well as a separate charge amplifier do not normally lend it to most predictive maintenance measurement systems today.



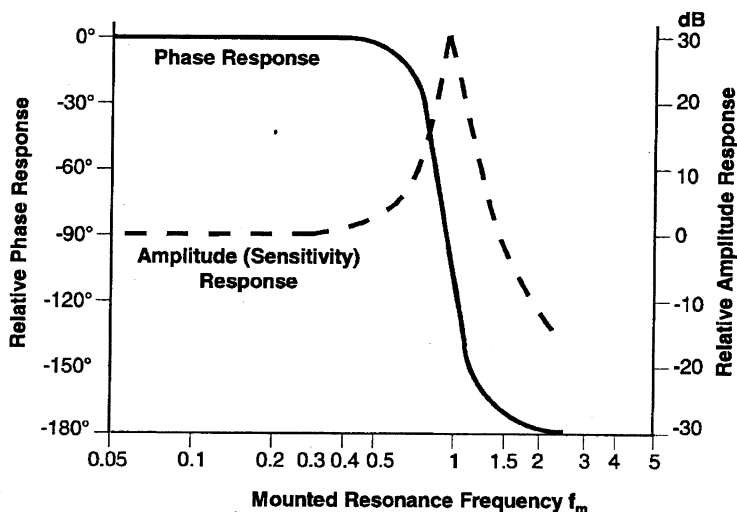
**FIGURE 5**  
**COMPARISON OF CHARGE MODE AND ICP VOLTAGE-MODE**  
**ACCELEROMETERS (REF. 3)**

Therefore, the most commonly used PMP accelerometer today is the low impedance, ICP type containing the built-in microelectronics which condition the signal from a crystal within the accelerometer to a low impedance voltage which is compatible with the readout device. A constant current power source is required for the transistor within the accelerometer to turn on and perform its measurement (again this constant current power source is normally available in most all FFT analyzers today). Advantages of the ICP accelerometer include fixed sensitivity, ability to operate reliably in dirty and moist environments, the use of ordinary signal transmission cables, the lack of requirement of a separate signal conditioner if one is available in the analyzer and typically a lower system cost. Disadvantages sometimes include limited temperature range due to the survivability of the built-in electronics (although this can sometimes be overcome by use of special ceramic construction materials). Another disadvantage which is important to point out is that if an ICP-type accelerometer is not hermetically sealed, the entry of moisture into the transducer can significantly degrade its low frequency performance. Reference 4 points out that:

"All accelerometers breathe, unless they are certified as hermetically sealed, and during this micro-breathing, moisture inevitably penetrates. As already mentioned, the low-frequency cutoff in a voltage amplifier depends on the accelerometer as well as cable resistance and capacitance. As moisture penetrates, the internal resistance decreases from the gigaohm range to the megaohm range. This changes amplifier performance dramatically. It has been observed that some transducers with built-in preamplifiers gave no useful signal below 40 Hz after only a few months of use on paper making machinery"<sup>4</sup>

Therefore, it is important that if one acquires an ICP-type accelerometer, acquire a hermetically sealed unit, particularly if it is to be used in humid environments. This is especially important in those cases where low-frequency machines are to be evaluated.

One final important point needs to be considered when choosing accelerometers. It is important that the user be aware of the phase response of the particular transducer and how it varies with frequency. This is illustrated by Figure 6 which shows an accelerometer with fairly constant phase response from approximately 5% up to about 60% of its mounted resonance frequency ( $f_m$ ). Note in Figure 6 that as the transducer approaches its mounted resonance, it begins to experience a



**FIGURE 6**  
**SCHEMATIC OF ACCELEROMETER SENSITIVITY AND PHASE RESPONSE**  
**AS A FUNCTION OF FREQUENCY (REF. 7)**

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dramatic phase change which approaches almost 180° when it passes through this resonance. Therefore, if one were using this accelerometer when performing a phase analysis on a machine, the user must be aware of the frequency range at which the phase response of the transducer itself undergoes this significant change. This can totally negate the accuracy of the phase analysis which would be important in helping clearly identify whether high vibration is caused by symptoms having similar vibration signatures such as unbalance, misalignment, bent shaft, eccentricity, etc. One must also be aware that if a magnet or an extension probe is used to mount this same accelerometer, this will significantly lower the natural frequency, and therefore, will significantly lower the frequency ranges in which accurate phase measurements can be made with this same accelerometer.

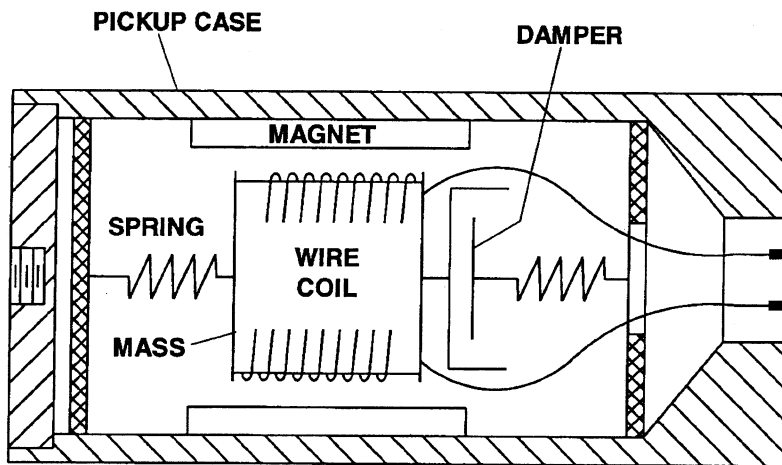
Note that the **APPENDIX** includes examples of many different accelerometers from various vendors, including their specifications. Comparing these, note that a very wide range of accelerometers exist ranging from those capable of measuring down to as low as .01 Hz (.6 CPM) up to over 60,000 Hz (3,600,000 CPM). In addition, note that voltage sensitivities will vary from as low as about 0.4 mV/g up to as high as 10,000 mV/g (10 V/g); and that accelerometer masses can vary from as low as 1 gram up to over 1000 grams. Therefore, in most predictive maintenance programs which are outfitted not only with normal speed machinery ranging from about 600 up to 3600 RPM, but also with low speed (particularly below 200 RPM) as well as high speed machinery (particularly above 10,000 RPM), it will very likely be necessary to have at least three different accelerometers - a low frequency, a general purpose and a high frequency transducer. Very likely, the general purpose accelerometer can probably be used to successfully make over 90% of the measurements. However, the accuracy of measurements on the very low or very high speed machines will suffer without the low and high frequency accelerometers (and therefore, results will prove far less effective).

#### 4.12 VELOCITY PICKUPS

As of today, velocity pickups have generally been replaced by accelerometers for most applications because of their expense, limited frequency range, and relatively heavier weight. However, one advantage of velocity pickups is that they do not have to be powered by an external power supply such as ICP current or a separate charge amplifier. Also, since velocity is the commonly accepted unit of vibration analysis for most PMP programs, their signal does not have to be integrated from acceleration to velocity (as is the case with accelerometers). Therefore, this eliminates integration noise which typically occurs in the first 2 to 4 lines of the FFT spectra taken by accelerometers integrated to velocity.

There are two types of velocity pickups - seismic velocity transducers and piezoelectric velocity pickups. Figure 7 shows an example seismic velocity transducer while Figures 9 and 10 show example piezoelectric velocity pickups.

Unlike the piezoelectric velocity transducers (and the piezoelectric accelerometers previously discussed), the seismic velocity pickup is self-generating and does not require an external power supply of any kind. Figure 7 shows that the seismic velocity pickup consists of a case in which a coil of wire is wound around a seismic mass which itself is supported by soft springs. This gives the unit a very low natural frequency on the order of only 10 Hz (600 CPM). In addition, a permanent magnet is firmly attached to the transducer case providing a strong magnetic field around the coil. In essence, whenever the velocity pickup is mounted on a bearing housing, the permanent magnet will vibrate while the spring-suspended coil of wire will remain stationary in space. When the moving magnetic field cuts through the stationary coil of wire, a voltage proportional to the relative velocity of the magnet and coil is generated, with the voltage determined by the number of turns of wire in the coil. The voltage generated is then transmitted out of the transducer into a vibration analyzer.



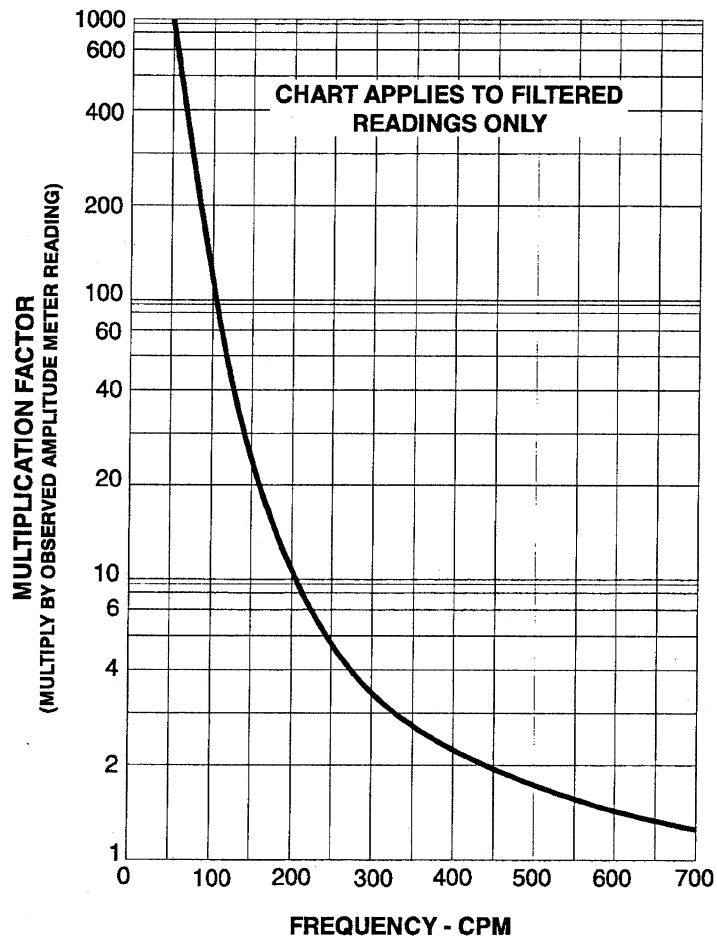
**FIGURE 7**  
**IRD 544 SEISMIC VELOCITY PICKUP (Ref. 5)**

|                                 |                                                      |
|---------------------------------|------------------------------------------------------|
| Frequency Range ( $\pm 10\%$ ): | 600-60,000 CPM (10-1000 Hz)                          |
| Sensitivity ( $\pm 10\%$ ):     | 764 mV RMS / in/sec PK<br>= 1080 mV Peak / in/sec PK |
| Temperature Range:              | -40° F to + 500° F<br>(-40° C to + 260° C)           |
| Impedance:                      | R = 2000 ohms; L = 0.7 Hy                            |
| Damping Factor:                 | 0.7 (critical)                                       |
| Transverse Response (Avg):      | Less than 5% (10-1000 Hz)                            |
| Magnetic Field Sensitivity:     | 0.15 in/sec /Gauss                                   |
| Weight:                         | 21oz (595 grams)                                     |

The suspended mass shown on Figure 7 is damped (either electrically or with synthetic oil) such that it remains motionless throughout the frequency range as the case vibrates around it. Therefore, the transducer's low frequency response is limited by its first natural frequency, with the sensitivity rolling off typically around 600 to 1200 CPM, (or 10 to 20 Hz). Therefore, this sensitivity will drop as the mass overcomes the damping effect at low frequencies and begins to move in phase with the vibration. Figure 8 shows what must be done if measurements are required by an example velocity pickup (IRD 544) if measurements are desired below 600 CPM (10 Hz). For example, if a measurement of .1 in/sec were recorded using this transducer at 300 CPM, Figure 8 shows that the actual level would be about 3.3 times higher than this (or about .33 in/sec). Note that with decreasing frequencies, significantly higher and higher multiplication factors must be applied to correct the vibration readings. Therefore, a seismic velocity pickup is not a good low frequency transducer.

In addition, the upper frequency response is limited on most velocity pickups to no more than about 60,000 to 120,000 CPM (1000 - 2000 Hz) due to the increasing difficulty of overcoming system inertia. Another problem with these units is that they typically are very large in size and weight [the example in Figure 7 weighs 21 ounces (or 595 grams)], which can undesirably contribute to mass loading of the machine or structure being evaluated.

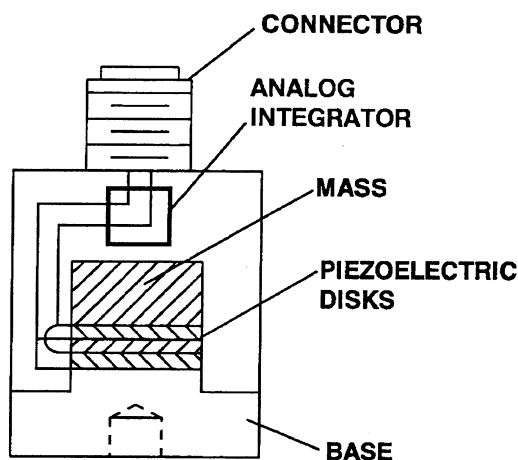
If much lower or higher frequency measurements with a direct velocity readout are required, a piezoelectric velocity transducer like that shown in either Figure 9 or Figure 10 is recommended. In these cases, Figure 9 shows again that a piezoelectric disk is mounted below a seismic mass,



**FIGURE 8**  
**CORRECTION FACTOR CHART FOR IRD 544 SEISMIC VELOCITY PICKUP**  
**WHEN MEASURING FREQUENCIES BELOW 600 CPM (10 Hz) (REF. 5)**

along with an internal analog integrator which converts the acceleration signal to velocity. This analog integration inside the transducer case will sometimes decrease the noise at low frequencies due to integration and other sources. This piezoelectric crystal also provides the unit a small amount of phase shift making it a much better transducer for phase analysis and balancing than the seismic velocity pickup. In addition, one of the problems with seismic transducers is that they are most often highly susceptible to magnetic interference. On the other hand, piezoelectric velocity pickups and piezoelectric accelerometers are virtually unaffected by magnetic fields (see comparative magnetic field sensitivities in Figures 7 and 9, respectively).

Finally, the frequency range of piezoelectric velocity pickups is much wider than that of seismic units as seen by comparing Figure 7 (seismic) with Figures 9 and 10 (piezoelectric). Note that the unit of Figure 9 has a frequency response ( $\pm 10\%$ ) of 60 up to 270,000 CPM (1 - 4500 Hz) while that of Figure 10 can measure as low as 150 CPM (2.5 Hz) up to 210,000 CPM (3500 Hz). Further examples of similar velocity transducers, along with their specifications, are contained within the **APPENDIX**.



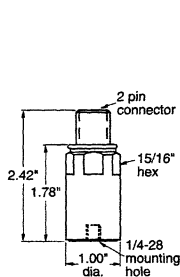
**FIGURE 9**  
**IRD 560 PIEZOELECTRIC VELOCITY PICKUP (Ref. 5)**

|                                 |                                                      |
|---------------------------------|------------------------------------------------------|
| Frequency Range ( $\pm 10\%$ ): | 60-270,000 CPM (1-4500 Hz)                           |
| Sensitivity ( $\pm 10\%$ ):     | 764 mV RMS / in/sec PK<br>= 1080 mV Peak / in/sec PK |
| Temperature Range:              | -40° F to + 250° F<br>(-40° C to + 121° C)           |
| Mounted Natural Freq.:          | 600,000 CPM min. (10,000 Hz)                         |
| Transverse Response :           | 5% max @ 600 CPM (10Hz), 5 g peak                    |
| Magnetic Field Sensitivity:     | .00001 in/sec /Gauss                                 |
| Weight:                         | 38 oz (1079 grams)                                   |

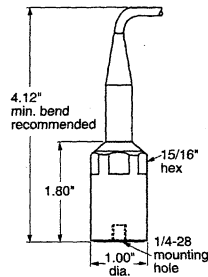


**FEATURES:**

- Intrinsic safety certified option
- Industrial ruggedness
- Eliminates distortion caused by high frequency signals
- Corrosion-resistant
- Internally integrated to velocity
- Ultra low-noise electronics for clear signals at very low vibration levels
- Miswiring protection



**793V Series  
Models**



**793VS Series  
Models**

## 793V Series Piezoelectric Velocity Transducers

| SPECIFICATIONS                                        | All Models                                             | 793V                         | 793V-5                   | Units                               |
|-------------------------------------------------------|--------------------------------------------------------|------------------------------|--------------------------|-------------------------------------|
| <b>DYNAMIC</b>                                        |                                                        |                              |                          |                                     |
| Sensitivity, $\pm 10\%$ , $25^\circ\text{C}$          | —                                                      | 100                          | 500                      | mV/in/sec                           |
| Velocity Range                                        | —                                                      | 50                           | 10                       | in/sec peak                         |
| Amplitude Nonlinearity                                | 1%                                                     |                              |                          |                                     |
| Frequency Response:                                   |                                                        |                              |                          |                                     |
| $\pm 10\%$                                            | —                                                      | 2.0 - 3,500                  | 8.0 - 3,500              | Hz                                  |
| $\pm 3\text{ dB}$                                     | —                                                      | 1.5 - 7,000                  | 5.0 - 7,000              | Hz                                  |
| Resonance Frequency, mounted, nom.                    | 15 kHz                                                 |                              |                          |                                     |
| Transverse Sensitivity, max.                          | 5 % of axial                                           |                              |                          |                                     |
| Temperature Response                                  | see graph                                              |                              |                          |                                     |
| <b>ELECTRICAL</b>                                     |                                                        |                              |                          |                                     |
| Power Requirement:                                    |                                                        |                              |                          |                                     |
| voltage source                                        | 18 - 30 VDC                                            |                              |                          |                                     |
| current regulating diode <sup>1</sup>                 | 2 - 10 mA                                              |                              |                          |                                     |
| Electrical Noise, equiv. in/sec, nominal:             |                                                        |                              |                          |                                     |
| Broadband 2.5 Hz to 25 kHz                            | —                                                      | 100                          | 40                       | $\mu\text{in/sec}$                  |
| Spectral 10 Hz                                        | —                                                      | 10                           | 4.0                      | $\mu\text{in/sec}/\sqrt{\text{Hz}}$ |
| 100 Hz                                                | —                                                      | 1.0                          | 0.4                      | $\mu\text{in/sec}/\sqrt{\text{Hz}}$ |
| 1000 Hz                                               | —                                                      | 0.2                          | 0.05                     | $\mu\text{in/sec}/\sqrt{\text{Hz}}$ |
| Absolute Phase Shift, nom. the greater of:            | $\tan^{-1} 2/f$ or $2^\circ$                           | $\tan^{-1} 5/f$ or $2^\circ$ |                          |                                     |
| Output Impedance, nominal 4mA supply, the greater of: | —                                                      | 5,000/f or 200 $\Omega$      | 25,000/f or 200 $\Omega$ |                                     |
| Bias Output Voltage, nominal                          | 10 VDC                                                 |                              |                          |                                     |
| Grounding                                             | case isolated, internally shielded                     |                              |                          |                                     |
| <b>ENVIRONMENTAL</b>                                  |                                                        |                              |                          |                                     |
| Temperature Range                                     | -50 to $120^\circ\text{C}$                             |                              |                          |                                     |
| Vibration Limit                                       | 250 g peak                                             |                              |                          |                                     |
| Shock Limit                                           | 2,500 g peak                                           |                              |                          |                                     |
| Electromagnetic Sensitivity, equiv. in/sec            | 25 $\mu\text{in/sec/gauss}$                            |                              |                          |                                     |
| Sealing                                               | hermetic                                               |                              |                          |                                     |
| Base Strain Sensitivity                               | 0.0005 in/sec/ $\mu\text{strain}$                      |                              |                          |                                     |
| <b>PHYSICAL</b>                                       |                                                        |                              |                          |                                     |
| Weight                                                | 145 grams                                              |                              |                          |                                     |
| Case Material                                         | 316L stainless steel                                   |                              |                          |                                     |
| Mounting                                              | $1/4$ - 28 tapped hole                                 |                              |                          |                                     |
| Output Connector                                      | MIL-C-5015 2-pin                                       |                              |                          |                                     |
| Connections Pin A                                     | signal, power                                          |                              |                          |                                     |
| Connections Pin B                                     | common                                                 |                              |                          |                                     |
| Cabling <sup>2</sup> : Mating Connector               | Wilcoxon R6 (Amphenol 97-3106A-10SL-4S)                |                              |                          |                                     |
| Recommended Cable                                     | J9T2, two conductor shielded, Tefzel jacket, 30 pF/ft. |                              |                          |                                     |

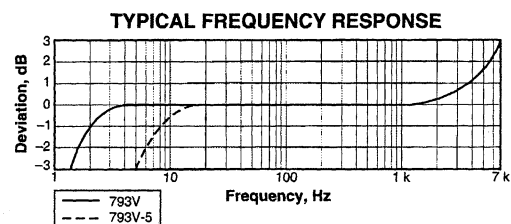
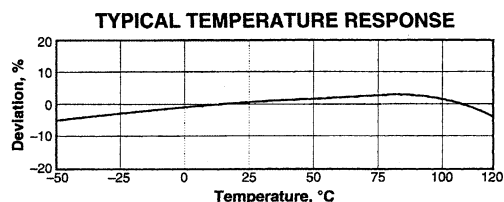
**NOTES:** <sup>1</sup> A maximum current of 6 mA is recommended for operating temperatures in excess of  $100^\circ\text{C}$ .

<sup>2</sup> Cabling for "S" option model is J61, coaxial, polyurethane jacket, 30 pF/ft.

**OPTIONS:** Intrinsic safety certification; stainless steel flexible hose; 10 ft. sealed integral cable "S" or "SG", temperature sensor "T", customer specified sensitivity, low frequency filtering, cable length, connector.

**ACCESSORIES SUPPLIED:** SF6 mounting stud, calibration data, CCD card

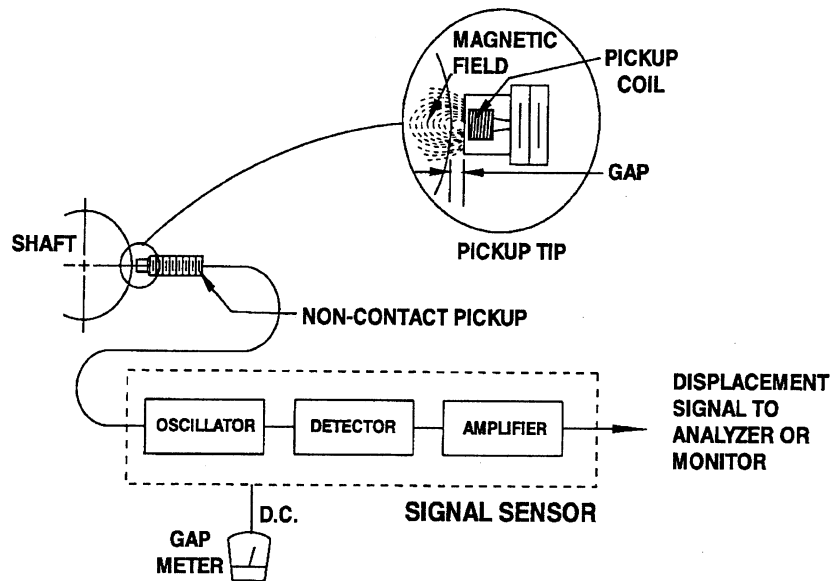
**ACCESSORIES AVAILABLE:** R6SL-J9T2-XX splash-proof cable assembly, magnetic mounting bases, SF8 cementing studs, power supplies, amplifiers, signal conditioners.



**FIGURE 10**  
**793V SERIES PIEZOELECTRIC VELOCITY TRANSDUCERS (Ref. 5)**

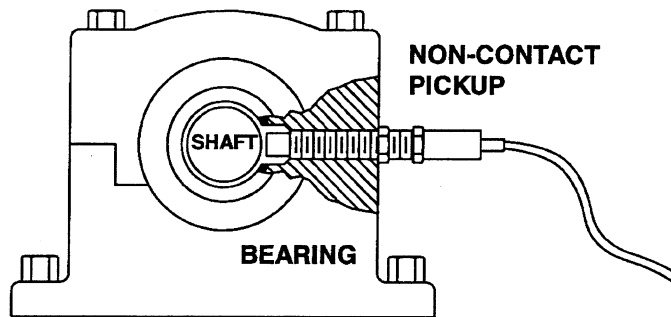
#### 4.13 NON-CONTACT EDDY CURRENT DISPLACEMENT PROBES

Figure 11 pictures a non-contact eddy current displacement probe system (sometimes called proximity probes). These systems are primarily used to measure shaft vibration, radial and axial shaft position and differential expansion between the case and rotor. They provide direct relative shaft vibration (shaft vibration relative to the non-contact mounting point on the machine housing). On the other hand, accelerometers and velocity pickups mounted on the machine housing measure absolute vibration. Eddy Current Probes are especially effective on large machinery with oil film plain bearings such as turbine/generators, compressors, large motors, etc. In these cases, there is often a significant signal loss by the time the shaft vibration passes through the oil film to the bearing, and then through potentially several metal interfaces out to the machine housing, often at distances of 2 feet or more. In these cases, shaft vibration can be 10 to 20 times higher than that for smaller machines with rolling element bearings. Therefore, the best possible vibration spectrum, particularly for events at 1X RPM up through about 6X RPM is that which can be acquired directly from a non-contact eddy current probe. This probe will measure vibration of the shaft relative to the mounting system of the probe itself (therefore, it is most important to anchor the probe as much as possible to get the truest picture of shaft vibration).



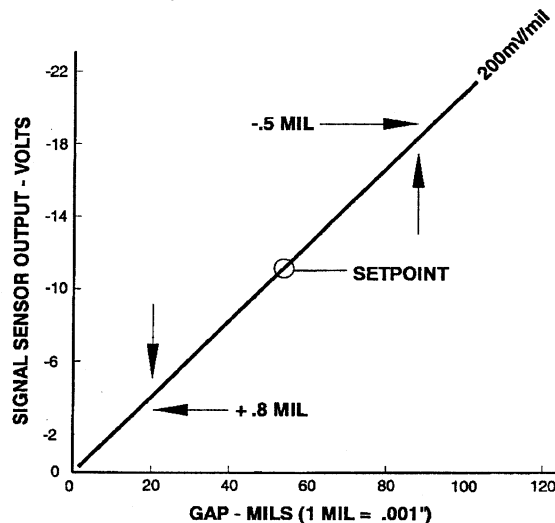
**FIGURE 11**  
**SCHEMATIC DIAGRAM OF AN EDDY CURRENT NON-CONTACT**  
**DISPLACEMENT PROBE SYSTEM (REF. 5)**

Figure 11 shows a schematic diagram of an eddy current non-contact probe system. Unlike piezoelectric accelerometers and velocity pickups, a non-contact transducer does not generate a voltage or electrical charge in response to vibration. Instead, a non-contact sensor requires external electronic circuitry to generate a very high frequency A.C. signal and detects oscillations in the A.C. signal caused by vibration of the shaft. This high frequency electrical signal is called the carrier signal, which is applied through a coaxial cable to a small coil of wire in the tip of the non-contact pickup. Figure 11 shows that this high frequency electrical signal applied to the coil which generates a magnetic field. Then the target surface, which will often be placed at a gap of



**FIGURE 12**  
**NON-CONTACT EDDY CURRENT PROBE MOUNTED IN A BEARING**  
**HOUSING (REF. 5)**

only 40 to 60 mils from the non-contact probe tip, will absorb some of the magnetic energy with the resulting absorption in effect placing an electrical load on the signal reducing its strength. Therefore, as the shaft moves relative to the pickup tip, the strength of the electrical signal changes proportional to the movement. The probe will provide an A.C. signal voltage proportional to the vibration and a D.C. signal proportional to the gap. Thus, the A.C. signal can be read by a permanent monitor or an analyzer to determine the amount of shaft vibration while the D.C. signal can be used to monitor the change in gap (particularly useful if targeted towards the shaft end to detect axial shaft movement and possible wear of bearings). Figure 12 shows a non-contact probe mounted in a bearing housing. Here again, the probe itself will measure the shaft displacement relative to the bearing and housing. Again, the signal from the probe can be taken directly into an analyzer using output connectors from the transducer. In these cases, be very careful! Do not connect the cable to the ICP port on a spectrum analyzer or data collector because the ICP current will cause the vibration readout to suddenly jump which can cause these machines to alarm, or even worse, trip off line.



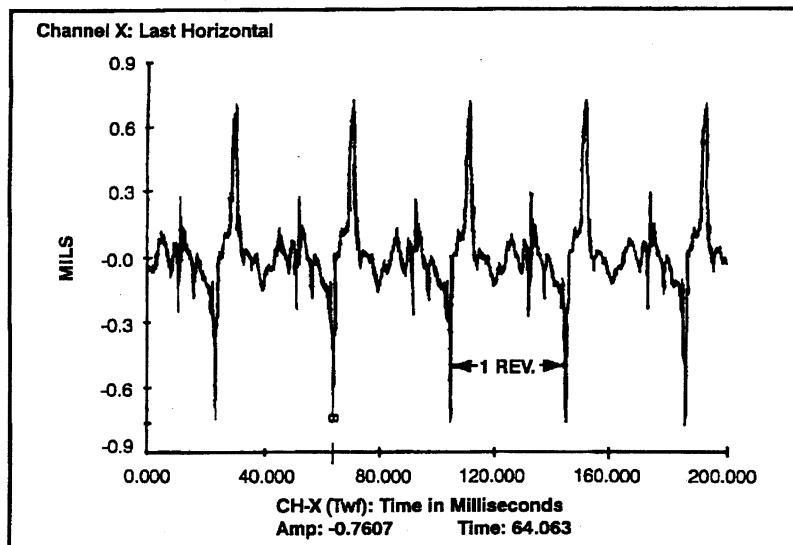
**FIGURE 13**  
**TYPICAL OUTPUT CURVE FOR AN EDDY CURRENT NON-CONTACT**  
**DISPLACEMENT PROBE SYSTEM**

Figure 13 shows a typical response curve for a non-contact probe and signal sensor. This curve represents the change in sensor output voltage as the distance between the tip of the pickup and the target material is changed. Looking closely at Figure 13, note that the relationship of the output voltage to the gap approximates a straight line. The slope of this line, millivolts per mil (mV/mil) determines the non-contact probe sensitivity. In general, the linearity of this system will be defined as that portion of the output voltage versus gap within which the slope does not vary by more than  $\pm 5\%$ , or for which the absolute gap error does not exceed  $\pm 1.5$  mils. In the system pictured in Figure 13, this system has a linear gap range from about 20 to 100 mils or a total of 80 mils. Therefore, the optimum "setpoint" for this non-contact pickup system would be selected at the midpoint (60 mils) of the specified linear gap range.

### Displacement Probe System

Of great importance when installing non-contact probes is providing a very smooth, uniform surface finish on the target. In fact, a common problem with non-contact probes is that the target material must be free of any surface imperfections such as scratches or high spots. When a scratch is present, the probe cannot discern between true vibration and the scratch depth, thereby causing an error in the readout. However, if one examines time waveforms and/or vibration spectra from a shaft having a scratch, it will be very noticeable such as the time waveform caused by a scratch on a target shown in Figure 14.

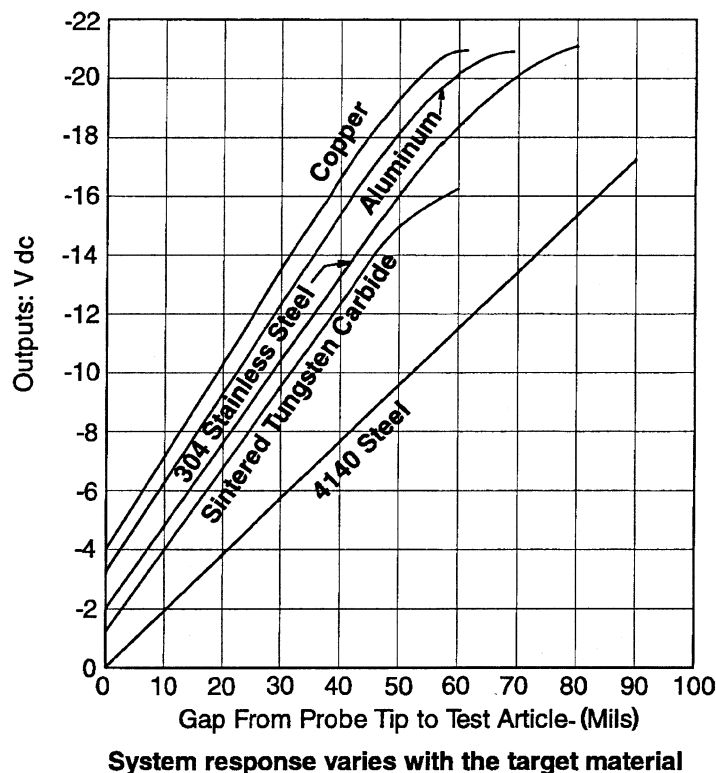
Another problem common to non-contact probe systems is runout - either mechanical or electrical runout. Mechanical runout is an error in measuring the position of the shaft center line with a displacement probe which is caused by out-of-roundness and surface imperfections (scratches, chain marks, dents, rust or other conductive buildup on the shaft, flat spots and engravings). It is most important to ensure that there is less mechanical runout in the shaft than the minimum vibration displacement which is to be measured. Readings should be taken at  $30^\circ$  -  $45^\circ$  rotational intervals around the shaft plotting runout versus angular position to ensure this and also to provide a baseline for comparison with electrical runout data.



**FIGURE 14**  
**TIME WAVEFORM TAKEN FROM A NON-CONTACT PROBE TARGETED**  
**ON A SCRATCHED SHAFT (REF. 10)**

Electrical runout is a signal error which occurs in eddy current displacement measurements when the shaft surface conductivity varies. That is, the target's surface itself has nonuniform electrical conductivity/resistivity/permeability properties. Another source of electrical runout is due to the presence of a local (spot) magnetic field(s) at a point(s) on the shaft surface. This error repeats exactly with each shaft revolution. Such shaft magnetism may result from magnetic-particle testing or simply from operating in a strong magnetic field which sometimes occurs with motors, generators or alternators. In addition, welding operations conducted nearby a machine can sometimes induce electrical runout into a shaft. Here again, it is impossible for the displacement probe to differentiate true vibration from either the mechanical or the electrical runout, depending on where exactly the electrical and mechanical runout is located on the shaft, the runout may either add or subtract from one another and from the total vibration. That is, the displacement readout will be a vectorial addition of electrical runout, mechanical runout and actual vibration. There are procedures in place for determining the presence of mechanical and electrical runout and for minimizing its effect. In the case of electrical runout, it often is necessary to correct this with a degaussing instrument.

It is very important that the non-contact probe be calibrated for the specific shaft material to which it will be targeted. Figure 15 shows typical response curves for many different target materials as well as comparing it to the gap set between the probe and the target. This chart is used to set up and calibrate the response of the proximity probe. It is important to notice that "roll-off" problems occur as the gap is increased. It is not recommended that probes be used in this roll-off range.



**FIGURE 15**  
**VOLTAGE RESPONSE VERSUS GAP FROM PROBE TIP TO SHAFT**  
**FOR VARIOUS TARGET MATERIALS (REF. 8)**

Finally, one of the remaining problems with a non-contact probe is frequency range. Typically, specifications for non-contact probe systems will claim a frequency response of from about 0 to 600,000 CPM (0 - 10,000 Hz). However, this does not actually occur in the real world due to practical reasons. For example, in the real world, since the probe is measuring displacement, it will tend to emphasize low frequency sources, but de-emphasize high frequency sources. For example, consider a centrifugal compressor outfitted with non-contact probes on each of its impellers which range in operating speeds from about 20,000 up to approximately 50,000 RPM. For all practical purposes, since the displacement amplitudes become so low with increasing frequencies, these displacement measuring systems will not be sufficiently sensitive to pick up problems with the gearing or with failing thrust bearings on these machines (which are commonly outfitted with about 6 pads in each thrust bearing). However, if problems with unbalance due to impeller buildup and/or erosion occur, or if a misalignment problem takes place, the system will readily respond whether it is targeting the low-speed or high-speed impeller. But, if, for example, a problem occurs with a thrust bearing of either stage, the displacement probe will not likely report any real problem since the thrust bearing frequencies will range from 120,000 CPM up to 300,000 CPM. In the case of these machines, a common alarm level is at about 0.7 mil (Pk-Pk) while a normal trip set point is at about 1.1 mil (Pk-Pk).

Therefore, calculations show that if the low-speed stage had a high velocity of 2 in/sec Pk shaft vibration at the 120,000 CPM frequency, this would correspond to an equivalent displacement of only .318 mil (Pk-Pk) (and only .127 mil at 300,000 CPM on the high-speed thrust bearing). Even worse, these systems will not be responsive to wear of the pinion teeth since the typical fundamental gear mesh frequency (GMF) for these machines is on the order of 1,000,000 CPM (16,667 Hz). Therefore, calculations show that an excessive vibration of 100 g's acceleration at this fundamental gear mesh frequency of about 1,000,000 CPM would correspond to an equivalent displacement of only .0071 mil (Pk-Pk), making it impossible for the system to detect this pinion tooth wear. Still, however, these machines should always be outfitted with such non-contact probe permanent vibration monitors in order to detect the actual shaft vibration and the onset of problems such as misalignment and unbalance to which they are, in fact, sensitive.

#### **4.14 SHAFT CONTACT DISPLACEMENT PROBES**

Shaft contact probes (shaft sticks and shaft riders) are probes that actually ride on the shaft surface of the machine to measure the vibration. They are an older technology which has generally been replaced by the non-contact eddy current probe in most cases. However, they are still in use for specific applications such as balancing or periodic vibration checks if a proximity probe is not installed and a direct measure of shaft vibration is required. These transducers are very limited in frequency response and are not capable of measuring vibration much above 12,000 CPM (200 Hz). Also, in general, they should not be used when shaft surface speeds exceed approximately 30,000 ft/min ( $\pm 10,000$  m/min).

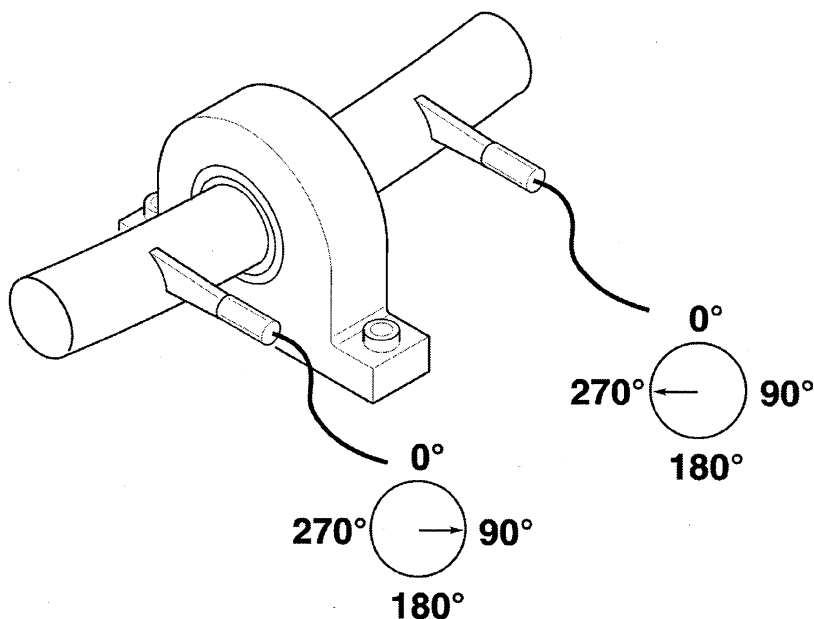
##### **4.141 Shaft Sticks**

Figure 16 is a picture of a shaft stick in use. It typically consists of a hardwood, fish-tailed" stick outfitted with a stud on its other end for attachment of a velocity pickup or accelerometer transducer. The fish-tail shape provides two points of contact required to help the shaft stick in contact with the shaft throughout its circumference and also allows it to be used with a variety of shaft diameters. Also, it is a good idea to taper the end of the stick to provide a minimum contact area with the shaft to reduce friction and to prevent chatter with the shaft. The shaft stick is hand-held to the shaft. Figure 17 shows the use of a shaft stick to measure both vibration amplitude and phase to confirm a bent shaft condition with readings taken on the shaft on each side of the bearing (if the shaft is bent, phase readings will likely differ by approximately 180° from that taken on the other side of the bearing).

As with the non-contact eddy current probe, the shaft stick is often used in applications where journal bearings are installed or where the housing or casing is very heavy compared to the shaft or rotor and vibrations are absorbed by the shaft clearances, lubrication, or heavy casings. However, the shaft stick has a much lower usable frequency range than does a non-contact probe due to its construction, and typically should only be used if a non-contact eddy current probe is not installed.



**FIGURE 16**  
**SHAFT STICK (REF. 5)**



**FIGURE 17**  
**PHASE READINGS USING SHAFT STICK ON BOTH SIDES OF**  
**BEARING HELP IN CONFIRMING A BENT SHAFT**

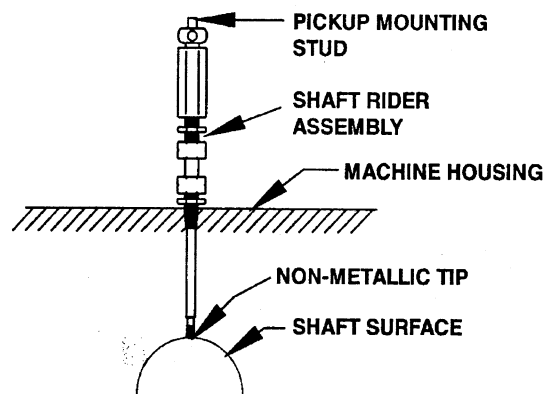
When using the shaft stick, use special care to follow these precautions:

- Hold the stick tightly to prevent it from “walking down” the shaft.
- Keep the stick perpendicular to the shaft.
- Do not mount the stick on shafts traveling faster than 10,000 RPM for safety reasons.
- Do not keep the stick on the shaft for extended periods of time which may burn the stick or score the shaft.
- Apply a continuous stream of lubricant to the shaft while taking the measurement to reduce friction, burning and scoring.
- Try to use the same pressure when taking PMP type measurements as this will affect your month to month vibration amplitudes.
- Make sure the shaft is finished, smooth and free of rust, pits or dents which can produce useless information as well as damage the stick.
- Do not rely on the shaft stick to monitor “high frequency” defects such as bearings, blade passes, gear mesh frequencies, and electrical problems.

#### 4.142 Shaft Riders

A shaft rider (pictured in Figure 18) is similar to the non-contact eddy current probe in that it is permanently mounted to the housing and extends into the casing to measure shaft vibration displacement. However, it differs from the proximity probe in that it actually rides on the shaft surface by typically using a spring mounted tip. On the other end of the shaft rider, a transducer is mounted. One of two methods of picking up the vibration is usually used. The first has an accelerometer or velocity transducer mounted on the end of the shaft rider measuring the vibration (this case is quite similar to using a probe with an accelerometer or velocity transducer). The second employs a method in which a spring loaded rod is driven by the vibration and moves a mass inside of coiled wires surrounded by magnets, thereby generating an electrical signal which is used to measure the vibration.

Once again, the shaft rider is an older technology which has gradually been replaced by the non-contact eddy current probe. Typically, the shaft rider was employed as a device to continually monitor vibration of the shaft for purposes of determining if a machine's vibration level should sound an alarm or shut down the machine. As with the shaft stick, its usable frequency range is up to about 11,000 CPM which is only useful for monitoring unbalance, misalignment, and other problems that occur at 1X or 2X RPM. Furthermore, the shaft speed must not exceed 3600 RPM to be able to detect these problems and should be used only on machines with a shaft surface speed below approximately 20,000 ft/min (6100 m/min). The shaft rider will not be able to detect gear mesh problems and other high frequency problems.



**FIGURE 18**  
**SHAFT RIDER**

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## 4.2 SELECTION CRITERIA FOR TRANSDUCERS

In order to help the reader better understand vibration transducers, a list of common transducer characteristics is presented in **TABLE I**, along with a summary of each type of transducer's typical capabilities. Table I, along with the information shared in Section 4.1 will help the analyst choose the right transducer for a particular job. Following below are descriptions of the items in Table I, along with other items which should be considered when selecting a transducer:

**1) TYPICAL SENSITIVITY RANGE** - Sensitivity is the capability of the transducer to determine the amplitude of vibratory motion (displacement, velocity, or acceleration) from the amplitude of the voltage signal. An example might be 100 mV/g (for an accelerometer), where "g" is 1 gravity unit (32.2 ft/sec<sup>2</sup> or 386 in/sec<sup>2</sup>). For example, if this 100 mV/g accelerometer generated 10 mV, it would convert this voltage to an amplitude of 0.1 g ( $10/100 = 0.1$ ).

**2) TYPICAL FREQUENCY RANGE** - Frequency range is the low and high end frequency measuring capability of the transducer. Each individual transducer has its own frequency range (which must be known by the user). However, each type of transducer has a relative range that is displayed in **TABLE I**. The low end of the frequency range is primarily controlled by the sensitivity of the transducer where the high end frequency limit is controlled by the transducer's mounted natural frequency. Typically, the transducer's frequency response might be given for various amplitude tolerances such as  $\pm 5\%$ ,  $\pm 10\%$ , and/or  $\pm 3\text{dB}$  (for example, see Figure 10).

**3) NATURAL FREQUENCY RANGE** - The natural frequency of the transducer is the limiting factor for the upper frequency range capability of the transducer. When a forcing frequency begins to approach the transducer natural frequency, the amplitude of vibration will be magnified up to as much as 50 times due to the resonant condition of the transducer. Therefore, it is very important that the highest frequency of interest is well below the natural frequency of the transducer. Normally, the effective accurate range for an accelerometer ( $\pm 5\%$ ) will be about 20% to 33% of its stud mounted natural frequency (thus, for an accelerometer with a 30,000 Hz natural frequency, its effective range will likely be limited to only about 6000 to 10,000 Hz, or 360,000 to 600,000 CPM).

**4) TYPICAL WEIGHT RANGE** - The weight of the transducer is important for the following two reasons. First, it must be light enough to carry around on a PMP route. Second, the transducer weight must only be a fraction of the weight of the housing to which it will be attached. If the transducer is heavy enough to affect the mass of the part being measured, it can have a significant effect on the part's natural frequency, and therefore, true frequency response.

**5) TYPICAL USABLE TEMPERATURE RANGE** - This is the minimum and maximum temperature that a transducer can withstand without significantly affecting its response capabilities. Make special note of this parameter when attaching permanent transducers inside of machinery that may have very high or very low temperatures.

**6) MEASUREMENT DIRECTION** - Most transducers measure only in the mounting direction (although they may respond to about 3% to 5% of the vibration from directions perpendicular to the orientation of the transducer). However, a few allow vibration to be collected in all three directions simultaneously (triaxial transducers).

**7) TRANSDUCER SIZE** - Some of the measurement locations on machinery require that the transducer be of a certain size or cross section to physically fit on a small surface for instance. Therefore, a small transducer must be used. However, due to the small size, the transducer may have a low voltage sensitivity restricting its capability to pick up low frequency information. Also, if the object that is vibrating is very small, the added mass of the transducer may change the natural frequency of the object, thereby changing its vibratory motion.

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**8) TRANSDUCER POWER SUPPLY** - Most transducers require a power supply to operate. There are 3 common methods to power a transducer:

- i) **ICP (Integrated Circuit Piezoelectric) Power Supply.** The vibration instrument sends the power through the cable attaching it to the transducer to power the transducer.
- ii) **Independent Power Supplies.** A battery or AC powered unit is used to send power to the transducer.
- iii) **Charge Amplifier.** A battery or AC powered unit is placed between the transducer and the vibration instrument to amplify the signal voltage.

**9) CABLING** - Cable length should be considered when purchasing your transducers. ICP powered transducers can generally use cables up to about 1000 feet in length. If the transducer is not ICP powered, a charge amplifier may be required at approximately 50 foot intervals of cable length (or less). Discuss the cabling requirements for the transducer with the transducer vendor.

**10) MOUNTING SENSITIVITY** - As will be described in **Sections 4.3 and 8**, there are many ways to mount transducers (hand-held probes, magnetic connectors, permanent stud mounts, etc.) and each has a significant effect on the ability of the transducer to measure the vibration accurately, as well as to reproduce the same spectra in subsequent measurements. This one fact is often critical to obtain accurate, repeatable data (and, therefore, to the success or failure of the entire PMP program).

**11) MAGNETIC INTERFERENCE** - Magnetic interference may affect the performance of the transducer, cable, and vibration instrument. Proximity probe type transducers have a magnetic field to determine the shaft gap from the transducer. If another magnetic field (introduced by electrical runout for instance) interferes with the proximity probe's field, erroneous vibration readings will be recorded. Typically, magnetic interference does not affect transducers with piezoelectric crystals. However, it can affect transducers which work on a spring-mass system (such as seismic velocity pickups). Also, cable movement generates a magnetic field and can distort the signal in the cable. Lastly, large generators or motors can produce a magnetic field which may affect spring-mass transducers, cables, and the vibration analyzer itself.

**12) SEALING METHOD** - Remember, all accelerometers will "breathe" if they are not certified as hermetically sealed. During this micro-breathing, moisture will inevitably penetrate those units which are not hermetically sealed which will change the transducer's performance dramatically, particularly at lower frequencies below approximately 3000 CPM (50 Hz). Pay attention to the specifications for your particular accelerometer. If its sealing is not listed, contact your vendor and ask this important question.

### 4.3 MOUNTING OF TRANSDUCERS

Typically, there are 5 different transducer mounting methods as listed in Table II and each has a typical frequency response range when used with transducers capable of these frequency response ranges.

**TABLE II**  
**TRANSDUCER MOUNT USABLE FREQUENCY RANGE**  
**FOR THE WILCOXON 726T (Ref. 9)**

| ACCELEROMETER MOUNTING                                   | MAXIMUM ACCEPTABLE FREQUENCY (CPM) | MOUNTING NATURAL FREQUENCY (CPM) |
|----------------------------------------------------------|------------------------------------|----------------------------------|
| 1) Stud Mount                                            | 975,000                            | 1,900,000                        |
| 2) Adhesive Mount with Hottinger Baldwin Messtechnik X60 | 540,000                            | None Observed                    |
| 3) Stud Mount on Rare Earth Magnet                       | 450,000                            | 724,500                          |
| 4) Mounted on Quick-Connect Stud Mount                   | 360,000                            | 609,000                          |
| 5) Hand-held Mount Using a 2" Probe                      | 48,000                             | 88,500                           |

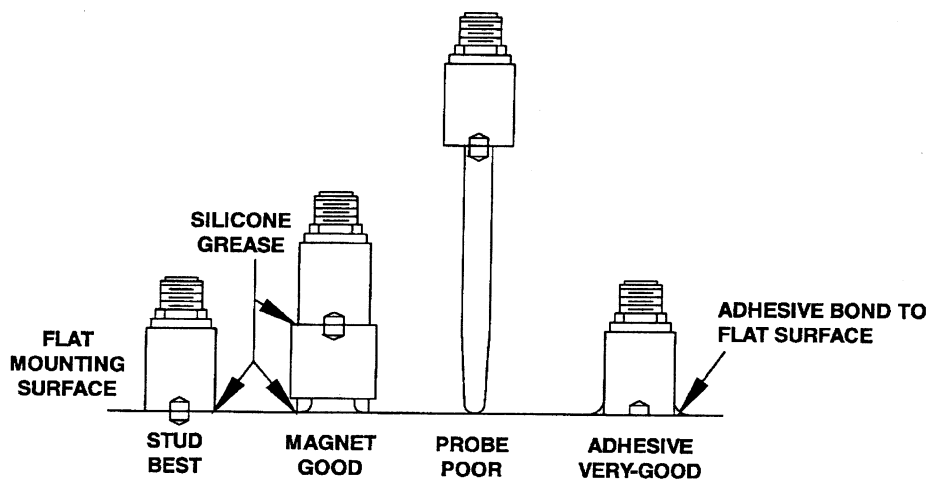
Table II is a summary of an article written by Computational Systems, Inc. (CSI)<sup>9</sup>. Many other factors come into play when the effectiveness of a transducer mounting type is considered. Refer to the referenced article for an in-depth look at transducer mounting effectiveness. Figure 19 shows illustrations of these mounting methods (Quick-Lock not shown).

#### 4.31 TRANSDUCER MOUNTING APPLICATIONS

Each particular transducer mounting has certain applications based on a number of factors. These applications will be discussed in detail for each mounting method.

##### STUD MOUNTING

Stud mounting is used for permanently mounted transducer applications since it provides the highest frequency response range and is an effective means of keeping the transducer in place. Sometimes an adhesive will be used in combination with stud mounting to prevent the transducer from working its way off of the stud mount. Typically, stud mounting is not practical for collecting PMP route data due to the length of time required to attach and detach the transducer. However, if very high frequency measurements (>3,000,000 CPM or 50,000 Hz) are required, stud mounts will have to be used at those data collection points. Also, stud mounting gives extremely repeatable data from measurement to measurement over periods of time.



**FIGURE 19**  
**MOUNTING METHODS ILLUSTRATED (REF. 1)**

#### **ADHESIVE MOUNT**

Adhesive mounts also have a very good frequency response range if the proper adhesive type is used and if the proper adhesive thickness is applied (if too thick, this will add damping and degrade the high frequency performance of the mount). However, these mounts may work loose from the machine over long periods of time. Therefore, care should be taken when using an adhesive mount in a permanently mounted transducer configuration. Adhesive mounts are sometimes used in a PMP route when high frequency data is to be collected (>750,000 CPM or 12,500 Hz); however, these are also time consuming. Adhesive mounts are very useful in a diagnostics situation when reliable high frequency data is to be analyzed since a stud mount is rarely found at the point a measurement is desired. Adhesive mounting also gives very repeatable data over a series of measurement surveys.

#### **MAGNETIC MOUNT**

Magnetic mounts are the most common used method in PMP programs, as well as when taking diagnostic data. Their response range is generally adequate for most PMP and diagnostic needs (ranging from about 120,000 to 450,000 CPM or 2000 - 7500 Hz). However, it is important to note that some machines such as centrifugal compressors have gear mesh frequencies and harmonics that start at 900,000 CPM (15,000 Hz) and go up to 4,000,000 CPM (almost 70,000 Hz). Data on these machines should be taken with an adhesive or stud mount. A magnetic mount also tends to give reliable repeatability over measurement surveys which is adequate for PMP purposes.

#### **QUICK-CONNECT**

Quick-Connect mounts are also ideal for collecting PMP route data since they provide easy mounting and dismounting, as well as allowing a relatively large frequency range that is able to detect most common machinery problems which occur in higher frequencies such as bearing defects. Also, repeatability between measurement surveys is consistent enough for PMP purposes. However, a quick-connect mount should not be used for detecting frequencies above approximately 420,000 CPM (7000 Hz) such as those found in high speed compressor gear mesh frequencies and harmonics.

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## HAND-HELD PROBE

This is the most undesirable mounting method. The usable frequency range is only up to a maximum of 30,000 to 60,000 CPM (500 - 1000 Hz), no matter the length, diameter or material of the probe! Depending on the length of probe used, the maximum accurate frequency measurable may be as low as 30,000 CPM. With these constraints, the only machinery problems detectable and repeatable will normally be those which occur at 1X, 2X, and 3X RPM for machinery running at 3600 RPM or less. Most importantly, bearing defect frequencies and their harmonics as well as bearing natural frequencies often lie above 30,000 CPM and will be missed entirely if using a hand-held probe. Furthermore, as demonstrated in the research covered in Reference 9, the repeatability of measurement amplitudes and maximum frequency ranges from survey to survey is not consistent and will vary depending on the probe position or amount of hand pressure applied. This impairs the analyst's ability to accurately trend problems and may cause him to miss some problems that occur at the top (or beyond) of the probe's measurable frequency range if insufficient hand pressure was applied during a survey. Handholding a transducer or probe is useful in hard-to-reach places such as where a screen prevents the use of a magnet or other mount type. Also, it is useful if safety is a concern and the analyst cannot safely reach into moving parts of the machine. However, be aware that using this method will miss very important information that is crucial to maintaining a successful PMP program. In the instances described above, it is wiser to attach a permanently mounted transducer in unsafe or hard to reach locations to obtain useful diagnostic data. **Chapter 8 - "COMMON PITFALLS IN EVERYDAY VIBRATION MEASUREMENT"** has an entire section (**8.16 Transducer Mounting and Probes**) devoted to problems that occur with this type of mounting method, as well as real examples of data collected where a bearing is obviously near failure when measured with a magnet or adhesive mount, but does not even show up with a probe type mount. Furthermore, this example shows that probes actually add "phantom" data in the bearing defect range which makes the bearing appear to be in its last stage of failure due to the natural frequency of the probe being excited even though there may be no bearing wear at all.

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**APPENDIX**  
**SPECIFICATIONS FOR VARIOUS TRANSDUCERS FROM A VARIETY OF MANUFACTURERS**

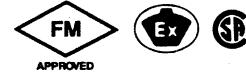
(Transducers are sorted alphabetically by vendor name, then numerically for  
each transducer model number for each vendor)

Sample Transducers are included from each of the following vendors:

- Bently Nevada based in Minden, NV.
- Entek IRD International based in Milford, OH.
- PCB Piezotronics, Inc. based in Depew, NY.
- Rion Co., LTD. based in Tokyo, Japan.
- Vibra Metrics, Inc. based in Hamden, CT.
- Wilcoxon Research based in Rockville, MD.

## 7200 Proximity Transducer Systems

5 mm and 8 mm



### Accurate, reliable measurements

The 7200 5 mm and 8 mm Proximity Transducer Systems are noncontacting, gap-to-voltage transducer systems that measure static as well as dynamic distances between the probe tip and the observed target. The general application is any requirement for an accurate, non-contacting displacement measurement. However, the most common use is for shaft position and vibration measurements on rotating and reciprocating machinery. They are designed for use with virtually any machine type including gas and steam turbines, compressors, pumps, centrifuges, electric motors and generators. The systems offer 80 mils (2 mm) of linear measuring range and are compatible with API 670 requirements.

A system consists of a proximity probe with integral coaxial cable, an extension cable, and Proximitor®. All 7200 proximity transducer systems are compatible with 3300 Monitoring Systems.

### Applications

The 7200 5 and 8 mm Systems have a frequency response of 0 Hz (DC) to 10 kHz and can be used to make the following types of measurements:

- Radial vibration for indicating bearing condition and such machine malfunctions as rotor imbalance and misalignment.
- Axial thrust position for determining thrust bearing wear or potential bearing failure.
- Shaft average radial position, for determining attitude angle, an indicator of rotor stability.
- Vibration amplitude and phase angle for plotting diagnostic information in Polar and Bodé formats.
- Eccentricity to measure the amount of rotor bow.

### How a proximity transducer system works

Operating on the eddy current principle, the proximity probe senses the distance between the probe tip and the observed surface. The Proximitor® generates a radio frequency signal, which is radiated through the probe tip into the

observed surface. Eddy currents are generated in the surface, and the loss of strength in the return signal is detected by the Proximitor®, which conditions the signal for linear display on a monitor.

### Improved proximity probes

An improved 8 mm version of the 7200 System is now standard which includes Bently Nevada's patented Cable Loc™ feature. This technique of securing the coaxial cable to the probe tip provides the strongest junction yet offered by Bently Nevada and increases the durability of the probe during installation and usage.

Two cable length options are also now available on 8 mm probes. They are 5 metres and 9 metres of integral cable. Since the entire lead-length is built into the probe, an extension cable is not used with these length options.

Nine probe configurations are available to accommodate English and metric thread requirements (See Table 1). The 5 mm probe tips are constructed of fiberglass. The 8 mm probe tips are constructed of polyphenylene sulfide, (PPS), a high performance plastic capable of withstanding unusually harsh, wet and/or chemical environments.

Both transducers are suitable for use in a confined space and offer a scale factor of 200 mV/mil (8 V/mm).

### Extension cables

The combination of probe with integral lead and extension cable is designed to achieve a system length of either five or nine metres from probe tip to Proximitor. Probes are available with integral leads of various lengths to match the average extension cable lengths. Probes and cables of the same length, and Proximitors with the same part number, are completely interchangeable.

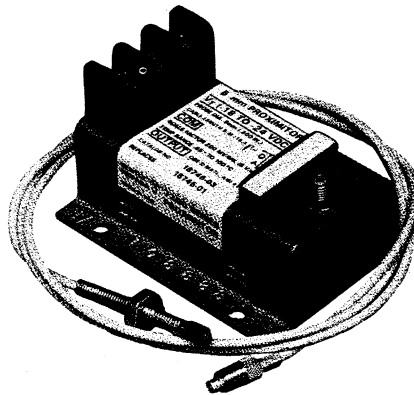
For added protection, stainless steel armor is available. It provides improved resistance to physical abuse.

### Proximitors

The 5 mm and 8 mm systems use the same type of Proximitor®. Both 5 and 9 metre Proximitors are available. A three-conductor, shielded cable is available to provide the signal output and power input between 7200 Proximitors and Bently Nevada monitors. Proximitors can be located up to 1,000 feet (305 metres) from standard Bently Nevada monitors for signal connections.

### Hazardous locations

Bently Nevada 7200 Proximity Transducers can be certified by CSA, BASEEFA and FM for use in specific hazardous locations. Refer to Data Sheet L1035 "Machinery Monitoring Systems for Hazardous Locations" for more details.



7200 Proximity Transducer System

## 7200 Proximity Transducer Systems

5 mm and 8 mm

### Specifications

Specifications were determined with a -24 Vdc power supply, 10 k $\Omega$  load, and an AISI 4140 steel target at +72°F (+22°C).

#### INPUT

**Power:** -17.5 Vdc to -26 Vdc at 12 mA maximum consumption.

**Leadwire Length:** 1,000 feet (305 metres) maximum between Proximitor® and 3300 monitor. Consult manual for frequency roll-off at longer lengths.

#### OUTPUT

**Calibrated Range:** 80 mils (2.0 mm). Begins at approximately 10 mils (0.25 mm) from probe face.

**Scale Factor:** 200 mV/mil (7.87 V/mm),  $\pm 4\%$  (measured in increments of 10 mils over the calibrated range) if calibrated as a system. Within  $\pm 9.5\%$  including interchangeability errors.

**Linearity:** Within 0.8 mils (0.02 mm) of a 200 mV/mil straight line if calibrated as a system. Within 2.3 mils (0.06 mm) of a 200 mV/mil straight line including interchangeability errors.

**Frequency Response:** 0 to 10 kHz (0 to 600,000 cpm); -5% at 10 kHz (600,000 cpm).

**Temperature Sensitivity:** Typically -3% change in scale factor at +150°F (+65°C) at 50 mils gap (1.27 mm).

**Minimum Target Size:** .6 inch (15 mm) Diameter.

#### ENVIRONMENTAL LIMITS

##### Operating Temperature:

**Proximitor®:** -60°F to +212°F (-51°C to +100°C).

**Probe and Extension Cable:** -30°F to +350°F (-34°C to +177°C).

**Relative Humidity:** Product produced before 01 January 1992: To 95%, non-condensing. Product produced after 01 January 1992: To 100%, condensing, non-submerged (with protection of coaxial connector).

**System Weight:** 1.3 lbs. (0.59 kg).

### Ordering Information

#### 5 mm and 8 mm Standard Mount Probe

A B C D  
□□□□ - □□ - □□ - □□ - □□

##### Option Descriptions

##### □□□□ Probe Part Number Option

Select from Table 1.

##### A □□ Unthreaded Length Option ①

Order in increments of 0.4 inches [04] for English thread, 10 mm [01] for metric threads.

##### English thread configurations:

*Maximum unthreaded length:*

8.8 inches [88]

*Minimum unthreaded length:*

0.0 inches [00]

**Example:** [04] = 0.4 inches

##### Metric thread configuration:

*Maximum unthreaded length:*

230 mm [23]

*Minimum unthreaded length:*

0.0 mm [00]

**Example:** [06] = 60 mm

##### B □□ Overall Case Length Option

Order in increments of 0.4 inches [04] for English thread, 10 mm [01] for metric threads.

##### English thread configurations:

*Maximum case length:*

9.6 inches [96]

*Minimum case length:*

0.8 inches [08]

**Example:** [24] = 2.4 inches

##### Metric thread configuration:

*Maximum length:*

250 mm [25]

*Minimum length:*

20 mm [02]

**Example:** [04] = 40 mm

##### C □□ Total Length Option

05 0.5 metre (20 inches).

*Tolerance:* +.13 metre, -.05 metre (+5 inches, -2 inches).

10 1.0 metre (39 inches).

*Tolerance:* +.25 metre, -.10 metre (+10 inches, -4 inches).

50 5 metre (195 inches).③

*Tolerance:* +.6 metre, -.5 metre (+24 inches, -20 inches).

90 9 metre (351 inches).

*Tolerance:* +1.0 metre, -.9 metre (+39 inches, -35 inches)

##### D □□ Connector Option

00 Without connector.

02 With miniature male coaxial connector.

#### 8 mm Reverse Mount Probe

A B C D  
21508 - 02 - 12 - □□ - □□

##### Option Descriptions

##### C □□ Total Cable Length Option

05 0.5 metre (20 inches).

*Tolerance:* +.13 metre, -.05 metre (+5 inches, -2 inches).

10 1.0 metre (39 inches).

*Tolerance:* +.25 metre, -.10 metre (+10 inches, -4 inches).

50 5 metre (195 inches).③

*Tolerance:* +.6 metre, -.5 metre (+24 inches, -20 inches).

90 9 metre (351 inches).

*Tolerance:* +1.0 metre, -.9 metre (+39 inches, -35 inches)

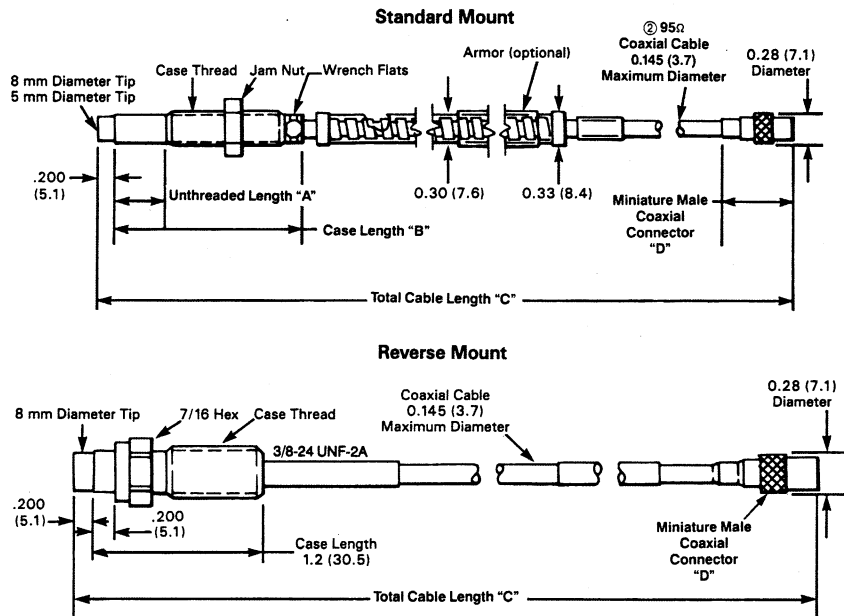
##### D □□ Connector Option

00 Without connector.

02 With miniature male coaxial connector.

## 7200 Proximity Transducer Systems

5 mm and 8 mm



5 mm and 8 mm Proximity Probe Configurations ④ ⑤

TABLE 1  
Probe Part Number Option Descriptions

| TIP SIZE         |          | CASE                  |         |                               |
|------------------|----------|-----------------------|---------|-------------------------------|
| PART NO.<br>□□□□ | DIAMETER | CONFIGURATION<br>TYPE | THREADS | STAINLESS STEEL<br>LEAD ARMOR |
| *21500-          | 5 mm     | Standard              | 1/4-28  | No                            |
| *21501-          | 5 mm     | Standard              | 1/4-28  | Yes                           |
| 21504-           | 8 mm     | Standard              | 3/8-24  | No                            |
| 21505-           | 8 mm     | Standard              | 3/8-24  | Yes                           |
| 21508-02-12      | 8 mm     | Reverse               | 3/8-24  | No                            |
| 22810-           | 8 mm     | Standard              | M10 × 1 | Yes                           |
| 22811-           | 8 mm     | Standard              | M10 × 1 | No                            |
| *22812-          | 5 mm     | Standard              | M8 × 1  | Yes                           |
| *22813-          | 5 mm     | Standard              | M8 × 1  | No                            |

\* Maximum case length for all M8 × 1 probes is 100 mm; and for 1/4-28 probes is 4 inches. Also, maximum overall length for 5 mm probes is 1 metre.

### NOTES:

- ① Unthreaded length must be at least 0.8 inches (20 mm) less than case length.
- ② Maximum diameter for 21500 and 21501 probes is 0.111 in. (3.4 mm).
- ③ For use with the 18745-03 five-metre Proximitor® only.
- ④ Letters inside quotation marks refer to probe ordering options.
- ⑤ Dimensions in parentheses are in millimetres.

## 7200 Proximity Transducer Systems

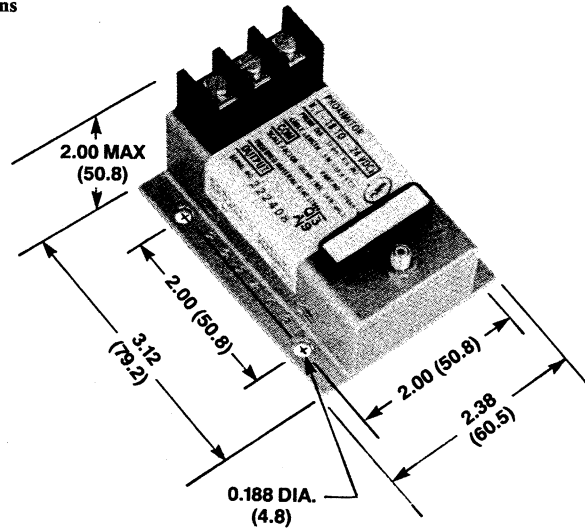
5 mm and 8 mm

### Proximitor®

18745-□□

- 03 For combined probe and extension cable system length of 5 metres (16.4 feet).  
04 For combined probe and extension cable system length of 9 metres (29.6 feet).

### Dimensions



Dimensions in parentheses are in millimetres.

### Extension Cable

A B  
21747 - □□□ - □□

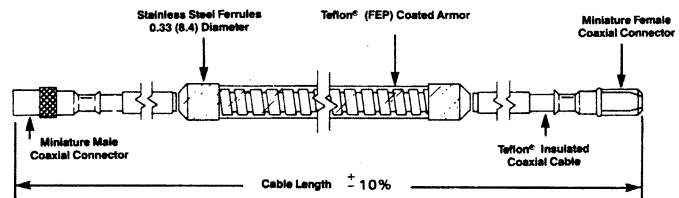
#### Option Description

- A □□ Cable Length Option ⑥  
040 4.00 metres (157 inches).⑦  
045 4.50 metres (177 inches).⑦  
080 8.00 metres (315 inches).  
085 8.50 metres (335 inches).

- B □□ Armor Option  
00 Without armor.  
01 With armor.

#### NOTES:

- ⑥ Extension cable physical length equals the electrical length,  $\pm 10\%$   
⑦ For use with the 18745-03 five-metre Proximitor® only.



## Velocity Transducer Systems

### 86205

**Our low cost solution for monitoring rolling element bearings on plant machinery**

Bently Nevada's 86205 case-mounted Velocity Seismoprobe® is a cost-effective solution for monitoring rolling element bearings on machines where most of the shaft vibration is transmitted to the case. On some machinery, it may not be possible to install proximity probes to observe actual shaft vibration. This can be due to a variety of reasons including installation cost or mounting restrictions. The 86205 can provide valuable condition data on general purpose or essential machinery such as:

- motors
- pumps
- fans
- air handlers
- centrifugal chillers
- mill rollers

#### Economical cost

In the past, it was too expensive to install permanent monitoring equipment on general purpose or balance-of-plant machinery. There was simply too much equipment involved. However, more timely machine condition data was needed. A cost-effective transducer was required to address these machines.

Bently Nevada's 86205 Velocity Seismoprobe® is the answer. Innovative design and manufacturing techniques make it possible to keep the cost of this two-wire transducer low. Now cost-effective transducers can be installed on large numbers of support machinery without sacrificing integrity.

#### Easy mounting methods

Installation cost is a major factor when deciding whether or not to place permanently-installed transducers on a large number of machines. Some applications, especially those involving machines with thin cases or housings, do not allow for drilling and tapping. In these applications, it may be more economical to adhesively-mount transducers on machine cases. An adhesive-mount base is available as an accessory for quick and easy installation.



86205 Velocity Seismoprobe®

#### Machine servicing requirements

When machine servicing is required, installed transducers need to be removed. In the past, this was difficult and damage to the transducer or cabling occurred. The 86205 Velocity Seismoprobe® is designed with a mechanic's needs in mind.

When machine servicing is required, the transducer housing cap, connector and cable assembly are easily disconnected by loosening two screws. This cable/cap assembly is then set aside. The 86205 can then be unscrewed from its base or stud and stored accordingly. Only the adhesive base or mounting stud remains on the machine. Mechanics can easily do their jobs without machine condition equipment in the way.

#### Use with Trendmaster® 2000 System

The 86205 is specifically designed for use with Bently Nevada's cost-effective, automated data collection and diagnostic system, Trendmaster® 2000 (data sheet L4110). Together, these two products are an effective method of collecting machine information. The result is fewer unplanned shutdowns due to undetected machine malfunctions such as bearing failure. When used with the Trendmaster® 2000, the (part number 86950-01) Velocity Transducer Interface Module must be ordered.



#### Designed for industrial use

The 86205 consists of a molded polyphenylene sulfide housing which contains a Seismoprobe® velocity transducer cartridge. It is available in both vertical and horizontal mounting configurations. The compression fit and O-ring seal make the transducer environmentally protected (NEMA 4X rated). A 5-pin insulation displacement connector is supplied with the transducer to connect it to the transducer cable (part number 85033). The transducer is also designed so the cap assembly can be attached in either direction. This results in a simplified connection during installation.

The 86205 Velocity Seismoprobe® includes a female 3/8-24 UNF threaded insert for mounting. An adaptor to convert to a female metric M6 x 1.0-6H thread, adhesive-mount base kits, and a variety of mounting studs are available as standard options.

For hazardous areas, the 86205 is approved by BSEEFA, Canadian Standards Association (CSA) and Factory Mutual (FM). BSEEFA approvals are for Zone 1, Group II C. CSA approvals are for Division 1 and 2, all Groups and Classes. FM approvals are for Divisions 1 and 2, Class I, Groups A through D.

Approvals are based on using the 86205 Velocity Seismoprobe® with the Trendmaster® 2000 system.

## Velocity Transducer Systems

### 86205

### Specifications

Specifications are at +72°F (+22°C) and 1 in/sec (25 mm/s) at 100 Hz (6,000 cpm) unless otherwise specified.

#### ELECTRICAL

**Frequency Response:** 10 Hz to 1000 Hz (600 cpm to 60,000 cpm), -3dB typical.

**Sensitivity:** 500 mV/in/sec (20 mV/mm/s) zero-to-peak  $\pm 5\%$  at 100 Hz (6,000 cpm) and 1 in/sec (25 mm/s) with a mounting tolerance of  $\pm 2^\circ$  for both vertical and horizontal orientations when operated with a 24.9 k $\Omega$  load.

**Orientation Sensitivity Change:**  $\leq 2\%$  at  $\pm 20^\circ$  (vertical) or  $\pm 10\%$  (horizontal).

**Input Load:** 24.9 k $\Omega \pm 1\%$ .

**Amplitude Linearity Error:**  $\pm 5\%$  at 100 Hz (6,000 cpm) from 0.1 to 4.0 in/sec (2.5 to 100 mm/s) zero-to-peak.

**Amplitude Range:** 32 mils (0.81 mm) peak-to-peak, maximum.

**Transverse Sensitivity:**  $<12\%$  typical over operating frequency.

**Coil Resistance:** 395  $\Omega \pm 5\%$ .

**Temperature Coefficient:** -0.04%/°F (-0.22%/°C).

**Polarity of Output Signal:** When case movement is toward the connector, the center male pin provides positive signal relative to the adjacent pins.

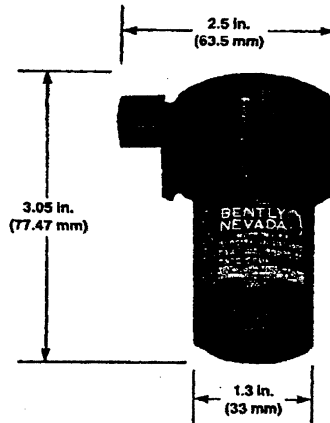
#### ENVIRONMENTAL LIMITS

##### Temperature Range:

**Operating:** -22°F to +180°F (-30°C to +82°C).

**Storage:** -30°F to +180°F (-34°C to +82°C).

**Relative Humidity:** 100%, nonsubmerged. (NEMA 4X requirement).



86205 Velocity Transducer Dimensions

#### PHYSICAL

**Reference Height:** 3.05 inches (77 mm) with cap assembly.

##### Case Body Diameter:

Cap: 1.8 inches (46 mm) nominal.

Body: 1.3 inches (33 mm) nominal.

**Reference Weight:** 7 ounces (200g)

**Case Material:** Molded polyphenylene sulfide with 304 SST threaded insert.

**Mounting:** 3/8-24 UNF-2B female threaded insert is standard. Adhesive-mount base kits and threaded mounting studs are available by ordering the standard Mounting Adapter option.

#### MECHANICAL

**Maximum Shock:** 50g (490 m/s<sup>2</sup>).

### Ordering Information

#### Velocity Seismoprobe®

A 5-pin cable connector and cover are included for cable connection.

A B

86205 - ☐ - ☐

#### Option Descriptions

##### A ☐ Transducer Orientation Option

- 01 Vertical ( $0^\circ \pm 20^\circ$ )
- 02 Horizontal ( $90^\circ \pm 10^\circ$ )

##### B ☐ Mounting Adapter Option

- 01 3/8-24 UNF
- 02 1/2-20 UNF
- 03 Adhesive kit, 3/8-24 UNF
- 04 M6  $\times$  1
- 05 Adhesive kit, M6  $\times$  1
- 06 1/4-28 UNF

#### ACCESSORIES

##### Transducer Cable

4-conductor cable with drain wire; 18 AWG (1.0 mm<sup>2</sup>); ordered in rolls (refer to ordering information below). Cable specifications are available in the Trendmaster® 2000 data sheet L4110. Maximum recommended cable length per transducer is 98 feet (30 metres).

A

85033 - ☐ - 00

#### Option Descriptions

##### A ☐ Cable Roll Length Option

- 01 500 ft (150 metres)
- 02 1000 ft (305 metres)

##### Cable Connectors

When ordering the transducer, a connector to the transducer cable is provided. The following spare parts are available.

- 00523103 5-pin, 18 AWG (1.0 mm<sup>2</sup>) connector for interfacing cable to transducer.
- 00523104 Cover for 00523103 connector.

## Velocity Transducer Systems

### 86205

#### Cable Connection Tools

85718-01 Crimp tool assembly for attaching connectors to cable. One is required for cable installation.

89138-01 Cable armor adaptor; 5/16" armor.

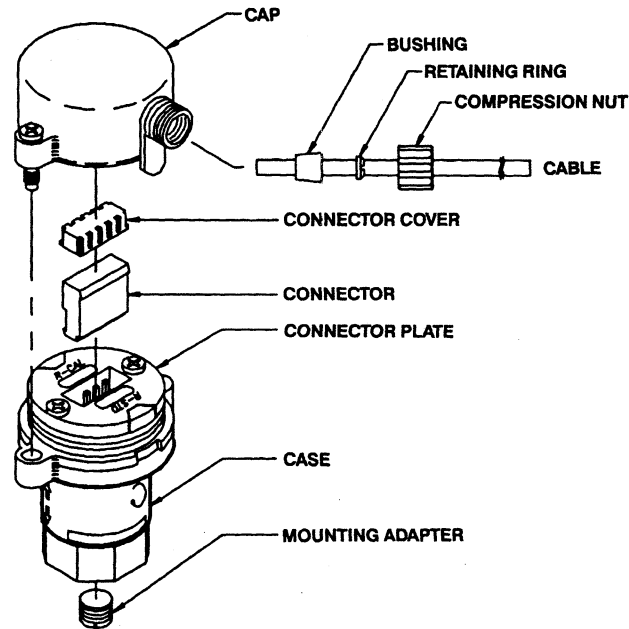
#### Adhesive-mount Base Kits

Adhesive-mount base kits are available for quick and easy transducer mounting. The kit contains materials (adhesive and bases) for adhesive-mount bases for Velocity Seismoprobes®.

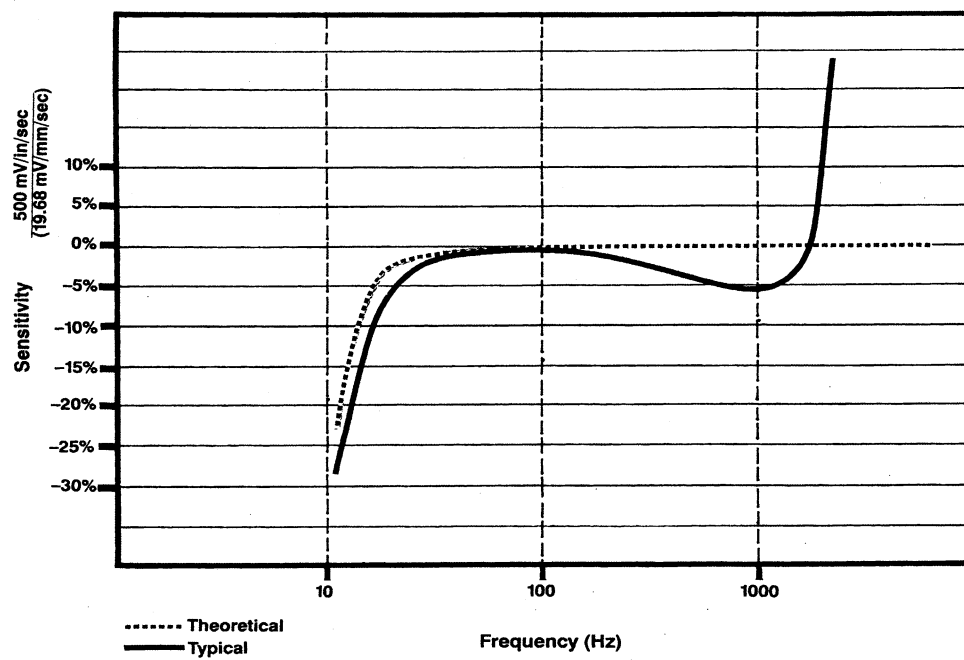
### CAUTION

If casing vibration measurements are being made for the overall protection of a machine, thought should be given to the usefulness of the measurement for each application. Most common machine malfunctions, such as unbalance, misalignment, etc., occur on the rotor and originate as an increase (or at least a change) in rotor vibration. For any casing measurement alone to be effective for overall machine protection, a significant amount of rotor vibration must be faithfully transmitted to the machine casing or mounting location of the transducer.

In addition, care should be exercised in the physical location of the velocity sensor on the bearing housing or machine casing. Improper installation may result in a decrease of the transducer amplitude and frequency response and/or the generation of false signals that do not represent vibration on that particular machine. For more information, refer to the appropriate Instruction Manuals and Application Notes.



# Velocity Transducer Systems 86205



86205 Velocity Transducer Frequency Response

## Dual Probe Transducer System

26530

### Measures absolute shaft motion on turbine generators

Large rotating machinery can often have substantial absolute shaft motion, which is the result of rotor vibration combined with excessive bearing housing vibration. Absolute vibration consists of radial vibration of the rotor and seismic vibration of the bearing housing or machine case. When significant bearing housing vibration exists, a Dual Probe Transducer System can provide additional information on the condition of the machine. Bently Nevada recommends the Dual Probe System when **more than one-third** of the shaft vibration is transmitted to the bearing housing.

The Dual Probe's primary function is to measure absolute motion on a machine. From this measurement, you're able to obtain vital diagnostic information on rotor position, vibration waveform, and phase angle.

Two types of measurements are required to determine absolute rotor motion: casing or foundation vibration and rotor motion relative to the bearing. The Dual Probe uses an eddy current proximity probe to measure rotor motion and a Velocity Seismoprobe® transducer to measure casing or foundation vibration.

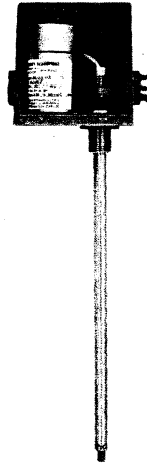
A Dual Probe Monitor vectorially sums the two signals to yield an absolute motion measurement. Prior to summation, the proximity and integrated velocity signals are available at the monitor for examination.

The Dual Probe Transducer System consists of a proximity probe, a Velocity Seismoprobe® transducer, adapter, side fittings, and a probe support/oil seal. It is designed to be weatherproof when properly installed. The Dual Probe is used in conjunction with the 3300/65 Dual Probe Monitor.

### Types of measurements

By using the Dual Probe to observe absolute motion during fixed running speed and startup and shutdown, new insights can be found on the machine's mechanical condition. You can measure:

- Rotor motion relative to bearing motion.



Dual Probe Transducer System

- Rotor absolute motion relative to free space.
- Rotor position relative to the bearing.
- Bearing cap velocity.

Many machinery vibration problems begin with rotor position changes within the bearing clearance. By making rotor position and rotor motion measurements, it's possible to identify a fault before it becomes evident from other measurement parameters.

For complete machinery monitoring diagnostics using orbit analysis and shaft centerline position information, two Dual Probe assemblies in an XY configuration are required per radial bearing. Thus, Dual Probe assemblies are often used in pairs on each bearing to allow Polar, Orbit and Timebase formats used in balancing and machine analysis. However, they can be used as a standard single channel absolute vibration measurement in a wide variety of applications.

Additional information on the Dual Probe is available in an Applications Note entitled **Bently Nevada Dual Probe Versus Shaft Rider** (AN006). Also available are reprints: **Evaluating TSI Instrumentation for Retrofit Program** (L8105) and **An Expanding Role for Turbine Supervisory Instrumentation** (L8108).

### Proximity probe

The Dual Probe Proximity Transducer is available in high performance plastic capable of withstanding unusually harsh, wet and/or chemical environments.

The probe coil diameter of both versions is 8 mm. The probes are offered in one-metre and half-metre lead lengths.

Other Bently Nevada Proximity Probes are available upon request. For more information, please contact your nearest Bently Nevada sales representative.

### Velocity Seismoprobe® Transducer

The Velocity Seismoprobe® Transducer is available in standard and high temperature versions. It is ordered according to the intended mounting angle.

### Adapter

The standoff adapter provides clearance between the probe housing and machine casing. It is ordered according to the length required and is available in 7/8-14 UNF, 1-11½ NPT, and 3/4-14 NPT mounting threads.

### Side fittings

Side fittings are used to connect conduit or cabling to the Dual Probe housing. Four configurations are available.

### Probe Support/Oil Seal

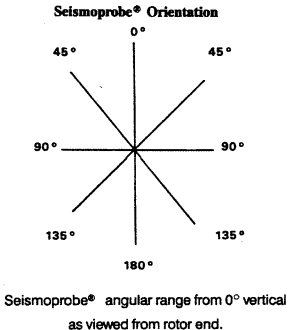
The Probe Support/Oil Seal is available as an option. It can protect the transducer system from oil and provides additional support for the probe sleeve.

Some type of sleeve support is required for sleeve lengths greater than 12 inches (305 mm). Without additional support, sleeve resonances may develop, causing inaccurate relative probe measurements or sleeve failure.

\_\_\_\_\_

### Relative Probe Adjustment

| Adapter Mounting Thread | Dimension A on Drawing        | Adjustment Range From Nominal Setting of Dimension B on Drawing |                          |
|-------------------------|-------------------------------|-----------------------------------------------------------------|--------------------------|
|                         |                               | Toward Target                                                   | Away From Target         |
| Without Adapter         | Not Applicable                | 1.02 inches<br>(25.9 mm)                                        | 1.03 inches<br>(26.2 mm) |
| 7/8-14 UNF              | Less than 2 inches<br>(51 mm) | 1.02 inches<br>(25.9 mm)                                        | 1.03 inches<br>(26.2 mm) |
| 1-11½ NPT               | Less than 2 inches<br>(51 mm) | 0.85 inch<br>(21.6 mm)                                          | 0.86 inch<br>(21.8 mm)   |
| 3/4-14 NPT              | Less than 2 inches<br>(51 mm) | 0.88 inch<br>(22.4 mm)                                          | 0.89 inch<br>(22.6 mm)   |



## Ordering Information

26530 -   -   -   -    -

**A B C D E**

**F G H**

-   -   -

### Option Descriptions

### A ☐ Proximity Probe Option

- 00 Probe not required  
05 7200 Series 8 mm probe  
12 3300 Series 8 mm probe  
19 3300 Series 5 mm probe.

### B ☐ ☐ Probe Lead Length Option

- 00 Probe not required  
05 0.5 metre  
10 1.0 metre

### C ☐ Adapter Thread Option

- 00 Adapter not required  
01 7/8-14 UNF  
02 1-11½ NPT  
03 3/4-14 NPT

**D    Adapter Length Option**

(Dimension A on dimension drawing)

Order in increments of 0.1 inch (2.5 mm).

**Minimum length:** 1.7 inches

**017** (43 mm)

*Maximum length: 2.0 inches*

020 (51 mm)

If you do not require an adapter, please enter .

### E ☐ ☐ ☐ Probe Penetration Option

(Dimension B on dimension drawing)

Order in increments of 0.1 inch  
(2.5 mm).

**Minimum length: 2.0 inches**

020 (51 mm)

Maximum length is dependent upon the Adapter Length (Dimension A).

**For 0.5 metre probe cable:**

Maximum length of Dimension A  
+ Dimension B is  
13.3 inches 133 (338 mm).

**For 1.0 metre probe cable:**

Maximum length of Dimension A + Dimension B is  
31.0 inches  $\boxed{3110}$  (787 mm).

**F ☐ ☐ Velocity Seismoprobe® ①**  
**Transducer Option**

- 00 Without Velocity Seismoprobe®
- 
- Transducer

**For 600 cpm Minimum Operating Frequency:**

| Seismoprobe®<br>Type         | Angular<br>Range |
|------------------------------|------------------|
| 21 Standard, 2 wire          | 0° ± 100°        |
| 22 High temperature, 2 wire, | 0° ± 100°        |

### G ☐ ☐ Fittings Option

- 00 Not required
- 01 One 3/4 NPT conduit fitting and one 3/4 NPT plug
- 02 Two 3/4 NPT conduit fittings
- 03 One 3/4 NPT plug, one 3/4 to 1/2 NPT reducer, and one cable seal grip with grommet
- 04 One 3/4 NPT plug

### H ☐ ☐ Probe Support/Oil Seal Option

- 00 Not required  
01 Required

**Notes:**

- ① **Standard temperature Seismoprobe®** rated  $-20^{\circ}\text{F}$  to  $+250^{\circ}\text{F}$  ( $-29^{\circ}\text{C}$  to  $+121^{\circ}\text{C}$ ). **High temperature Seismoprobe®** rated  $-20^{\circ}\text{F}$  to  $+400^{\circ}\text{F}$  ( $-29^{\circ}\text{C}$  to  $+204^{\circ}\text{C}$ ).

## Dual Probe Transducer System

### 26530

#### ACCESSORIES

40467 -   -   Individual Sleeve

##### Option Descriptions

To order an individual sleeve, add Adapter Length (Dimension A) and Probe Penetration (Dimension B) plus 2.4 inches (61 mm) for length inside housing.

Order in increments of 0.1 inches

Maximum Dimension A: 2.0 inches  
    (51 mm)

Minimum Total Sleeve Length: 4.4 inches  
    (112 mm)

Maximum Total Sleeve Length: 33.0 inches  
    (838 mm)

The recommended maximum Total Sleeve Length is:

For 0.5 metre probe cable: 13.3 inches  
    (338 mm)

For 1.0 metre probe cable: 31.0 inches  
    (787 mm)

40468 -   -   Individual Standoff Adapter

##### Option Descriptions

A   Base Thread Option

01 7/8 - 14 UNF

02 1 - 11 1/2 NPT

03 3/4 - 14 NPT

B   Length Option

Order in increments of 0.1 inches.

Minimum Length for Base Thread Option:

-01: 1.70 inches

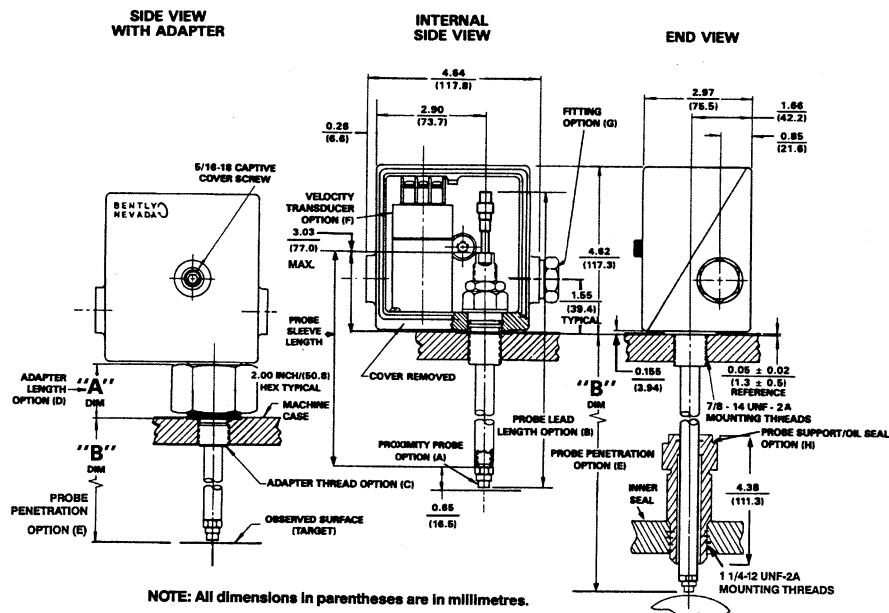
-02: 1.90 inches

-03: 1.80 inches

Maximum length: 2.0 inches  
(51 mm)

37948-01 Individual Probe Support/Oil Seal made from Ertalon.

TW8029326 Instruction Manual



## Model 911

Accelerometer

EFFECTIVE: Preliminary

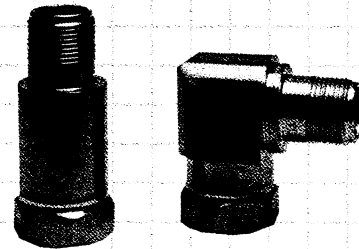
### DESCRIPTION

The IRD 911 is a solid-state seismic accelerometer for permanent mounting on machinery. It is manufactured in a fully welded 316L SSTL housing with a hermetic sealed connector, that will withstand harsh industrial environments including high temperatures and splashing fluids.

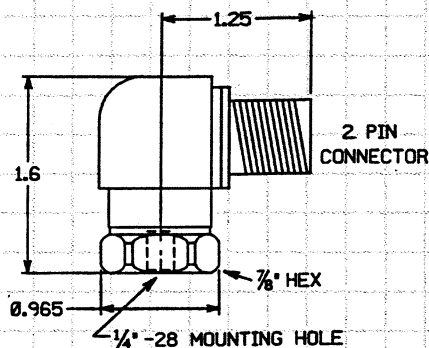
The IRD 911's wide frequency and amplitude ranges accommodate most types of rotating machines. The small size permits installation near bearings.

The IRD 911 has an internal, low-output-impedance amplifier capable of transmitting a vibration signal accurately through several thousand feet of cable. Ideal for large installations with centralized signal processing equipment. A D.C. bias voltage on the output facilitates remote testing for cable continuity and proper operation.

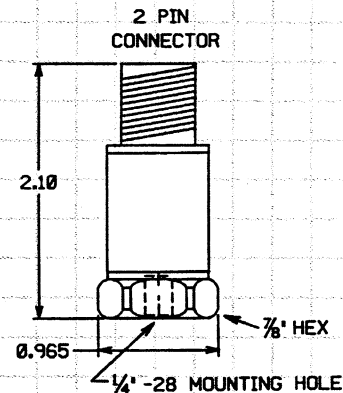
The IRD 911 can be mounted directly on the machine, via a 1/4 28 screw. The IRD 911 accelerometer is available in a straight or a right angle mounted connector configuration.



### OUTLINE DIMENSIONS



Model 911R



Model 911S



| Specifications for                        | 911 Accelerometer                                                   | Units                                    |
|-------------------------------------------|---------------------------------------------------------------------|------------------------------------------|
| Sensitivity, nom                          | 100                                                                 | mV/g                                     |
| Frequency Response<br>$\pm 10\%$<br>- 3dB | 1.5 - 7k<br>1                                                       | Hz<br>Hz                                 |
| Dynamic Range                             | 001 - 75                                                            | g peak                                   |
| Resonant Frequency,<br>mounted, nom       | 22                                                                  | kHz                                      |
| Transverse Sensitivity, avg               | 5                                                                   | %                                        |
| Temperature Range                         | -40 to +300<br>-40 to +148                                          | $^{\circ}\text{F}$<br>$^{\circ}\text{C}$ |
| Temperature Coefficient                   | .025                                                                | % / $^{\circ}\text{F}$                   |
| Magnetic Field<br>Susceptibility          | 100                                                                 | $\mu\text{g}/\text{Gauss}$               |
| Base Strain Sensitivity                   | .005 max                                                            | g/ $\mu\text{strain}$                    |
| Construction (case and armor)             | 316L-series stainless steel                                         |                                          |
| Seal                                      | Hermetic                                                            |                                          |
| Environmental Protection                  | Meets NEMA 4x, 12, 13, IEC IP66                                     |                                          |
| Grounding                                 | Case isolated from circuit                                          |                                          |
| Power                                     | +18 to +26 VDC source through<br>1 to 4 mA Constant - current Diode |                                          |
| Output Impedance                          | 100                                                                 | ohms                                     |
| Output Bias                               | +10 to +18                                                          | VDC                                      |

#### ORDERING INFORMATION

(Qty) P/N02402 - PVC Twisted Pair Shielded  
 (Qty) P/N 04366 - Mating Cable Connector  
 (Qty) P/N37691 - Model 911S Accelerometer (Straight Connector)  
 (Qty) P/N37692 - Model 911R Accelerometer (Right Angle Connector)  
 (Qty) P/N30539 - Tefzel Twisted Pair Shielded  
 (Qty) P/N 04636 - Rubber Insulated Shielded Pair



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FORM # 5202

## Model 943

### Accelerometer and Accessories

EFFECTIVE: December 1992

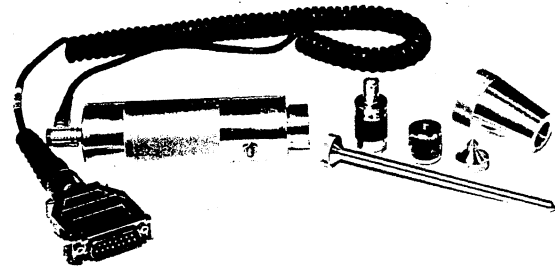
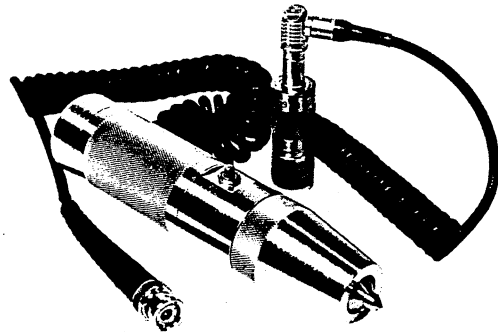
#### DESCRIPTION

The IRD 943 is a solid-state seismic accelerometer specifically designed for use with portable analyzers or data collectors to measure vibration levels on mechanical equipment. The 943 design is based on the proven performance of the 942 transducer used in permanent applications, and it exceeds the performance of the 970 transducer for portable applications. The primary advantage of the 943 are its small and light weight packaging and the applications flexibility.

A complete kit of accessories is available for the 943, including a small rare-earth magnet, 2 probe options, and a handle with built-in STORE button. The 943 can be stud-mounted or used with a magnet or probe in collecting data. The remote STORE function speeds up the data collection process by providing "hands-free" operation of the data collector and allows complete concentration on the measurement itself.

The 943 incorporates the industry standard ICP interface circuit and has a sensitivity of 100 mv/g. For walk-around data collection routes combining permanently installed 942's and hand held measurements, the 943 provides a distinct advantage where there is no need for switching cables, input ports or transducer sensitivity.

The 943 incorporates a low-output-impedance, constant-current driven amplifier with low noise floor and utmost accuracy across the entire frequency range. The wide frequency and amplitude ranges of the 943 accommodate most vibration measurement applications for predictive maintenance, including frequencies in the range of 120 CPM to 720,000 CPM and ultrasonic Spike Energy™ signals. The small size of the 943 permits its use in remote and tight spaces. All 943 components are designed for the highest reliability in withstanding harsh industrial environments, including high temperatures and splashing fluids.



The 943 transducer and accessories are ideally suited for the IRD FAST TRACK, 838, 885, and 890 portable instruments. The following cable compatibility matrix is provided for ordering convenience.

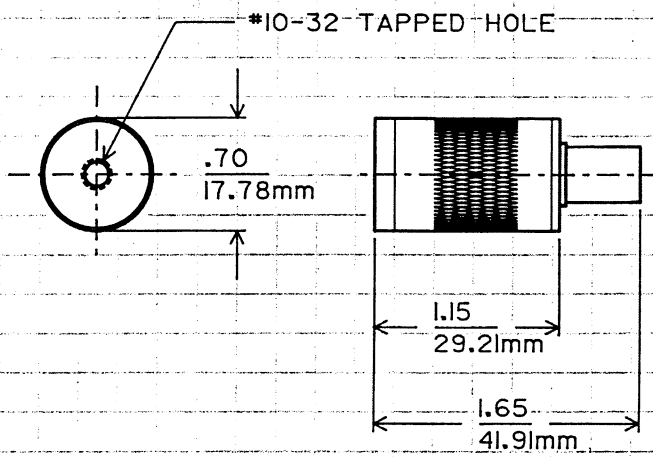
| Cable<br>Part Numbers | INSTRUMENT    |       |       |       |
|-----------------------|---------------|-------|-------|-------|
|                       | FAST<br>TRACK | 838*  | 885   | 890   |
| w/o remote<br>STORE   | 36365         | 36365 | 36365 | 36365 |
| with remote<br>STORE  | 35976         | -     | 36733 | 36733 |

\*Note: The remote STORE function is not supported by the 838 instrument



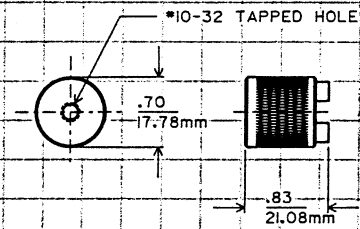
## OUTLINE DIMENSIONS

943 Accelerometer P/N 35530

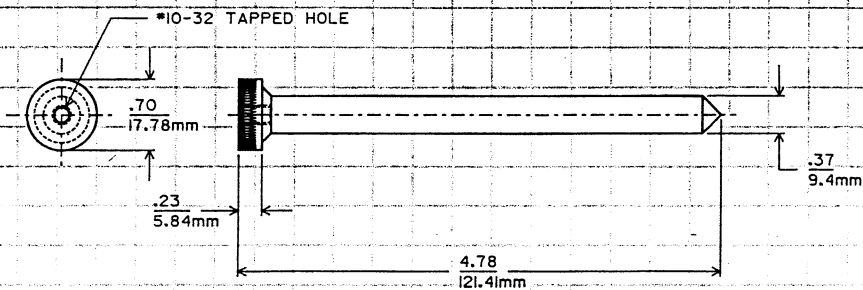


The 943 accelerometer can be used with either a magnetic holder (P/N 35536) or 5 inch probe (P/N 35540) as shown below.

Magnetic Holder P/N 35536

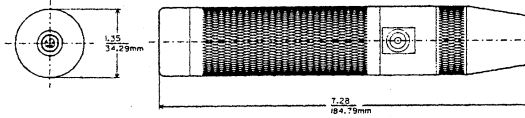


5 inch probe P/N 35540

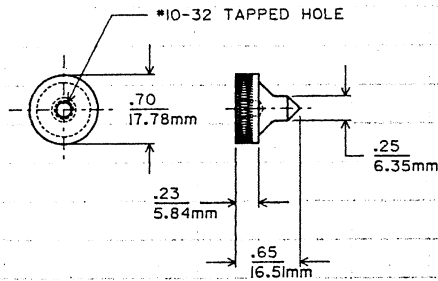


The 943 can also be used in combination with handle (P/N 35541) and choice of 5 inch probe or 1/2 inch probe P/N 35973) as shown below.

#### 943 Handle with STORE button P/N 35541



#### 1/2 inch probe P/N 35973

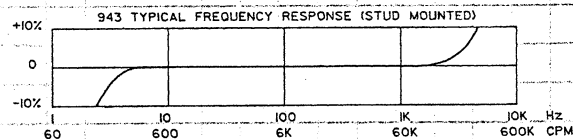


#### Assembly Instructions for the Use of the Handle

The handle nose cone is removed, the 943 accelerometer is snapped into place, either probe is attached to the 943, and the nose cone is re-installed to lock all components together. The transducer cable is then attached to the end of the handle. The integral STORE button in the handle operates with compatible data collectors (such as the IRD FAST TRACK).

| Specifications for                        | 943 Accelerometer                                                                                                 | Units                        |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------|
| Sensitivity, nom                          | 100                                                                                                               | mV/g                         |
| Acceleration range                        | 0-50                                                                                                              | g peak                       |
| Resonant Frequency, stud mounted, nominal | 25 k                                                                                                              | Hz                           |
| Frequency Response, stud mounted          | 3.5-7 k                                                                                                           | Hz $\pm$ 10%                 |
| Frequency Response, stud mounted          | 2-12 k                                                                                                            | Hz $\pm$ 3 dB                |
| Temperature Range                         | -60 to +250<br>-52 to +121                                                                                        | $^{\circ}$ F<br>$^{\circ}$ C |
| Transverse Response, avg                  | 5                                                                                                                 | %                            |
| Magnetic Field Sensitivity                | 100                                                                                                               | $\mu$ g/gauss                |
| Base Strain Sensitivity, stud mounted     | .01                                                                                                               | g/ $\mu$ strain              |
| Shock Limit                               | 1000                                                                                                              | g peak                       |
| Output Impedance                          | 100                                                                                                               | ohm                          |
| Seal                                      | Epoxy                                                                                                             |                              |
| Case Material                             | 304 Stainless Steel                                                                                               |                              |
| Connector                                 | Gold plated contacts; chrome plated brass shell                                                                   |                              |
| Power Requirement                         | 1-4 mA constant current;<br>+ 16 to + 24 VDC                                                                      |                              |
| Grounding                                 | Case connected to circuit common                                                                                  |                              |
| Weight (including connector)              | 1.55<br>44                                                                                                        | oz<br>grams                  |
| Environmental Protection                  | Waterproof, dustproof, corrosion-resistant per NEMA 4X; Submersible to 50 psi per NEMA 6 and IEC IP67 (100 hours) |                              |

#### 943 Typical Frequency Response (stud mounted)



**MAGNET P/N 35536 SPECIFICATIONS**

Frequency @ +3dB response 5000 Hz (300 k CPM)

Case Material 303 Stainless Steel

Dimensions Diameter .70 in (17.8 mm)  
Length .75 in (19.5 mm)  
Weight 1.06 oz (30 g)

**PROBE P/N 35973 SPECIFICATIONS**

Frequency @ +3dB response 1400 Hz (84 k CPM)

Material 303 Stainless Steel

Dimensions Diameter (Shaft) .25 in (6.35 mm)  
Length .65 in (16.5 mm)  
Weight .53 oz (15 g)

**PROBE P/N 35540 SPECIFICATIONS**

Frequency @ +3dB response 1000 Hz (60 k CPM)

Material 303 Stainless Steel

Dimensions Diameter (Shaft) .36 in (9.1 mm)  
Length 4.77 in (121.3 mm)  
Weight 1.48 oz (42 g)

**HANDLE P/N 35541 SPECIFICATIONS**

Switch Momentary Push Switch, Hermetically Sealed

Material (structural parts) Aluminum

Dimensions Diameter 1.35 in (34.3 mm)  
Length 6.87 in (175 mm)  
Weight 6.07 oz (172 g)

**CABLE P/N 35976 SPECIFICATIONS**

Connectors Right Angle 943; Submin-D FAST TRACK Data Collector

Cable Retractable; Neoprene Jacket; 72 in (1829 mm) extended

**CABLE P/N 36365 SPECIFICATIONS**

Connectors Right Angle 943; BNC

Cable Retractable; Neoprene Jacket; 72 in (1829 mm) extended

**ORDERING INFORMATION**

P/N 36394 - Complete Kit for IRD FAST TRACK includes:  
Accelerometer, Cable (P/N 35976), 2 probes,  
magnet & handle with built-in store button

Individual components can be ordered from the chart below:

P/N 35530 - Model 943 Accelerometer  
P/N 35976 - 943 Cable to FAST TRACK (w/ remote STORE),  
18 in (45.7cm) to 6ft (1.83m) \*\*  
P/N 36365 - 943 BNC instrument cable (no remote STORE),  
18 in (45.7 cm) to 6 ft (1.83M)  
P/N 36733 - 943 Cable to 885, 890 (w/ remote STORE),  
18 in (45.7 cm) to 6 ft (1.83M)  
P/N 35973 - 943 Probe 1/2 in (1.3cm)  
P/N 35540 - 943 Probe 5 in (12.7 cm)  
P/N 35536 - 943 Magnetic holder  
P/N 35541 - 943 Handle with STORE button



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FORM #5112

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**Model 970**  
Vibration Pickup

**SPEC SHEET: 41-06-04**  
**EFFECTIVE: May 1992**

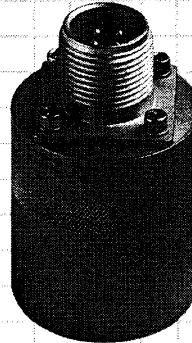
**DESCRIPTION**

The IRD Model 970 is a rugged low impedance accelerometer type vibration pickup for use with portable instruments on all types of rotating machinery.

The Model 970 has built-in electronics to provide low impedance output and eliminate the need for special low noise cables. The Model 970 is sealed against dust and moisture and operates up to a temperature of 250° F (121°C).

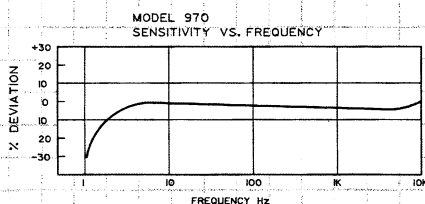
A single 1/4-28 mounting stud is used for attaching the pickup to a flat machine surface. Insulated Mounting Adapters are available for those cases where ground potential problems may exist between the pickup mounting surface and instrument.

The Model 970 connects directly to IRD Model 810, 840, 880 and other IRD instruments designed to accept the 970. Operating power is obtained from the instrument.



**SENSITIVITY**  
50 mV/g Nominal

**FREQUENCY RESPONSE**



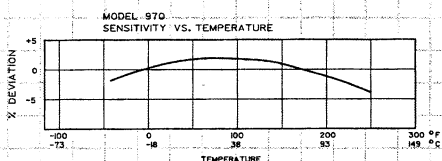
**MOUNTED RESONANT FREQUENCY**  
27K Hz Minimum

**MAXIMUM USABLE AMPLITUDE (Undistorted)**  
180 g peak up to 1 k Hz, 120 g peak at 5K Hz, or 70 g peak at 10 K Hz with  $\pm 15$  v power supply and 25 ft. of 50 pF/ft. cable. Longer cables may be used with some reduction in maximum amplitude and/or high frequency response. (Maximum amplitude is reduced proportionally at lower supply voltage)

**MAXIMUM VIBRATION (without damage)**  
1000 g peak sinusoidal vibration

**MAXIMUM SHOCK (without damage)**  
2000 g peak (10 ms half-sine)

**TEMPERATURE RESPONSE**



**ORIENTATION**  
Any

**TRANVERSE RESPONSE**  
5% Maximum, 3% Typical

**MAGNETIC FIELD SENSITIVITY**  
Less than .001 equiv. g/Gauss at 60 Hz

**RESIDUAL NOISE**  
Less than .0002 equiv. g broadband - transducer output.

**GROUNDING**  
Case isolated from signal ground

**POWER SUPPLY**  
Regulated  $\pm 8$  Vdc @  $\pm 1.2$  max max. to  $\pm 15$  Vdc @  $\pm 1.75$  ma max.

**HOUSING**  
300 series stainless steel, waterproof, dust proof, welded case

**CONNECTOR**  
4 Pin MS Type 12s-10P

**WEIGHT**  
7.7 oz. (220 gr.)

**MOUNTING**  
1/4-28 Tapped Hole, 1/4" Deep

**INSTALLATION NOTES**

1. Mounting surface must be flat and smooth, mounting stud must be square with surface and must not exceed 1/4" length.
2. Install with light coating of silicone grease on mating surface, tighten to 30-50 inch-pounds torque.
3. Connector Pin A + DCV power supply, Pin B - DCV power supply, Pin C is signal and power supply common, Pin D signal output.

**OPTIONAL ACCESSORIES**

- Insulated Mounting Stud
- Magnetic Holders P/N 4332, 24746
- Straight Probe P/N 1103.

**ORDERING INFORMATION**

(Qty) P/N 19697 Model 970 Vibration Pickup

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L05012

FORM #1853D

## Model 970i

### Accelerometer and Accessories

EFFECTIVE: February 1996

#### DESCRIPTION

The IRD Model 970i is a rugged low impedance accelerometer vibration pickup for use with portable instruments on all types of industrial equipment. The Model 970i has built in electronics to provide a low impedance output and eliminate the need for special low noise cables. The 970i design is based on the proven performance of the IRD Model 970 accelerometer. The 970i provides full database compatibility for users upgrading to newer instrumentation that supports industry standard accelerometers which operate on an instrument supplied constant current DC power source. The Model 970i was designed to provide a very similar frequency response as the standard Model 970 including Spike Energy response. The Model 970i is sealed against dust and moisture and operates up to a temperature of 250° F ( 121° C ).

A single 1/4-28 mounting stud is used for attaching the accelerometer to a flat machine surface. Insulated mounting adapters are available for those cases where ground potential problems may exist between the transducer mounting surface and the instrument.

The Model 970i connects directly to the dataPAC series Data Collectors and any other instruments which operate with industry standard constant current DC power source. Operating power and current for the accelerometer are supplied externally, typically from the instrument.

#### APPLICATIONS

Well suited as a rugged general purpose portable instrument transducer to measure vibration on a wide variety of machines. The 970i can be used on most machines with one times running speeds between 300 and 15000 RPM (5-250 Hz). The 970i can be temporarily or permanently stud mounted to improve high frequency response.

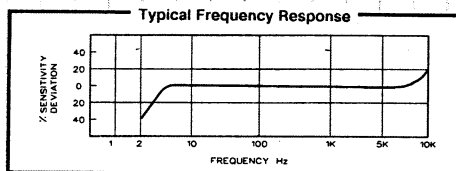


## IRD Mechanalysis

## TRANSDUCER CHARACTERISTICS

Sensitivity

50 mw/g,  $\pm 10\%$

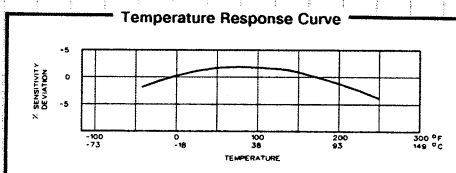


### Frequency Response:

$\pm 5\%$  from 6 Hz to 2 KHz  
 $\pm 10\%$  from 5 Hz to 10 KHz  
 -3dB at 2.5 Hz and + 3 dB at 15 KHz min.  
 Mounted Resonance 27 KHz  
 Maximum useable amplitude 75 Gs  
 (when powered from + 18 to + 24 VDC)

### ENVIRONMENTAL

Maximum Vibration without damage  
 1000 g peak sinusoidal vibration  
 Maximum Shock  
 1000 g peak ( 10 ms half sine )



Temperature response (5% deviation from -30° to + 250° F)  
 Orientation Any  
 Transverse Response 5% maximum, 3% typical  
 Magnetic Field Sensitivity Less than .001 equiv.  
 g/Gauss at 60 Hz.

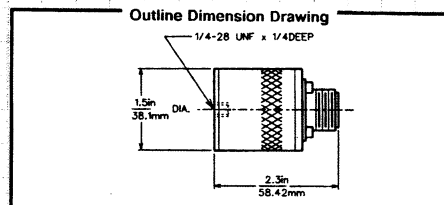
### ELECTRICAL

Residual noise Less than 0.0004 equiv. g broadband - transducer output.  
 Case isolated from signal ground  
 Grounding + 15 to + 24 VDC @ constant current from 2mA to 8mA  
 Power supply + 7 to + 11 VDC  
 Signal output bias voltage

## PHYSICAL

Housing

300 series stainless steel,  
 waterproof, dustproof, welded  
 case  
 Connector 2 pin MS type (Mating connector IRD P/N 4366 )  
 Weight 7.7 oz ( 220 gr. )  
 Mounting 1/4-28 Tapped hole, 1/4" depth



### Installation Notes

1. Mounting surface must be flat and smooth, mounting stud must be square with surface and must not exceed 1/4" length.
2. Install with light coating of silicone grease on mating surface, tighten to 30 - 50 inch-pounds torque.
3. Connector pin A is signal ( + ) and connector pin B is return ( - ).

### OPTIONAL ACCESSORIES

Insulated mounting stud - P/N 10308  
 Magnetic Holders:  
 Large P/N 4332  
 Small P/N 24746  
 Straight Probe P/N 1103

### CABLING TO IRD INSTRUMENTS

Any dataPAC series instrument:  
 • 6ft Coiled Cable - P/N 36960  
 • 25ft Extension cable (Lemo to Lemo connector) - P/N 37135  
 • 12ft Straight Cable-BNC connector for Fast Track, 890, 885 and Smart Meter/FS - P/N39480  
 • 50ft Straight Cable BNC connector for Fast Track, 890, 885 and Smart Meter/FS - P/N 39479  
 • 6 ft Coiled Cable BNC connector for Fast Track, 890, 885 and Smart Meter/FS - P/N 39450  
 • 25ft BNC to BNC Extension Cable - P/N 37388

## ORDERING INFORMATION

(Qty) P/N 38946 Model 970i Vibration Pickup



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 400 059  
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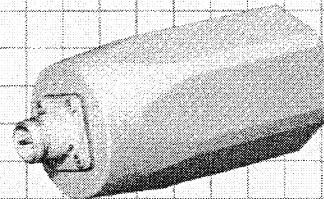
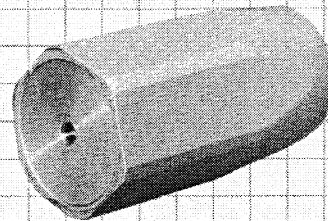
## Model 544

Vibration Pickup - P/N 4526

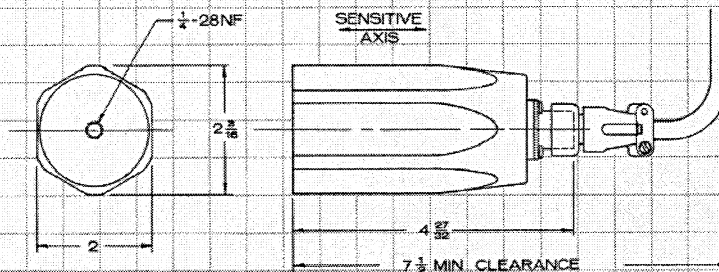
SPEC SHEET: 41-01  
EFFECTIVE: Feb 1990

### DESCRIPTION

The IRD Model 544 vibration pickup is a seismic velocity transducer (a moving coil in a permanent magnetic field with eddy-current damping). The Model 544 is suitable for a wide variety of manual and continuous monitoring applications due to its high output, low impedance, wide frequency response and rugged construction. The anodized aluminum housing is waterproof, dustproof and acceptable for use in Division 2 hazardous areas. A variety of accessories are available.



### OUTLINE DIMENSIONS



### INSTALLATION

1. The sensitive axis may be oriented in any direction.
2. The mounting surface must be flat and smooth (minimum diameter = 2-3/8 in).
3. The mounting stud axis must be perpendicular to the mounting surface.
4. Hand-tighten only.
5. Connector pin A is both signal and housing ground; connector pin B is signal ground.

### ORDERING INFORMATION

(Qty.) P/N 4526 Model 544 vibration pickup



**FREQUENCY RESPONSE**

14 to 1,000 Hz within  $\pm 8\%$ ; 10 Hz 20% (average) measured at 70° F (21° C)

**SENSITIVITY**

1080  $\pm 10\%$  mV/in/s (42.52 mV/mm/s)

**TEMPERATURE RANGE**

-40 to 500° F (-40 to 260° C)

**IMPEDANCE**

R = 2 k $\Omega$ ; L = 0.7 H at 70° F (21° C)

**DAMPING FACTOR**

0.7 (critical) at 70° F (21° C)

**MOUNTING ORIENTATION**

Any

**TRANSVERSE RESPONSE**

Less than 5% (average) 10 to 1,000 Hz

**MAGNETIC FIELD SENSITIVITY**

0.15 in/s/Gauss

**HOUSING**

Waterproof, dustproof

**CONNECTOR**

2-pin (GS-02-10SL-4P)

**WEIGHT**

21 oz

**GROUNDING**

Case and signal ground common

**MOUNTING**

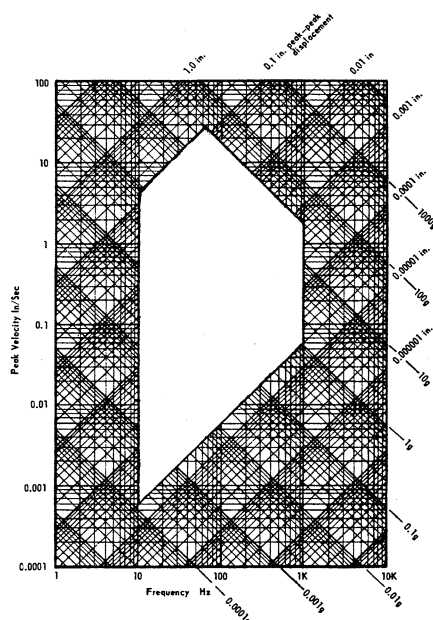
1/4 - 28 tapped hole, 1/4 in depth

**MAXIMUM AMPLITUDE**

125 mils pk-pk to 70 Hz; 30 g above 70 Hz

**ACCESSORIES**

- Magnetic shields (reduce sensitivity to AC magnetic fields by 100:1) - P/N 10449
- Explosion-proof enclosure (for installation in Division 1, Class I, Group D or Class II, Groups E,F,G hazardous areas) - P/N 4224
- Safety barrier (for intrinsically safe installations in Division 1, Class I, Groups B,C,D hazardous areas) - P/N 7685
- Straight cable connector (rated for continuous operation at 250° F or 121° C) - P/N 4366
- Straight cable connector (rated for continuous operation at 500° F or 260° C) - P/N 15039
- Right-angle connector (rated for continuous operation at 250° F or 121° C) - P/N 14797



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FORM # 1249F

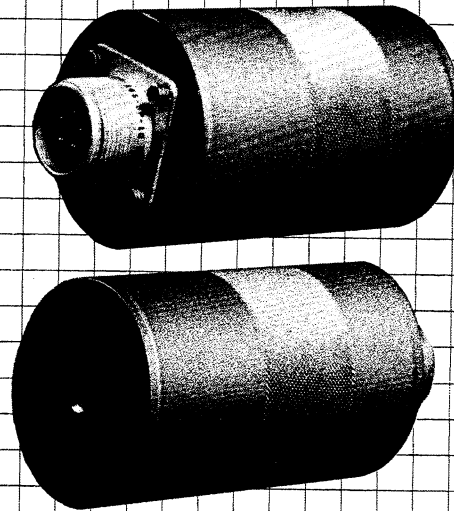
**Model 560**  
Vibration Pickup P/N 24957

SPEC SHEET: 41-20  
EFFECTIVE: Sept 1990

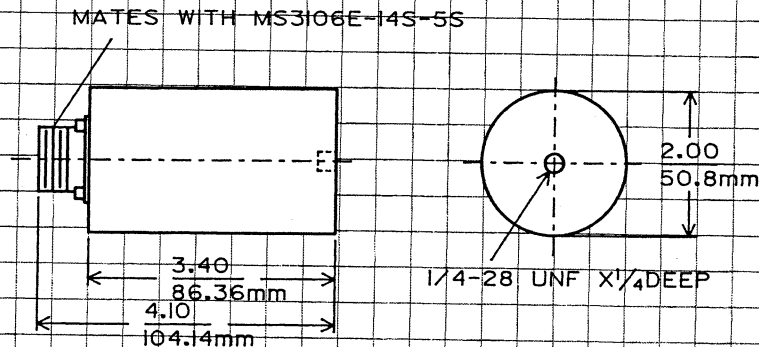
**DESCRIPTION**

The IRD 560 is a unique piezoelectric seismic velocity transducer with a flat-frequency response from 1 Hz to 4.5 kHz, designed specifically for low-frequency applications. The high upper-frequency limit accommodates both general and low-frequency machinery vibration measurements. Built-in electronics provide high output sensitivity and low impedance. Low residual noise and low phase shift make it an ideal transducer for low-speed balancing applications.

The IRD 560 is stainless steel and waterproof, with a 1/4-28 tapped hole for stud mounting.



**OUTLINE DIMENSIONS**



**INSTALLATION NOTES:**

1. Mounting surface must be flat and smooth.
2. Install on mating surface with light coating of silicone grease.
3. Use 1/4-28 stud mounting for permanent installation; magnetic holder (P/N 4332) for temporary installation.
4. Connector: Pin A + DCV power supply; Pin B - DCV power supply; Pin C signal and power supply common; Pin D signal output; Pin E not connected.

**ENTEK IRD**  
INTERNATIONAL

**SPECIFICATIONS AT +75°F (+24°C)**

**FREQUENCY RANGE**

Flat  $\pm 10\%$  (1 Hz to 4.5 kHz)

**MAXIMUM USABLE AMPLITUDE (undistorted)**

5 in/s (127 mm/s) peak

**RESIDUAL NOISE**

.0007 in/s (.0178 mm/s) equivalent broad-band

**WEIGHT**

38 oz (1,079 grams)

**SIZE**

2 in diameter (50.8 mm)

4 in height (101.6 mm) (including receptacle)

**TEMPERATURE RANGE**

-40°F to 250°F (-40°C to 121°C)

**OUTPUT UNITS**

Velocity

**VELOCITY SENSITIVITY**

1,080 mV/in/s  $\pm 5\%$  at 100 Hz, (764 mVrms/in/s-pk)

42.5 mV/mm/s  $\pm 5\%$  at 100 Hz (30 mVrms/mm/s-pk)

**MAGNETIC FIELD SENSITIVITY**

$10^{-5}$  in/s/Gauss ( $2.54 \times 10^{-4}$  mm/s/Gauss equivalent)

**TRANSVERSE RESPONSE**

5% maximum at 10 Hz, 5 g peak

**MAXIMUM CONTINUOUS VIBRATION**

250 g pk

**MAXIMUM SHOCK**

500 g pk

**MOUNTING**

Centered hole (1/4-28 x 1/4 in deep)

**MOUNTED NATURAL FREQUENCY**

10 kHz minimum

**WATER RESISTANCE**

Waterproof

**CONNECTOR**

5-pin top mounting

**GROUNDING**

Case isolated from signal ground

**POWER INPUT**

$\pm 15$  Vdc at  $\pm 5$  mA maximum

**CABLE DRIVE CAPABILITY**

No clipping of 1 kHz 5 in/s (127 mm/s) signal over 500 ft (25,000 pF) cable

**PHASE SHIFT**

10° at 7 Hz, 30° at 3 Hz maximum

**ELECTRONICS**

Built-in amplifier

**ORDERING INFORMATION**

(Qty) P/N 24957, IRD 560 Vibration Pickup

(Qty) P/N 24953, Mating Connector

(MS 3106E-14S-5S) for IRD 560

(Qty) P/N 25136, 12 ft cable



[www.entekird.com](http://www.entekird.com)

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**FORM #2008C**

## Model 580

Vibration Pickup P/N 28852

SPEC SHEET: 41-30  
EFFECTIVE: July 1991

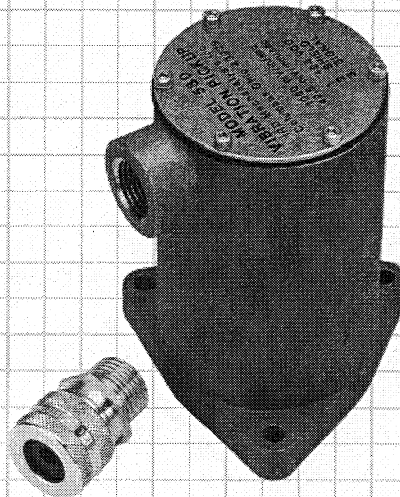
### DESCRIPTION

The Model 580 Vibration Pickup is a solid-state, seismic velocity pickup for permanent installation on industrial machinery. It has no moving parts. The waterproof case provides for an exceptionally long, trouble-free service life under adverse conditions, such as in mills, refineries, power plants, and mines. The broad frequency range, 120 to 120,000 CPM makes this pickup suitable for nearly any machine.

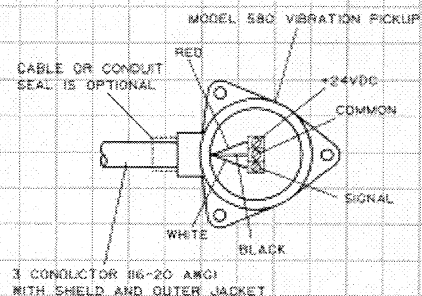
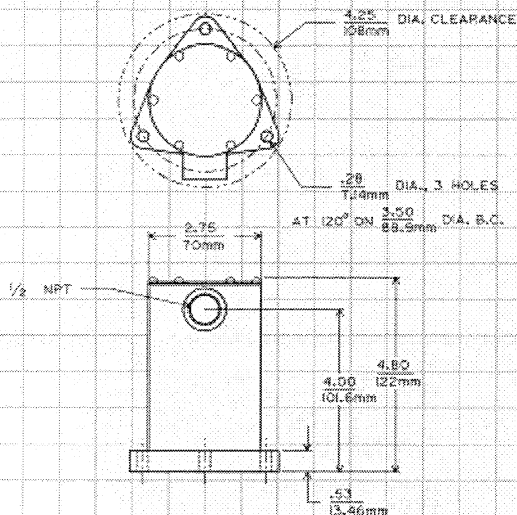
The cast stainless steel housing is designed to withstand mechanical abuse and is resistant to most chemicals. Electrical connections to the pickup are located within the housing protected from the environment. A threaded boss is provided on the case for cable installation through conduit or a sealing gland.

The case is electrically insulated from the internal components, so the Model 580 pickup can be used on insulated bearing machines or where good electrical grounding is not practical.

The Model 580 pickup is Factory-Mutual approved for use in hazardous areas (See Specifications).



### OUTLINE DRAWING



NOTE: DO NOT CONNECT SHIELD WIRE. CUT FLUSH WITH CABLE JACKET AND INSULATE.

IRD Mechanalysis

**OUTPUT SENSITIVITY**  
1080mV/in/s (42.5 mV/mm/s)

**OUTPUT IMPEDANCE**  
10 ohm

**OUTPUT OFFSET**  
+ 8 to + 13 Volts D.C.

**FREQUENCY RESPONSE**  
2Hz to 2K Hz (120 to 120,000 CPM), flat within  $\pm 10\%$  @ 70°F.  
4Hz to 1.5K Hz (240 to 90,000 CPM), flat within  $\pm 5\%$  @ 70°F.

**AMPLITUDE RANGE**  
0-6 in/s (0-152 mm/s)

**ORIENTATION**  
Omnidirectional

**TEMPERATURE RANGE OPERATING**  
-40° to +200°F. (-40° to + 93°C.)

**STORAGE**  
-55° to +250°F. (-47° to + 121°C.)

**SHOCK**  
500 g peak maximum

**WITHSTAND VOLTAGE**  
400 VDC or peak AC maximum, case to circuit

**APPROVALS**  
Approved by Factory Mutual

**INTRINSICALLY SAFE** for  
Class I, II, III  
Division 1  
Groups A, B, C, D, E, G when used with safety barriers (IRD Dwg. C98239)

**NON-INCENDIVE** (Without Safety Barrier) for  
Class I, II, III  
Division 2  
Groups A, B, C, D, & G when installed per (IRD Dwg C98239)

**RADIATION**  
10<sup>7</sup> RAD total dose

**MAGNETIC FIELD SENSITIVITY**  
1.5 x 10<sup>-4</sup> in/sec equivalent @ 60 Hz

**SEAL**  
Waterproof to 30 psi (approx. 70 ft head of water) when installed with waterproof pipe or conduit fittings.

**CONSTRUCTION**  
Type 316 stainless steel case; Neoprene cover gasket, stainless steel hardware.

**POWER**  
+ 18 to + 26 VDC @ 10-mA

**CONNECTIONS**  
Internal 3-screw barrier strip; Clamp terminals accept AWG 16 to 20 wire.

**CABLE ENTRY**  
1/2 NPT threaded boss in case. One (1) compression-type rubber seal fitting included for use with cable up to .250 in. (64 mm) diameter. Seal fitting not for use in radiation or corrosive environments.

**DIMENSIONS**  
4 7/8 in (124 mm) high x 2 3/4 in. (70 mm) body diameter; weight 5.8 lb. (2.46 Kg)

**MOUNTING**  
(3) holes for 1/4 in. (6mm) diameter bolts, 120° apart on 3 1/2 in. (89 mm) diameter circle. Foot print is 4 1/4 in. (108 mm) diameter circle.

### ORDERING INFORMATION

(Qty) P/N 28852, Model 580 Vibration Pickup.

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L05022

FORM #2139C

## 1900 SERIES PROBES

### FEATURES

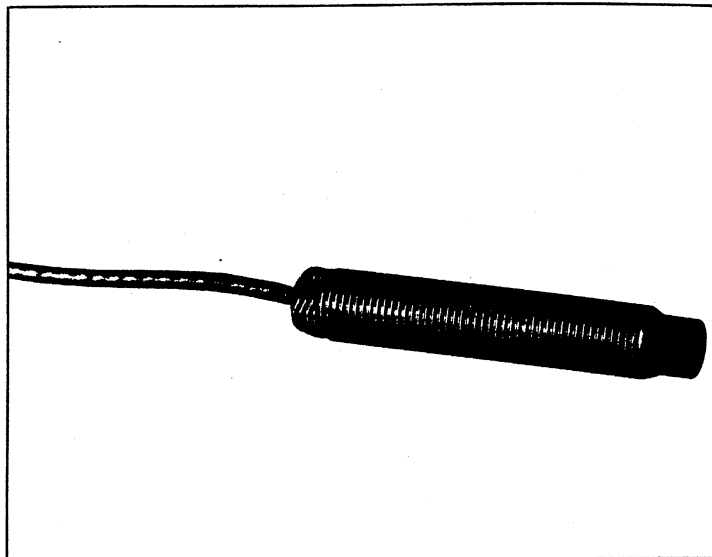
TOTALLY SEALED  
PROBES

INTEGRAL STAINLESS  
STEEL WIRE ARMORED  
CABLE

200°C OPERATION

CABLES LOCKED TO  
PROBE

HIGH THERMAL SHOCK  
CAPABILITY



### GENERAL OVERVIEW

A probe system consists of an eddy current probe, an oscillator/demodulator (known as the Probe driver) and an interconnecting cable. These three items form a resonant tuned circuit.

The radio frequency generated by this arrangement is present at the probe tip where a coil radiates a corresponding, low energy, electromagnetic field. This field induces stray electrical (eddy) currents in the surface of the target material placed within its path. The energy loss into the target is detected as a change in the modulated signal and produces a voltage proportional to the gap between the probe tip and the target.

In practical systems the target is the shaft of the machine and the probe is mounted on, or through, the casing. In this way the shaft vibration, relative to the casing, is measured.

The 1900 series of probe systems represents the most significant development of eddy-current probe systems in recent years. They have introduced new standards in terms of probe sealing, cable pull strengths and integrally armored cables, not seen in this type of product in the past.

### APPLICATION

The 1900 series are used for measuring single and two plane relative vibration on shafts of rotating machinery. They also find application in measuring shaft eccentricity, axial position of shafts, case expansion movement, differential expansion and other instances where a non-

contacting, relative measurement must be made.

### MACHINE MOUNTING

Mounting of probes is critical and must be done correctly. The easiest method is to use IRD 29000 probe holders which allow for micrometer adjustment of probes and safe removal of probes whilst machines are running. See the 29000 data sheet for this item.

With each shipment of probe/driver sets, a User Guidance Note (TD-128) is provided which details various methods of installation and suggests methods of avoiding some of the most common difficulties.

### PROBE SEALING

To prevent product, lubricant or gases from passing into unwanted areas on the machine being monitored, it is essential that the probe is completely sealed. The IRD 1900 series have set new standards in this area and standard production probes will withstand 100psi (7 bar). The type approval tests included prolonged high pressure tests interspersed with thermal shocks of 220°C.

The probe is sealed into the body and the probe tip sealed to the cable in such a manner that increased pressure tends to produce an improved seal.

36

1900 SERIES 5 & 8MM DIAMETER VIBRATION PROBES



## 1900 SERIES PROBES

### CABLE LOCKING

Installation of probes particularly in gas and steam turbines often involves feeding cables through oil ducts or other passage ways. Accidental damage can occur if cables are pulled too hard and industry has seen a history of probes damaged in this way.

The IRD 1900 series have a unique cable which has its steel wire armoring locked into the probe body. This design prevents load being borne by the signal cores, and allows the cable to withstand a continuous pull of 50lbs without damage or change to the probe characteristics.

### ARMORED CABLE

The cable used for the IRD 1900 series probes was designed by IRD engineering group especially for this new generation of products. The cable which is only available on IRD probes is a triaxial type with a multi-strand steel center core, an inner screening braid and stainless steel woven armor. The outer sheath is a bright orange PTFE (Teflon) and is quite exceptionally tough. It will not cause failure of the probe if the outer sheath is perforated and the armor is grounded.

Note the exceptionally small bend radius of 12.5 mm which is quite remarkable for a cable of this complexity of construction.

### API 670 COMPLIANCE

The IRD probe/driver systems comply fully with the requirements of API 670 and indeed exceed that specification in many areas.

### INTRINSIC SAFETY

Full BASEEFA and CENELEC certification is available for the complete range of probes giving certifications to EEx ia IIC T6. See also the section on INTRINSIC SAFETY CERTIFICATIONS; IMPLICATIONS AND APPLICATIONS towards the end of this catalog. Intrinsic safety is of such importance that uncertainty must not exist. IRD are keen to ensure users of its systems fully understand the implications and are supported throughout the design and specification phase of new installations.

### PROBE DRIVERS

These are typical of the quality engineering approach of all IRD products. The probe drivers are in stainless steel

**TABLE OF PROBE MODELS ETC**

This table shows the various models in the range. Essentially they are defined by tip diameter, body thread and type, and whether the cable is fitted with flexible stainless steel conduit. Note the preferred body lengths in the right hand column, and use these if possible. These standard preferred lengths are usually available on a shorter delivery than non standard items. IRD manufacture probes to non standard dimensions for a number of machinery manufacturers, and will be pleased to continue that service. Discuss any special requirements with Technical Sales.

Only specify conduit on probes if really necessary; remembering that the probe cable is stainless steel wire armored, and only needs additional protection outside of the machine, which is generally provided by fixed conduits.

| MODEL | TIP DIA | MOUNTING<br>TYPE      THREAD |         | S/S CONDUIT | PREFERRED BODY<br>LENGTH |          |
|-------|---------|------------------------------|---------|-------------|--------------------------|----------|
|       |         |                              |         |             | UNTHREADED               | THREADED |
| 1901  | 5       | STD                          | 1/4-28  | NO          | 0                        | 25mm     |
| 1902  | 5       | STD                          | 1/4-28  | YES         | 0                        | 25mm     |
| 1903  | 5       | STD                          | M8 x 1  | NO          | 0                        | 25mm     |
| 1904  | 5       | STD                          | M8 x 1  | YES         | 0                        | 25mm     |
| 1905  | 8       | STD                          | 3/8-24  | NO          | 0                        | 30mm     |
| 1906  | 8       | STD                          | 3/8-24  | YES         | 0                        | 30mm     |
| 1907  | 8       | STD                          | M10 x 1 | NO          | 0                        | 30mm     |
| 1908  | 8       | STD                          | M10 x 1 | YES         | 0                        | 30mm     |
| 1909  | 8       | REVERSE                      | 3/8-24  | NO          | 5                        | 30mm     |
| 1910  | 8       | REVERSE                      | 3/8-24  | YES         | 5                        | 30mm     |



**1900 SERIES 5 & 8MM DIAMETER VIBRATION PROBES**

**37**

## 1900 SERIES PROBES

castings and are fully encapsulated. They can withstand to 24 hours totally submerged to a depth of 1 meter and will resume working when dried off. IRD however do not recommend this as an operational activity.

## ORDERING INFORMATION

(See the table on earlier page)

## SPECIFICATION

### 5 AND 8 MM PROBES

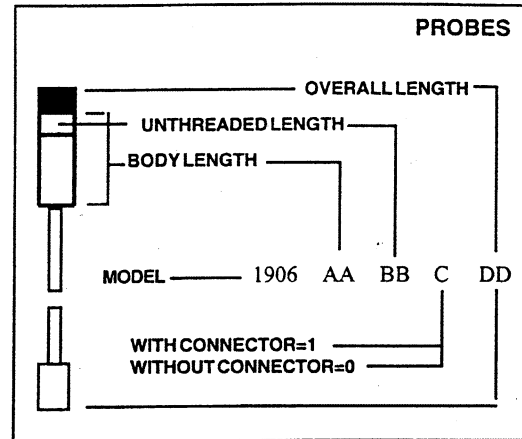
|                             |                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------|
| <b>Signal Sensitivity</b>   | 200mV/mil                                                                              |
| <b>Useful Range</b>         | 10 - 100 mils                                                                          |
| <b>Accuracy</b>             | ± 5% of gap                                                                            |
| <b>Linearity</b>            | within 1 mil of best straight line                                                     |
| <b>Voltage at 0.050 gap</b> | 10volts ± 1volt                                                                        |
| <b>Frequency Response</b>   | dc to 10KHz                                                                            |
| <b>Temperature Range</b>    | -55°C to +200°C                                                                        |
| <b>Cable Bend Radius</b>    | 12.5mm minimum                                                                         |
| <b>Cable Pull Strength</b>  | 50lbs maximum, Straight pull on cable with probe body retained and in line with cable. |
| <b>Probe Seal</b>           | 7 bar applied anywhere on the probe.                                                   |
| <b>Connector</b>            | Miniature coaxial                                                                      |

### PROBE DRIVERS

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| <b>Dimensions</b>        | To API 670 see illustration                                     |
| <b>Temperature Range</b> | -55 to +100°C                                                   |
| <b>Construction</b>      | Stainless steel casting with encapsulated electronics.          |
| <b>Weight</b>            | 310g                                                            |
| <b>Waterproofing</b>     | Fully encapsulated to withstand 1 meter immersion for 24 hours. |

### Intrinsic Safety Details

Approved to CENELEC EEx ia IIC T3 - T6 Certificates;  
 EX 872266X EEx ia IIC T3 (T.amb = 165°C)  
 EX 872265X EEx ia IIC T4 (T.amb = 100°C)  
 EEx ia IIC T5 (T.amb = 65°C)  
 EEx ia IIC T6 (T.amb = 50°C)



Note that body thread is implicit in model number.

AA= Body length in mm.(multiples of 5mm\*\*)

BB= Unthreaded length in mm.(multiples of 5mm\*\*)

C= 1 For fitted connector.

C= 0 For no connector fitted.

DD= Total length of probe & cable.

\*\* For notes on preferred standards see table on the previous page.

Select DD to suit the required position of the connector in relation to the conduit arrangements on the machine, but note the preferred standards of 0.5, 1.0 and 1.5 meters.

Probes are fitted with an insulating boot which mates with a matching boot on the Extension cable. These should be pushed together to ensure electrical isolation of the connector body from the ground.

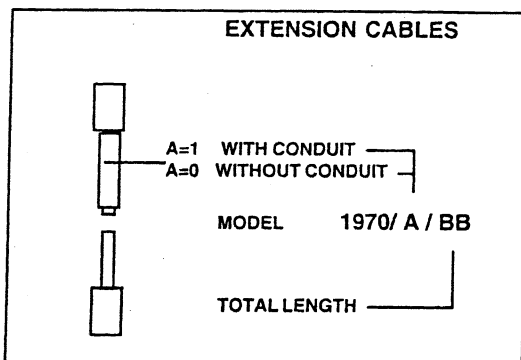
## 1900 SERIES PROBES

### EXTENSION CABLES

When ordering extension cables ensure that the total length of probe and extension cable total 5 or 9 meters, and that this also matches the Probe Driver selection. Preferred lengths of extension cables are therefore 3.5, 4.0, 4.5, 7.5, 8.0, 8.5 meters

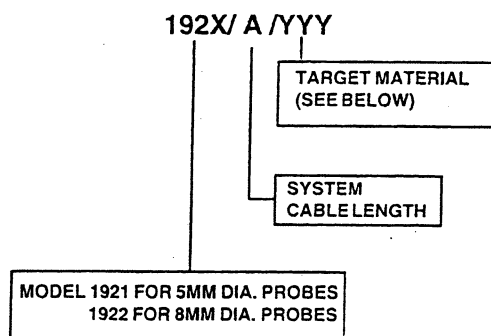
Probes and extension cables are not normally supplied with flexible stainless steel conduit as the cable has an internal stainless steel wire armoring. If flexible stainless steel conduit is preferred to standard rigid conduit on the machine it can be ordered from this section.

HOWEVER THE PREFERRED METHOD OF PROTECTING CABLES IS TO FIT A PERMANENT CONDUIT SYSTEM ON THE MACHINE AND THE NECESSARY COMPONENTS FOR THIS CAN BE SELECTED FROM THE 'MACHINE MOUNTED STAINLESS STEEL CONDUIT' SECTION IN THIS CATALOG.

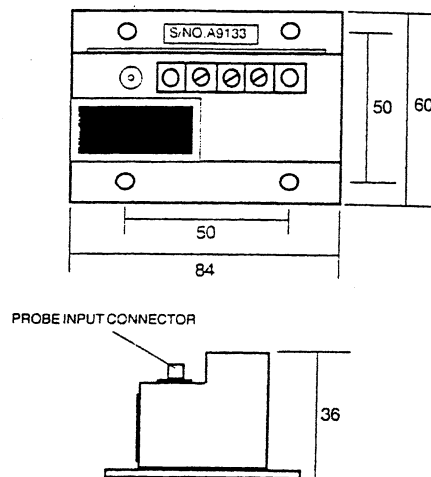


### PROBE DRIVERS

These are ordered under the following format:



### DIMENSIONS OF DRIVER



Body and base plate is stainless steel. Unit fully encapsulated.

All standard probes are calibrated against a standard target material of En19 (#4140 Steel). Where operation against a specified target material is required, the Probe Driver will be calibrated accordingly.

|          |                           |
|----------|---------------------------|
| YYY= 001 | EN19 T(#4140 Steel)       |
| 002      | A-479-410-CL12            |
| 003      | INCONEL 625               |
| 004      | EN25                      |
| 005      | EN29A(T) (Use EN40B)      |
| 006      | EN40B                     |
| 008      | 42CMR04 (Use EN19T)       |
| 009      | BS4360-50 (Use EN14A)     |
| 010      | BS970 709M40T (Use EN19T) |
| 011      | AISA 1045 (Use EN8)       |
| 012      | BS970 070 M20 (Use EN3B)  |
| 013      | EN8                       |
| 014      | EN3B                      |
| 015      | EN56B or BS970 420 S29R   |
| 016      | EN14A                     |
| 018      | EN9                       |

If a material not listed above is to be used, consult Technical Sales for assistance. It may have been added to the list since the publication of this catalog.

A small charge is made for calibrating probe drivers to any material other than En19(#4140 Steel).



**1900 SERIES 5 & 8MM DIAMETER VIBRATION PROBES**

**39**

**FLR**  
PIEZOTRONICS

RE

**QUARTZ  
ROMETERS  
-in amplifiers<sup>(1)</sup>  
es 300**

**PIEZOTRONICS**  
301A Vibration  
Calibration Std in  
394A03 System  
(see specific data sheet)

STOCK

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTES:</b> (# denotes exceptional characteristics)</p> <p>(1) Frequency response per chart and to transient response (otherwise as specified)</p> <p>(2) Discharge time constant relates to low TC should be 25 times event duration. Use 4848 or 4848B02 Series power units to utilize the full transducer discharge TC.</p> <p>(3) Standardized to tolerance indicated -</p> <p>(4) Supplied with attached 10 ft. ribbon wire terminating 10-32 cap screw.</p> <p>(5) Supplied with attached 10 ft. ribbon wire terminating 10-32 cap screw.</p> <p>(6) Accessory isolation pads and studs are available</p> <p>(7) Calibration furnished to 5%, optional TC should be furnished to 10% frequency range</p> <p>(8) Field repairable solder terminal connection</p> | <p><b>OPTIONS:</b> Special sensitivities, ranges, electrical isolation, shock sealing, low and high temperature, high pressure, resistive</p> <p>TC based and limited output, lower charge TC, TC based and limited output, lower</p> <p>noise, lower temperature electronics to -140°C or higher temperature to +325°F</p> <p>connector, connector, solder connector adaptor and thermal jackets.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTES:</b> (# denotes exceptional characteristics)</p> <p>(1) Frequency response per chart and to transient response (see chart) otherwise</p> <p>(2) Discharge time constant relates to low TC should be 25 times event duration. Use 4484B or 4484B02 Series power units to utilize the full transducer discharge TC.</p> <p>(3) Standardized to tolerance indicated -</p> <p>(4) Supplied with attached 10 ft. ribbon wire terminating 10-32 cap screw.</p> <p>(5) Supplied with attached 10 ft. ribbon wire terminating 10-32 cap screw.</p> <p>(6) Accessory isolation pads and studs are available</p> <p>(7) Calibration furnished to 5%, optional TC should be furnished to 10% frequency range</p> <p>(8) Field repairable solder terminal connection</p> | <p><b>OPTIONS:</b> Special sensitivities, ranges, electrical, magnetic sealing, low and high frequency response, longer life, lower frequency response, longer life, charge TC, biased and limited output, lower noise, lower temperature electronics to -440°C or higher temperature to +325°F, connector, solder, solder connector adaptor and thermal jackets.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



INDUSTRIAL  
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# INDUSTRIAL ICP® ACCELEROMETER SPECIFICATIONS

Model No.

326A01

Revisions:

-B- Rev # 4488 *ZL*

## DYNAMICS

|                                                                             | ENGLISH                            | S.I.                               |
|-----------------------------------------------------------------------------|------------------------------------|------------------------------------|
| Sensitivity ( $\pm 5\%$ ) <sup>(A)</sup>                                    | 100 mV/g                           | 10,2 mV/m/s <sup>2</sup>           |
| Measurement Range                                                           | $\pm 50$ g                         | $\pm 490$ m/s <sup>2</sup>         |
| Resolution                                                                  | 0.0001 g                           | 0,001 m/s <sup>2</sup>             |
| Frequency Range: ( $\pm 5\%$ ) <sup>(B)</sup> <sup>(C)</sup> <sup>(D)</sup> | 30 - 300,000 cpm (0.5 - 5,000 Hz)  | 30 - 300 000 cpm (0.5 - 5 000 Hz)  |
| ( $\pm 10\%$ ) <sup>(E)</sup> <sup>(F)</sup>                                | 18 - 420,000 cpm (0.3 - 7,000 Hz)  | 18 - 420 000 cpm (0.3 - 7 000 Hz)  |
| ( $\pm 3dB$ ) <sup>(G)</sup>                                                | 12 - 600,000 cpm (0.2 - 10,000 Hz) | 12 - 600 000 cpm (0.2 - 10 000 Hz) |
| Resonant Frequency <sup>(H)</sup>                                           | 1,200 kcpm (20 kHz)                | 1 200 kcpm (20 kHz)                |
| Transverse Sensitivity                                                      | $\leq 5\%$                         | $\leq 5\%$                         |
| Amplitude Linearity <sup>(I)</sup>                                          | $\pm 1\%$                          | $\pm 1\%$                          |

## ENVIRONMENTAL

|                         |                   |                                   |
|-------------------------|-------------------|-----------------------------------|
| Shock Limit (peak)      | 5,000 g           | 49 050 m/s <sup>2</sup>           |
| Temperature Range       | -65° to +250°F    | -54° to +121°C                    |
| Temperature Coefficient | See Graph         | See Graph                         |
| Strain Sensitivity      | 0.0005 g/ $\mu$ s | 0,0049 m/s <sup>2</sup> / $\mu$ s |

## ELECTRICAL

|                                          |                          |                                             |
|------------------------------------------|--------------------------|---------------------------------------------|
| Settling Time <sup>(J)</sup>             | 60 sec                   | 60 sec                                      |
| Excitation Voltage                       | 18 to 28 volts           | 18 to 28 volts                              |
| Excitation Current                       | 2 to 20 mA               | 2 to 20 mA                                  |
| Electrical Noise Broadband               | 100 $\mu$ g              | 981 $\mu$ m/s <sup>2</sup>                  |
| Electrical Noise Spectral: 1 Hz          | 10.0 $\mu$ g/ $\sqrt$ Hz | 98,0 ( $\mu$ m/s <sup>2</sup> )/ $\sqrt$ Hz |
| 10 Hz                                    | 2.0 $\mu$ g/ $\sqrt$ Hz  | 20,0 ( $\mu$ m/s <sup>2</sup> )/ $\sqrt$ Hz |
| 100 Hz                                   | 0.6 $\mu$ g/ $\sqrt$ Hz  | 6,0 ( $\mu$ m/s <sup>2</sup> )/ $\sqrt$ Hz  |
| 1000 Hz                                  | 0.2 $\mu$ g/ $\sqrt$ Hz  | 2,0 ( $\mu$ m/s <sup>2</sup> )/ $\sqrt$ Hz  |
| Output Impedance                         | <100 ohms                | <100 ohms                                   |
| Output Bias                              | 8 to 14 volts            | 8 to 14 volts.                              |
| Full Scale Output Voltage <sup>(K)</sup> | $\pm 5$ volts            | $\pm 5$ volts                               |
| Discharge Time Constant                  | $\geq 1.0$ sec           | $\geq 1.0$ sec                              |
| Case Isolation                           | $\geq 10^4$ ohms         | $\geq 10^4$ ohms                            |
| Protection                               | RFI / ESD                | RFI / ESD                                   |

## MECHANICAL

|                                         |                 |                 |
|-----------------------------------------|-----------------|-----------------|
| Size: Hex x Height                      | 1 3/16 x 2 3/16 | 30,2 x 55,6 mm  |
| Weight                                  | 5.3 oz          | 150 gm          |
| Mounting Thread (Female) <sup>(L)</sup> | 1/4 - 28        | Not Applicable  |
| Mounting Torque                         | 2 to 5 ft-lb    | 2,7 to 6,8 N-m  |
| Sensing Element                         | Shear - Ceramic | Shear - Ceramic |
| Case Material                           | 316L ST. ST.    | 316L ST. ST.    |
| Sealing                                 | Welded Hermetic | Welded Hermetic |
| Connector Type (2 - Pin)                | Mil-C-5015      | Mil-C-5015      |
| Connector Position                      | Top             | Top             |
| Mating Connector                        | MS 3106         | MS 3106         |

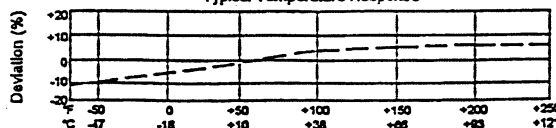
### Supplied Accessories:

Model 081B20 Mounting Stud  
Model 08SA31 Protective/Thermal Jacket  
Calibration Certificate (10 to 5,000 Hz)

### Note:

- (A) Conversion Factor: 1g = 9.81 m/s<sup>2</sup>
- (B) 1 Hz = 60 cpm (cycles per minute)
- (C) Mounted Resonance (nominal)
- (D) Zero Base Best Straight Line
- (E) Within 1% of output bias
- (F) For FSO at 14V bias, minimum of 20V excitation is required
- (G) 1/4-28 thread has no equivalent in SI systems

Typical Temperature Response



Specifications subject to change without notification.

App'd: U.F. 9/11/93 Engr. *[Signature]* 9/11/93 Sales: *[Signature]* 9/11/93 Spec No. 326-1010-80



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# INDUSTRIAL ICP® ACCELEROMETER SPECIFICATIONS

Model No.

**326A02**

Revision:

-A- Rev #4031

## DYNAMICS

Sensitivity ( $\pm 5\%$ ) <sup>[A]</sup>

Measurement Range

Resolution

Frequency Range: ( $\pm 5\%$ ) <sup>[B]</sup>

( $\pm 10\%$ )

( $\pm 3\text{dB}$ )

Resonant Frequency <sup>[C]</sup>

Transverse Sensitivity

Amplitude Linearity <sup>[D]</sup>

## ENVIRONMENTAL

Shock Limit (Peak)

Temperature Range

Temperature Coefficient

Strain Sensitivity

## ELECTRICAL

Settling Time <sup>[E]</sup>

Excitation Voltage

Excitation Current

Electrical Noise Broadband

Electrical Noise Spectral: (1 Hz)

(10 Hz)

(100 Hz)

(1,000 Hz)

Output Impedance

Output Bias

Full Scale Output Voltage

Discharge Time Constant

Case Isolation

Protection

## MECHANICAL

Size: Hex x Height

Weight

Mounting Thread (Female) <sup>[F]</sup>

Mounting Torque

Sensing Element

Case Material

Sealing

Connector Type (2-pin)

Connector Position

Mating Connector

Supplied with Accelerometer:

Model 081B20 Mounting Stud

Model 085A31 Protective/Thermal Jacket

Calibration Certificate (10 to 2,000 Hz)

Notes:

[A] Conversion Factor:  $1\text{g} = 9.81\text{ m/s}^2$

[B]  $1\text{ Hz} = 60\text{ cpm}$  (cycles per minute)

[C] Mounted Resonance (nominal)

[D] Zero Based, Best Straight Line

[E] Within 1% of Output Bias

[F] 1/4-28 has no equivalent in S.I. units.

## ENGLISH

500 mV/g

$\pm 10\text{ g}$

0.00001 g

30 - 120,000 cpm (0.5 - 2,000 Hz)

18 - 240,000 cpm (0.3 - 4,000 Hz)

12 - 360,000 cpm (0.2 - 6,000 Hz)

600 kcpm (10 kHz)

$\leq 7\%$

$\leq 1\%$

5,000 g

- 65 to +250 °F

See Graph

0.0005 g/ $\mu\text{z}$

60 sec

18 to 28 volts

2 to 20 mA

10  $\mu\text{g}/\text{Hz}$

2.0  $\mu\text{g}/\text{Hz}$

0.3  $\mu\text{g}/\text{Hz}$

0.2  $\mu\text{g}/\text{Hz}$

0.1  $\mu\text{g}/\text{Hz}$

<100 ohms

8 to 11 volts

$\pm 5\text{ volts}$

$\geq 1.0\text{ sec}$

$\geq 10^6\text{ ohms}$

RFI / ESD

1 3/16 x 2 3/16 in

7.4 oz

1/4-28

2 to 5 ft-lb

Ceramic

316L St. Stl.

Hermetic

Mil-C-5015

Top

MS3106

## S.I.

51 mV/m/s<sup>2</sup>

$\pm 98\text{ m/s}^2$

0.0001 m/s<sup>2</sup>

30 - 120 000 cpm (0,5 - 2 000 Hz)

18 - 240 000 cpm (0,3 - 4 000 Hz)

12 - 360 000 cpm (0,2 - 6 000 Hz)

600 kcpm (10 kHz)

$\leq 7\%$

$\leq 1\%$

49050 m/s<sup>2</sup>

- 54 to +121 °C

See Graph

0,0049 m/s<sup>2</sup>/ $\mu\text{z}$

60 sec

18 to 28 volts

2 to 20 mA

98 ( $\mu\text{m/s}^2$ )/Hz

20 ( $\mu\text{m/s}^2$ )/Hz

2,9 ( $\mu\text{m/s}^2$ )/Hz

2,0 ( $\mu\text{m/s}^2$ )/Hz

1,0 ( $\mu\text{m/s}^2$ )/Hz

<100 ohms

8 to 11 volts

$\pm 5\text{ volts}$

$\geq 1,0\text{ sec}$

$\geq 10^6\text{ ohms}$

RFI / ESD

30,2 x 55,6 mm

210 gm

Not Applicable

2,7 to 6,8 N-m

Ceramic

316L St. Stl.

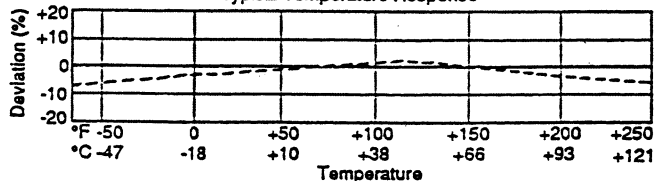
Hermetic

Mil-C-5015

Top

MS3106

Typical Temperature Response



Specifications subject to change without notification.

App'd:

D. L. H. 4/7/93

Engr:

7/1/93

Sales:

1/1/93

Spec. No.

326-1020-80



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# INDUSTRIAL ICP® ACCELEROMETER SPECIFICATIONS

Model No.

**326A03**

Revision:

## DYNAMICS

|                                               | ENGLISH                           | S.I.                             |
|-----------------------------------------------|-----------------------------------|----------------------------------|
| Sensitivity ( $\pm 5\%$ ) <sup>[A]</sup>      | 1000 mV/g                         | 102 mV/m/s <sup>2</sup>          |
| Measurement Range                             | $\pm 5$ g                         | $\pm 49$ m/s <sup>2</sup>        |
| Resolution                                    | 0.000005 g                        | 0.000049 m/s <sup>2</sup>        |
| Frequency Range: ( $\pm 5\%$ ) <sup>[B]</sup> | 30 - 120,000 cpm (0.5 - 2,000 Hz) | 30 - 120 000 cpm (0.5 - 2000 Hz) |
| ( $\pm 10\%$ )                                | 18 - 240,000 cpm (0.3 - 4,000 Hz) | 18 - 240 000 cpm (0.3 - 4000 Hz) |
| ( $\pm 3$ dB)                                 | 12 - 360,000 cpm (0.2 - 6,000 Hz) | 12 - 360 000 cpm (0.2 - 6000 Hz) |
| Resonant Frequency <sup>[C]</sup>             | 600 kcpm (10 kHz)                 | 600 kcpm (10 kHz)                |
| Transverse Sensitivity                        | $\leq 7\%$                        | $\leq 7\%$                       |
| Amplitude Linearity <sup>[D]</sup>            | $\leq 1\%$                        | $\leq 1\%$                       |

## ENVIRONMENTAL

|                         |                  |                                   |
|-------------------------|------------------|-----------------------------------|
| Shock Limit (Peak)      | 5,000 g          | 49 050 m/s <sup>2</sup>           |
| Temperature Range       | - 65 to +250 °F  | - 54 to +121 °C                   |
| Temperature Coefficient | See Graph        | See Graph                         |
| Strain Sensitivity      | 0.0005g/ $\mu$ ε | 0.0049 m/s <sup>2</sup> / $\mu$ ε |

## ELECTRICAL

|                                   |                  |                                    |
|-----------------------------------|------------------|------------------------------------|
| Settling Time <sup>[E]</sup>      | 60 sec           | 60 sec                             |
| Excitation Voltage                | 18 to 28 volts   | 18 to 28 volts                     |
| Excitation Current                | 2 to 20 mA       | 2 to 20 mA                         |
| Electrical Noise Broadband        | 5.0 $\mu$ g/√Hz  | 49 ( $\mu$ m/s <sup>2</sup> )/√Hz  |
| Electrical Noise Spectral: (1 Hz) | 1.0 $\mu$ g/√Hz  | 10 ( $\mu$ m/s <sup>2</sup> )/√Hz  |
| (10 Hz)                           | 0.2 $\mu$ g/√Hz  | 2.0 ( $\mu$ m/s <sup>2</sup> )/√Hz |
| (100 Hz)                          | 0.1 $\mu$ g/√Hz  | 1.0 ( $\mu$ m/s <sup>2</sup> )/√Hz |
| (1,000 Hz)                        | 0.05 $\mu$ g/√Hz | 0.5 ( $\mu$ m/s <sup>2</sup> )/√Hz |
| Output Impedance                  | <100 ohms        | <100 ohms                          |
| Output Bias                       | 8 to 11 volts    | 8 to 11 volts                      |
| Full Scale Output Voltage         | $\pm 5$ volts    | $\pm 5$ volts                      |
| Discharge Time Constant           | $\geq 1.0$ sec   | $\geq 1.0$ sec                     |
| Case Isolation                    | $\geq 10^8$ ohms | $\geq 10^8$ ohms                   |
| Protection                        | RFI / ESD        | RFI / ESD                          |

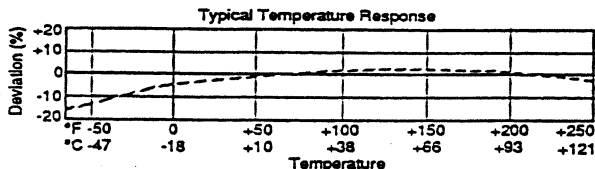
## MECHANICAL

|                                         |                    |                |
|-----------------------------------------|--------------------|----------------|
| Size: Hex x Height                      | 1 3/16 x 2 3/16 in | 30,2 x 55,6 mm |
| Weight                                  | 7.4 oz             | 210 gm         |
| Mounting Thread (Female) <sup>[F]</sup> | 1/4-28             | Not Applicable |
| Mounting Torque                         | 2 to 5 ft-lb       | 2,7 to 6,8 N-m |
| Sensing Element                         | Ceramic            | Ceramic        |
| Case Material                           | 316L St. Stl.      | 316L St. Stl.  |
| Sealing                                 | Hermetic           | Hermetic       |
| Connector Type (2-pin)                  | Mil-C-5015         | Mil-C-5015     |
| Connector Position                      | Top                | Top            |
| Mating Connector                        | MS3106             | MS3106         |

Supplied with Accelerometer:  
Model 081B20 Mounting Stud  
Model 085A31 Protective /Thermal Jacket  
Calibration Certificate (10 to 2,000 Hz)

### Notes:

- [A] Conversion Factor: 1g = 9.81 m/s<sup>2</sup>
- [B] 1 Hz = 60 cpm (cycles per minute)
- [C] Mounted Resonance (nominal)
- [D] Zero Based, Best Straight Line
- [E] Within 1% of Output Bias
- [F] 1/4-28 has no equivalent in S.I.units.



Specifications subject to change without notification.

App'd:

D. LALLY 1/6/93

Engr:

R. L. LALLY 1/6/93

Sales:

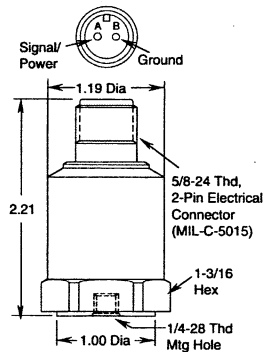
1/6/93

Spec. No.

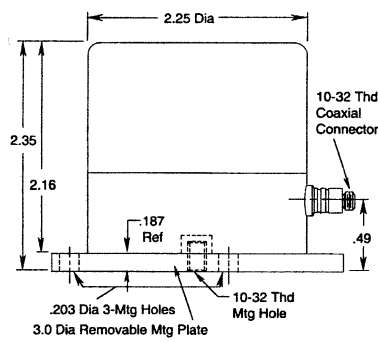
**326-1030-80**

Dimensions shown in inches except where noted.

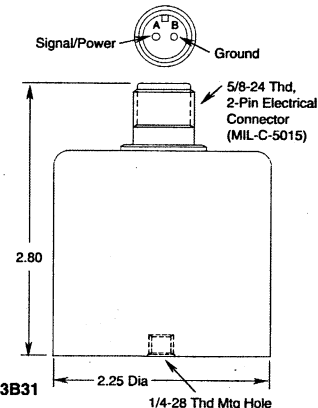
## English Selection Guide



393A03 &amp; 393B12



393C



393B31

| MODEL NUMBER                    | Unit             | GENERAL PURPOSE |               | HIGH SENSITIVITY |                |
|---------------------------------|------------------|-----------------|---------------|------------------|----------------|
|                                 |                  | 393A03          | 393C          | 393B12           | 393B31         |
| Voltage Sensitivity (1)         | mV/g             | 1 000           | 1 000         | 10 000           | 10 000         |
| Frequency Range ( $\pm 5\%$ )   | Hz               | 0.5 to 2 000    | 0.025 to 800  | 0.15 to 1000     | 0.1 to 200     |
| ( $\pm 10\%$ )                  | Hz               | 0.3 to 4 000    | 0.01 to 1 200 | 0.1 to 1500      | 0.07 to 300    |
| Resonant Frequency              | kHz              | $\geq 10$       | $\geq 3.5$    | $\geq 10$        | $\geq 1.0$     |
| Amplitude Range                 | $\pm g$ pk       | 5               | 1.5           | 0.5              | 0.5            |
| Resolution (broadband)          | g pk             | 0.000005        | 0.0001        | 0.000005         | 0.000001       |
| Mechanical Shock Limits         | $\pm g$ pk       | 5 000           | 100           | 200              | 40             |
| Temperature Range               | $^{\circ}F$      | -65 to +250     | -65 to +200   | -50 to +180      | 0 to +150      |
| Temperature Coefficient         | $\%/^{\circ}F$   | $\leq 0.05$     | $\leq 0.03$   | $\leq 0.05$      | $\leq 0.08$    |
| Amplitude Linearity             | %                | $\pm 1$         | $\pm 1$       | $\pm 1$          | $\pm 1$        |
| Transverse Sensitivity          | %                | $\leq 5$        | $\leq 5$      | $\leq 5$         | $\leq 5$       |
| Base Strain Sensitivity         | g/ $\mu\epsilon$ | $\leq 0.0005$   | $\leq 0.001$  | $\leq 0.0005$    | $\leq 0.005$   |
| Excitation Voltage              | VDC              | 20 to 30        | 18 to 30      | 20 to 30         | 24 to 28       |
| Constant Current Excitation     | mA               | 2 to 20         | 2 to 20       | 2 to 20          | 2 to 10        |
| Output Impedance                | ohm              | $\leq 200$      | $\leq 300$    | $\leq 300$       | $\leq 500$     |
| Output Bias Voltage             | VDC              | 8 to 14         | 3 to 6        | 8 to 14          | 8 to 14        |
| Discharge Time Constant         | second           | $\geq 1$        | $\geq 20$     | $\geq 3.5$       | $\geq 5$       |
| Ground Isolation (2)            | ohm              | $\geq 10^8$     | $\geq 10^8$   | $\geq 10^8$      | $\geq 10^8$    |
| Sensing Element                 | type             | ceramic         | quartz        | ceramic          | ceramic        |
| Element Configuration           | structure        | tri-shear       | compression   | tri-shear        | flex-tensional |
| Connector                       | type             | Mil-C-5015      | 10-32 coax    | Mil-C-5015       | Mil-C-5015     |
| Sealing                         | type             | weld/herm       | weld/herm     | weld/herm        | weld/herm      |
| Mounting Thread (3)             | size             | 1/4-28          | 10-32         | 1/4-28           | 1/4-28         |
| Housing                         | material         | stainless stl   | stainless stl | stainless stl    | stainless stl  |
| Weight                          | gram (oz)        | 210 (7.4)       | 1 000 (35.3)  | 200 (7.05)       | 635 (22.4)     |
| SUPPLIED ACCESSORIES: (3, 4, 5) |                  |                 |               |                  |                |
| Mounting Stud (6)               |                  | 081B20          | 081B05        | 081B20           | 081B20         |
| Protective / Thermal Boot (7)   |                  | 085A31          | N/A           | 085A31           | N/A            |
| OPTIONAL ACCESSORIES: (4)       |                  |                 |               |                  |                |
| Magnetic Mounting Base          |                  | 080A54          | 080A21        | 080A54           | N/A            |
| Triaxial Mounting Adaptor       |                  | 080A57          | 080M16        | 080A57           | N/A            |
| CABLING: (4)                    |                  |                 |               |                  |                |
| Mating Cable Connectors         | type             | E, M, P         | A, H, K, W    | E, M, P          | E, M, P        |
| Recommended Stock Cable         | series           | 002, 012, 024   | 002           | 002, 012, 024    | 002, 012, 024  |

## NOTES:

- Models 393A03 and 393B31 have a sensitivity tolerance of  $\pm 5\%$ . Models 393C and 393B12 have a sensitivity tolerance of  $\pm 10\%$ .
- Model 393C is base isolated; remaining models utilize case isolated design.
- Model 393C also supplied with Model 080A88 three hole mounting flange.
- See page cables/accessories section beginning on page 78 for a complete description of the supplied and optional models.
- NIST traceable calibration certificate from 10 Hz to +5% frequency point supplied with each accelerometer.
- Model 081B05 mounting stud is threaded 10-32 to 10-32. Model 081B20 mounting stud is threaded 1/4-28 to 1/4-28.
- Protective / thermal boot reduces thermal transient errors at low frequencies.

## Piezoelectric Acceleration Pickups

High performance pickups must be used to measure vibration accurately. The piezoelectric acceleration pickup is small, lightweight and can even be used

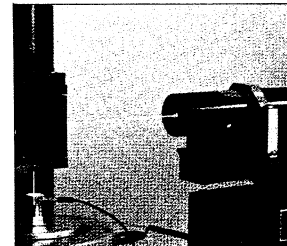
for measuring high frequency vibrations. When the object to be measured is small or if the object is a machine part, the vibration measurement must be made

with this type of pickup. RION products various models of acceleration pickups ranging in size from miniature to large output models including triaxial models.

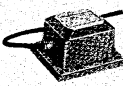
| Model                                   | PV-32A                                                                            | PV-34                                                                             | PV-35                                                                             | PV-36                                                                              | PV-37A                                                                              | PV-44                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| APPEARANCE                              |  |  |  |  |  |  |
| WEIGHT, g                               | 33                                                                                | 36                                                                                | 30                                                                                | 19                                                                                 | 18                                                                                  | 29                                                                                  |
| DIMENSIONS, mm (Hex x H)<br>(W x D x H) | 25 x 25 x 25<br>(W x D x H)                                                       | 17 x 22                                                                           | 17 x 23                                                                           | 14 x 20                                                                            | 14 x 19                                                                             | 16 x 20.5                                                                           |
| STRUCTURE                               | Compression                                                                       | Compression                                                                       | Compression                                                                       | Compression                                                                        | Compression                                                                         | Compression                                                                         |
| CHARGE SENSITIVITY,<br>PC/G             | 13                                                                                | 88                                                                                | 85                                                                                | 17                                                                                 | 15                                                                                  | 75                                                                                  |
| VOLTAGE SENSITIVITY,<br>mV/G            | 11                                                                                | 80                                                                                | 77                                                                                | 15                                                                                 | 14                                                                                  | 70                                                                                  |
| CAPACITANCE, pF (cable incl.)           | 1,100                                                                             | 1,100                                                                             | 1,100                                                                             | 1,100                                                                              | 1,100                                                                               | 1,100                                                                               |
| FREQUENCY RESPONSE, Hz<br>(±1.0 dB)     | 1~5,000                                                                           | 1~8,000                                                                           | 1~8,000                                                                           | 1~15,000                                                                           | 1~8,000                                                                             | 1~10,000                                                                            |
| MOUNTED RESONANCE<br>FREQUENCY, kHz     | 20                                                                                | 30                                                                                | 30                                                                                | 50                                                                                 | 30                                                                                  | 28                                                                                  |
| WORKING TEMPERATURE<br>RANGE, °C        | -50~+160                                                                          | -50~+160                                                                          | -50~+160                                                                          | -50~+160                                                                           | -50~+160                                                                            | -50~+260                                                                            |
| ACCELERATION RANGE, G                   | 400                                                                               | 500                                                                               | 500                                                                               | 500                                                                                | 500                                                                                 | 400                                                                                 |
| CABLE & CONNECTOR TYPE                  | VP-51C                                                                            | VP-51A                                                                            | VP-51A                                                                            | VP-51A                                                                             | VP-51A                                                                              | VP-51B                                                                              |
| MOUNTING                                | M6 Stud                                                                           | M6 Stud                                                                           | M6 Stud                                                                           | M6 Stud                                                                            | M6 Stud                                                                             | 10-32 Stud                                                                          |

| Model                                   | PV-65                                                                               | PV-81                                                                               | PV-84                                                                               | PV-85                                                                                | PV-86                                                                                 | PV-87                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| APPEARANCE                              |  |  |  |  |  |  |
| WEIGHT, g                               | 26                                                                                  | 405                                                                                 | 80                                                                                  | 23                                                                                   | 23                                                                                    | 115                                                                                   |
| DIMENSIONS, mm (Hex x H)<br>(W x D x H) | 16 x 22.5                                                                           | 42 x 42 x 42<br>(W x D x H)                                                         | 24 x 25                                                                             | 17 x 18.5                                                                            | 17 x 18.5                                                                             | 24 x 30.5                                                                             |
| STRUCTURE                               | Shear                                                                               | Shear                                                                               | Shear                                                                               | Shear                                                                                | Shear                                                                                 | Shear                                                                                 |
| CHARGE SENSITIVITY,<br>PC/G             | 60                                                                                  | 1,000                                                                               | 160                                                                                 | 58                                                                                   | 58                                                                                    | 320                                                                                   |
| VOLTAGE SENSITIVITY,<br>mV/G            | 17                                                                                  | 500                                                                                 | 170                                                                                 | 64                                                                                   | 64                                                                                    | 350                                                                                   |
| CAPACITANCE, pF (cable incl.)           | 3,700                                                                               | 2,000                                                                               | 900                                                                                 | 900                                                                                  | 900                                                                                   | 900                                                                                   |
| FREQUENCY RESPONSE, Hz<br>(±1.0 dB)     | 1~10,000                                                                            | 0.5~6,000                                                                           | 1~4,000                                                                             | 1~8,000                                                                              | 1~8,000                                                                               | 1~3,000                                                                               |
| MOUNTED RESONANCE<br>FREQUENCY, kHz     | 23                                                                                  | 4                                                                                   | 13                                                                                  | 22                                                                                   | 20                                                                                    | 7                                                                                     |
| WORKING TEMPERATURE<br>RANGE, °C        | -50~+260                                                                            | -50~+160                                                                            | -50~+160                                                                            | -50~+160                                                                             | -50~+160                                                                              | -50~+160                                                                              |
| ACCELERATION RANGE, G                   | 400                                                                                 | (40)                                                                                | 200                                                                                 | 500                                                                                  | 500                                                                                   | 100                                                                                   |
| CABLE & CONNECTOR TYPE                  | VP-51B                                                                              | VP-51A                                                                              | VP-51A                                                                              | VP-51A                                                                               | VP-51A                                                                                | VP-51A                                                                                |
| MOUNTING                                | M6 Stud                                                                             | M6 Stud                                                                             | M6 Stud                                                                             | M6 Stud                                                                              | M6 Stud                                                                               | M6 Stud                                                                               |

| Model                                     | PV-90A                                                                            | PV-92                                                                             | PV-93                                                                             | PV-95                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| APPEARANCE                                |  |  |  |  |
| WEIGHT, g                                 | 1.2                                                                               | 6.3                                                                               | 28                                                                                | 9                                                                                   |
| DIMENSIONS, mm (Hex x H)                  | 6 x 10                                                                            | 19.5 x 11 x 7<br>(W x D x H)                                                      | 21 x 21 x 16<br>(W x D x H)                                                       | 14 x 15                                                                             |
| STRUCTURE                                 | Shear                                                                             | Shear                                                                             | Shear                                                                             | Shear                                                                               |
| CHARGE SENSITIVITY,<br>pC/G               | 1.8                                                                               | 2.6                                                                               | 5.4                                                                               | 5.5                                                                                 |
| VOLTAGE SENSITIVITY,<br>mV/G              | 2.4                                                                               | 3.5                                                                               | 8                                                                                 | 8                                                                                   |
| CAPACITANCE, pF (cable incl.)             | 750                                                                               | 750                                                                               | 670                                                                               | 670                                                                                 |
| FREQUENCY RESPONSE, Hz<br>( $\pm 1.0$ dB) | 1~25,000                                                                          | 1~25,000                                                                          | 1~8,000                                                                           | 1~10,000                                                                            |
| MOUNTED RESONANCE<br>FREQUENCY, kHz       | 70                                                                                | 70                                                                                | 45                                                                                | 50                                                                                  |
| WORKING TEMPERATURE<br>RANGE, °C          | -50~+160                                                                          | -50~+160                                                                          | -50~+160                                                                          | -50~+160                                                                            |
| ACCELERATION RANGE, G                     | 500                                                                               | 500                                                                               | 1,000                                                                             | 1,000                                                                               |
| CABLE & CONNECTOR TYPE                    | VP-51D                                                                            | VP-51D x 3                                                                        | VP-51C                                                                            | VP-51A                                                                              |
| MOUNTING                                  | M3 Stud                                                                           | Adhesive                                                                          | M6 Stud                                                                           | M6 Stud                                                                             |



Vibration pickup Calibration

| Model                               | PV-20                                                                              | PV-22                                                                              | PV-62                                                                               | PV-83A                                                                               | PV-88A                                                                               |
|-------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| APPEARANCE                          |  |  |  |  |  |
| STRUCTURE                           | Shear                                                                              | Shear                                                                              | Shear                                                                               | Shear                                                                                | Shear                                                                                |
| VOLTAGE SENSITIVITY,<br>V/1,000 Gal | 1                                                                                  | 5                                                                                  | *35 mV/G                                                                            | 0.6                                                                                  | 0.6                                                                                  |
| FREQUENCY RESPONSE,<br>Hz           | 0.1~1,000                                                                          | 0.1~30                                                                             | 1~100                                                                               | 1~90                                                                                 | 1~90                                                                                 |
| ACCELERATION RANGE,<br>Gal          | 1~5,000                                                                            | 0.3~1,000                                                                          | 0.3~3,000                                                                           | 0.03~3,000                                                                           | 0.03~3,000                                                                           |
| WORKING TEMPERATURE<br>RANGE, °C    | 0~+40                                                                              | 0~+40                                                                              | -20~+60                                                                             | 0~+40                                                                                | 0~+40                                                                                |
| WEIGHT, g                           | 560                                                                                | 2,500                                                                              | 400                                                                                 | 740                                                                                  | 430                                                                                  |
| DIMENSIONS, mm                      | 60Wx98Dx25H                                                                        | 70 Dia. x 300                                                                      | 210 Dia. x 12                                                                       | 84 Dia. x 61                                                                         | 64 Dia. x 58                                                                         |
| POWER SUPPLY                        | $\pm 6$ VDC                                                                        | $\pm 6$ VDC                                                                        | —                                                                                   | 9 VDC                                                                                | 9 VDC                                                                                |
| WITH INTERNAL<br>ELECTRONICS        | Charge<br>Amplifier<br>3-direction                                                 | Charge<br>Amplifier<br>3-direction                                                 | NO<br>*Charge Sensitivity<br>(30 pC/G)<br>3-direction                               | Pre Amplifier<br>Low Pass Filter<br>3-direction                                      | Pre Amplifier<br>Low Pass Filter<br>1-direction                                      |

# Vibra-Metrics® Accelerometer Specifications

| APPLICATION                                                                          | MINIATURE/HIGH FREQUENCY        | GENERAL PURPOSE                  | LOW COST                        | INDUSTRIAL              | HIGH TEMPERATURE        | UNDER WATER             | LOW FREQUENCY           |
|--------------------------------------------------------------------------------------|---------------------------------|----------------------------------|---------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Specifications<br>(All values typical @ 75°F, 75% RH,<br>unless otherwise specified) |                                 |                                  |                                 |                         |                         |                         |                         |
| Model                                                                                | 9001A                           | 1000                             | 1018/19                         | 3000                    | 1020S                   | 1022                    | 1026                    |
| Sensitivity<br>(A = 5%, B = ±1dB)                                                    | A, 8.5                          | A, 10                            | A, 10                           | A, 10                   | A, 10                   | A, 10                   | A, 10                   |
| Frequency Response                                                                   | ±5% Hz<br>±3dB Hz               | 3-15,000<br>1-40,000             | 3-10,000<br>1-30,000            | 3-10,000<br>1-11,000    | 3-10,000<br>1-12,000    | 3-10,000<br>1-11,000    | 3-10,000<br>1-11,000    |
| Turn-On-Time<br>(2% of final bias)                                                   | 5                               | 5                                | 5                               | 5                       | 5                       | 5                       | 5                       |
| Amplitude Range (±1%)                                                                | 550                             | 500                              | 500                             | 500                     | 500                     | 500                     | 500                     |
| Resonant Frequency                                                                   | 80                              | 90                               | 60                              | 50                      | 50                      | 50                      | 50                      |
| Noise                                                                                | 1500<br>100<br>25<br>10         | 1500<br>100<br>25<br>10          | 1500<br>100<br>25<br>10         | 1500<br>100<br>25<br>10 | 1500<br>100<br>25<br>10 | 1500<br>100<br>25<br>10 | 1500<br>100<br>25<br>10 |
| Transverse Sensitivity<br>(typical)                                                  | <5%                             | <5%                              | <5%                             | <5%                     | <5%                     | <5%                     | <5%                     |
| Output Impedance                                                                     | OHM                             | 1000                             | 1000                            | 1000                    | 1000                    | 1000                    | 1000                    |
| EMI/RFI Protection                                                                   | dB                              | 20                               | 52                              | 52                      | 52                      | 52                      | 52                      |
| ESD Protection                                                                       | M ohm                           | yes                              | yes                             | yes                     | yes                     | yes                     | yes                     |
| Ground Isolation                                                                     |                                 | yes                              | yes                             | yes                     | yes                     | yes                     | yes                     |
| Bias Volts (nom.)                                                                    | 7                               | 7                                | 7                               | 7                       | 7                       | 7                       | 7                       |
| Power                                                                                | Vdc, 2-4ma                      | 15-30                            | 15-30                           | 15-30                   | 15-30                   | 15-30                   | 15-30                   |
| Temperature Operating<br>(°C/F)                                                      | Minimum<br>Maximum              | -40/-40<br>121/250               | -40/-40<br>121/250              | -40/-40<br>121/250      | -40/-40<br>121/250      | -40/-40<br>121/250      | -40/-40<br>121/250      |
| Size                                                                                 | height (in/cm)<br>flats (in/cm) | 30/8<br>25/6                     | 45/11<br>3/7                    | 86/22<br>5/13           | 112/7<br>10/25          | 97/6<br>13/33           | 97/6<br>13/33           |
| Weight                                                                               | grams (without cable)           | 7                                | 2                               | 16                      | 10                      | 35                      | 50                      |
| Seal                                                                                 |                                 | epoxy                            | epoxy                           | epoxy                   | epoxy                   | epoxy                   | epoxy                   |
| Connector (S=Side, T=Top)                                                            |                                 | S, cable<br>to 10-32             | S, 10-32                        | S, 10-32                | S, 10-32                | S, 10-32                | S, 10-32                |
| Mounting Stud                                                                        |                                 | None                             | 6-32                            | 10-32                   | 10-32                   | 10-32                   | 10-32                   |
| Design                                                                               |                                 | iso. comp.                       | iso. comp.                      | iso. comp.              | iso. comp.              | iso. comp.              | iso. comp.              |
| Sensor Highway® Compatible                                                           |                                 | yes                              | yes                             | yes                     | yes                     | yes                     | yes                     |
| Other Versions Available                                                             |                                 | *9002A<br>Low Noise<br>*100 mV/g | *Case Gold<br>*50 KHz<br>*Shock | *3000X<br>7 KHz         |                         |                         |                         |

1. As a part of continual product improvement, specifications subject to change without notice.  
2. Registered trademarks of Vibra-Metrics, Inc. include: Sensor Highway, IsoShield, IUREM.

3. Calibration supplied with accelerometers is 10-10 Hz.

\*\* Available on selected models. Consult factory for details.

Bulletin No. VM172

**LIFETIME WARRANTY ON 8000 SERIES\*\***

\* Patent No. 5,473,941

\*\* Patent No. 5,473,941

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Vibra-Metrics, Inc.

# VIBRA•METRICS ACCELEROMETER SELECTOR GUIDE

| GENERAL<br>AREA OF<br>APPLICATION              | Lab/<br>Lab Use — Ideal where small size, high sensitivity and broad frequency response are required for accurate structural testing. |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     | General Purpose — Laboratory Shake Testing, Acceleration, Velocity, Displacement Control, Production Screening, useful for most vibration measurements. |                                                                                     |                                                                                     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                |                                                      |  |  |  |  |  |                                                                      |  |  |
| MODEL NUMBER                                   | 9001A                                                                                                                                 | 1000                                                                              | 1001                                                                              | 2006                                                                              | TRIAx 3000                                                                        | TRIAx 3130                                                                          | 1018                                                                                                                                                    | 1019                                                                                | 1020S                                                                               |
| <b>SPECIAL FEATURES</b>                        |                                                                                                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                                                                                         |                                                                                     |                                                                                     |
| Low Impedance for Long Lines                   | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Physical Shock Protection                      | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Electric Overload Protection                   | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Electrically Isolated                          | —                                                                                                                                     | •                                                                                 | —                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | —                                                                                   | •                                                                                   |
| Internal Faraday Shielding                     | —                                                                                                                                     | —                                                                                 | —                                                                                 | •                                                                                 | —                                                                                 | —                                                                                   | —                                                                                                                                                       | —                                                                                   | —                                                                                   |
| Glass Fused Connector                          | —                                                                                                                                     | —                                                                                 | —                                                                                 | —                                                                                 | —                                                                                 | —                                                                                   | —                                                                                                                                                       | —                                                                                   | —                                                                                   |
| Removable Stud                                 | —                                                                                                                                     | —                                                                                 | —                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | —                                                                                                                                                       | •                                                                                   | —                                                                                   |
| Ultrasonic Range (above 20KHz)                 | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | —                                                                                 | —                                                                                   | •                                                                                                                                                       | •                                                                                   | —                                                                                   |
| Very Low Noise                                 | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| <b>SPECIAL OPTIONS</b>                         |                                                                                                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                                                                                         |                                                                                     |                                                                                     |
| 50 Ohm Output                                  | •                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| ± 2% Sensitivity version                       | —                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Magnetic Mount available                       | —                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | —                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Low Frequency 0.5 Hz version                   | —                                                                                                                                     | •                                                                                 | •                                                                                 | •                                                                                 | •                                                                                 | —                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| No Stud (for epoxy mounting)                   | Std.                                                                                                                                  | •                                                                                 | •                                                                                 | —                                                                                 | —                                                                                 | —                                                                                   | —                                                                                                                                                       | —                                                                                   | —                                                                                   |
| Splashproof                                    | •                                                                                                                                     | •                                                                                 | •                                                                                 | —                                                                                 | —                                                                                 | •                                                                                   | •                                                                                                                                                       | •                                                                                   | •                                                                                   |
| Pre-filter                                     | —                                                                                                                                     | —                                                                                 | —                                                                                 | —                                                                                 | —                                                                                 | —                                                                                   | —                                                                                                                                                       | —                                                                                   | •                                                                                   |
| <b>OPERATING SPECIFICATIONS</b>                |                                                                                                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                                                                                         |                                                                                     |                                                                                     |
| Sensitivity mv/g @ 100 Hz ± 5%                 | 8-9                                                                                                                                   | 10                                                                                | 10                                                                                | 10                                                                                | 10                                                                                | 10                                                                                  | 10                                                                                                                                                      | 10                                                                                  | 100                                                                                 |
| Frequency Range Hz ± 5%                        | 3 - 18K                                                                                                                               | 3 - 10K                                                                           | 3 - 10K                                                                           | 3 - 10K                                                                           | 3 - 10K                                                                           | 3 - 10K                                                                             | 3 - 10K                                                                                                                                                 | 3 - 10K                                                                             | 3 - 10K                                                                             |
| Amplitude Linearity ± 2% (g)                   | 600                                                                                                                                   | 480                                                                               | 480                                                                               | 480                                                                               | 480                                                                               | 480                                                                                 | 480                                                                                                                                                     | 480                                                                                 | 48                                                                                  |
| Max. Accel w/o Clipping (± g/VDC) Pk.          | 750/15                                                                                                                                | 500/15                                                                            | 500/15                                                                            | 500/15                                                                            | 500/15                                                                            | 500/15                                                                              | 500/15                                                                                                                                                  | 500/15                                                                              | 50/15                                                                               |
| Max. Shock w/o Damage (g)                      | 10000                                                                                                                                 | 10000                                                                             | 10000                                                                             | 10000                                                                             | 5000                                                                              | 5000                                                                                | 10000                                                                                                                                                   | 10000                                                                               | 5000                                                                                |
| Resonant Frequency Nom. (KHz)                  | 80                                                                                                                                    | 90                                                                                | 90                                                                                | 60                                                                                | 50                                                                                | 50                                                                                  | 60                                                                                                                                                      | 60                                                                                  | 25                                                                                  |
| Noise, g (rms), Wide Band                      | .0012                                                                                                                                 | .0012                                                                             | .0012                                                                             | .0012                                                                             | .0015                                                                             | .0015                                                                               | .0012                                                                                                                                                   | .0012                                                                               | .0015                                                                               |
| Transverse Sens (%) Max.                       | 5                                                                                                                                     | 5                                                                                 | 5                                                                                 | 5                                                                                 | 5                                                                                 | 5                                                                                   | 5                                                                                                                                                       | 5                                                                                   | 5                                                                                   |
| Sens. Temperature (%)<br>(Nominal-Static Soak) |                                                                                                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                                                                                         |                                                                                     |                                                                                     |
| -65°F to +100°F/38°C                           | +5                                                                                                                                    | +5                                                                                | +5                                                                                | +5                                                                                | +5                                                                                | +5                                                                                  | +5                                                                                                                                                      | +5                                                                                  | +5                                                                                  |
| 100°F to 150°F/66°C                            | -5                                                                                                                                    | -5                                                                                | -5                                                                                | -5                                                                                | -5                                                                                | -5                                                                                  | -5                                                                                                                                                      | -5                                                                                  | -5                                                                                  |
| 100°F to 250°F/120°C                           | -8                                                                                                                                    | -8                                                                                | -8                                                                                | -8                                                                                | -8                                                                                | -8                                                                                  | -8                                                                                                                                                      | -8                                                                                  | -8                                                                                  |
| Thermal Shock g/°C Nominal                     | 4                                                                                                                                     | 4                                                                                 | 4                                                                                 | 4                                                                                 | 4                                                                                 | 3                                                                                   | 2                                                                                                                                                       | 2                                                                                   | 0.5                                                                                 |
| Base Strain g/micro Strain, nom.               | .005                                                                                                                                  | .005                                                                              | .005                                                                              | .005                                                                              | .0015                                                                             | .003                                                                                | .003                                                                                                                                                    | .003                                                                                | .008                                                                                |
| Output Impedance - Ohms                        | 1000                                                                                                                                  | 1000                                                                              | 1000                                                                              | 1000                                                                              | 1000                                                                              | 1000                                                                                | 1000                                                                                                                                                    | 1000                                                                                | 1000                                                                                |
| Part Number                                    | 9353031                                                                                                                               | 9353018                                                                           | 9353027                                                                           | 9354533                                                                           | 9354323                                                                           | 9354707                                                                             | 9353173                                                                                                                                                 | 9353174                                                                             | 9353205                                                                             |
| <b>PHYSICAL SPECIFICATIONS</b>                 |                                                                                                                                       |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                     |                                                                                                                                                         |                                                                                     |                                                                                     |
| Weight - Grams (Nominal)                       | 0.75                                                                                                                                  | 2                                                                                 | 1.9                                                                               | 6                                                                                 | 10                                                                                | 7                                                                                   | 16                                                                                                                                                      | 16                                                                                  | 35                                                                                  |
| Size - Height in/cm<br>Flats in/cm             | 0.32/.81<br>0.25/0.63                                                                                                                 | 0.44/1.12<br>0.28/0.71                                                            | 0.44/1.12<br>0.28/0.71                                                            | 0.72/1.83<br>0.438/1.11                                                           | 1.08/2.74<br>0.98/2.48                                                            | 0.61/1.55<br>0.25/0.63                                                              | 0.86/2.18<br>0.50/1.27                                                                                                                                  | 0.82/2.08<br>0.50/1.27                                                              | 1.20/3.05<br>0.75/1.91                                                              |
| Wiring                                         | Coax                                                                                                                                  | Coax                                                                              | Coax                                                                              | Coax                                                                              | 4 Wire<br>Shielded                                                                | 3 Coax                                                                              | Coax                                                                                                                                                    | Coax                                                                                | Coax                                                                                |
| Power - Voltage<br>Current (ma)                | 15 - 30<br>1 - 4                                                                                                                      | 15 - 30<br>1 - 4                                                                  | 15 - 30<br>1 - 4                                                                  | 15 - 30<br>1 - 4                                                                  | 15 - 30<br>1 - 4/Axis                                                             | 15 - 30<br>1 - 4/Axis                                                               | 15 - 30<br>1 - 4                                                                                                                                        | 15 - 30<br>1 - 4                                                                    | 15 - 30<br>1 - 4                                                                    |
| Grounding                                      | Case<br>Ground                                                                                                                        | Isolated                                                                          | Case<br>Ground                                                                    | IsoShield                                                                         | Isolated                                                                          | Isolated                                                                            | Isolated                                                                                                                                                | Case<br>Ground                                                                      | Isolated                                                                            |
| Connector - Location                           | 6' integral<br>cable side                                                                                                             | 6' integral<br>cable side                                                         | 6' integral<br>cable side                                                         | top                                                                               | side                                                                              | integral<br>side                                                                    | side                                                                                                                                                    | side                                                                                | side                                                                                |
| Type                                           | 10 - 32                                                                                                                               | 10 - 32                                                                           | 10 - 32                                                                           | 10 - 32                                                                           | 4 - Pin                                                                           | 10 - 32                                                                             | 10 - 32                                                                                                                                                 | 10 - 32                                                                             | 10 - 32                                                                             |
| Mounting Stud                                  | None                                                                                                                                  | 6 - 32                                                                            | 6 - 32                                                                            | 10 - 32                                                                           | 10 - 32                                                                           | 10 - 32                                                                             | 10 - 32                                                                                                                                                 | 10 - 32                                                                             | 1/4 - 28                                                                            |
| Material - Base                                | Titanium                                                                                                                              | Titanium                                                                          | Titanium                                                                          | Titanium                                                                          | Titanium                                                                          | 303 SS                                                                              | 303 SS                                                                                                                                                  | 303 SS                                                                              | Titanium                                                                            |
| Material - Case                                | Titanium                                                                                                                              | Titanium                                                                          | Titanium                                                                          | Titanium                                                                          | Titanium                                                                          | Titanium                                                                            | 303 SS                                                                                                                                                  | 303 SS                                                                              | Titanium                                                                            |
| Seal                                           | epoxy                                                                                                                                 | epoxy                                                                             | epoxy                                                                             | epoxy                                                                             | epoxy                                                                             | epoxy                                                                               | epoxy                                                                                                                                                   | epoxy                                                                               | epoxy                                                                               |

P71291

Calibrations are traceable to N.I.S.T. to 10KHz and conform to MIL-STD-45662A and MIL-I-45208A.

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1014 SHERMAN AVENUE

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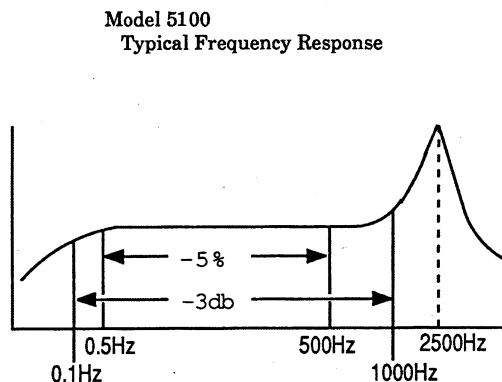
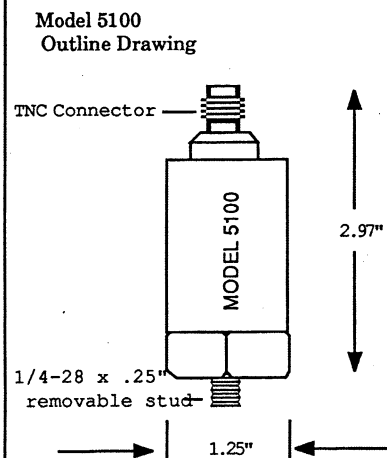
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## MODEL 5100 ACCELEROMETER

### LOW FREQUENCY - HIGH OUTPUT - SHEAR DESIGN

Industrial accelerometer designed for industrial mills and plants with low RPM machinery that require low frequency and high sensitivity specifications. The 5100 features rugged top connector, internal isolation, splash-proof construction along with the ability to measure 5 RPM and 10 $\mu$ g vibrations. Internal electronics will allow cable runs up to 1000 feet. Shear Design virtually eliminates base strain, thermal transients, and other phenomenon that are accentuated at low frequencies.

| Operating Specifications      |                     | Operating Specifications (cont'd.) |                                       |
|-------------------------------|---------------------|------------------------------------|---------------------------------------|
| Sensitivity $\pm 5\%$ @ 100Hz | 500 mv/g            | Noise Floor (rms)                  | 20 microvolts                         |
| Frequency Range $\pm 5\%$     | 0.5 Hz to 500 Hz    | Power Requirements                 | 15-30 vdc, 1-4 ma<br>constant current |
| $\pm 3$ db                    | 0.1 Hz to 700 Hz    | Physical Specifications            |                                       |
| Amplitude Range               | $\pm 10$ g          | Size (inches)                      | 2.97" high x 1.25" dia                |
| Transverse Sensitivity        | 7% maximum          | Weight                             | 9.5 oz.                               |
| Case Noise Rejection          | -50 db              | Connector                          | Top TNC                               |
| Maximum Shock                 |                     | Material                           | 303 Stainless Steel                   |
| without damage                | 5,000 g             | Mounting                           | 1/4 - 28 Stud (removeable)            |
| Mounted Resonant              |                     | Weatherseal                        |                                       |
| Frequency (nom.)              | 2.3 KHz             | Rating                             | Splash-Proof                          |
| Bias Voltage (nom.)           | 7 v                 | Temperature                        | -65° to +250°F                        |
| Output Impedance              | 1,000 $\Omega$      |                                    |                                       |
| Isolation                     | Internally Isolated |                                    |                                       |



Accelerometers and instrumentation for vibration monitoring, quality control, modal analysis and structural testing.

89

## Internally Amplified Vibration Sensors

### INDUSTRIAL, GENERAL PURPOSE ACCELEROMETERS

*These rugged, general purpose sensors are ideal for machinery monitoring and predictive maintenance applications. They are used on machinery monitoring systems worldwide.*

|                                                                                              | Sensitivity,<br>mV/g | Acceleration<br>Range, g | Frequency<br>Response, Hz<br>±5%, ±10% | Temperature<br>Range, °C | Weight,<br>grams | Case Material   | Mounting               |
|----------------------------------------------------------------------------------------------|----------------------|--------------------------|----------------------------------------|--------------------------|------------------|-----------------|------------------------|
| 756<br>     | 100                  | .0007 - 50               | 3 - 1,000<br>2 - 5,000                 | -50 to 90                | 180              | stainless steel | three #8 x 1/4" screws |
| 762<br>     | 10                   | .0015 - 500              | 3 - 10,000<br>2 - 12,000               | -50 to 120               | 80               | stainless steel | 1/4-28 tapped hole     |
| 766<br>     | 100                  | .00055 - 50              | 3 - 10,000<br>2 - 12,000               | -50 to 120               | 80               | stainless steel | 1/4-28 tapped hole     |
| 768*<br>   | 500                  | .00002 - 10              | 3 - 4,000<br>2 - 6,000                 | -50 to 120               | 90               | stainless steel | 1/4-28 tapped hole     |
| 776<br>   | 100                  | .0002 - 40               | 3 - 2,500<br>2 - 5,000                 | 0 to 85                  | 32               | stainless steel | 1/4-28 tapped hole     |
| 793<br>   | 100                  | .0007 - 50               | 3 - 5,000<br>2 - 7,000                 | -50 to 120               | 110              | stainless steel | 1/4-28 tapped hole     |
| 793L*<br> | 500                  | .00001 - 10              | 0.6 - 700<br>0.4 - 1,000               | -50 to 120               | 135              | stainless steel | 1/4-28 tapped hole     |
| 793LS<br> | 500                  | .00001 - 10              | 0.6 - 700<br>0.4 - 1,000               | -50 to 120               | 135              | stainless steel | 1/4-28 tapped hole     |
| 793S<br>  | 100                  | .0007 - 50               | 3 - 5,000<br>2 - 7,000                 | -50 to 120               | 110              | stainless steel | 1/4-28 tapped hole     |
| 796<br>   | 100                  | .0008 - 50               | 3 - 5,000<br>2 - 7,000                 | -50 to 120               | 100              | stainless steel | 1/4-28 captive screw   |

\* Ultra low noise sensor.

## Internally Amplified Vibration Sensors

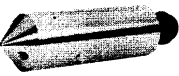
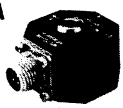
### LIGHTWEIGHT, COMPACT SENSORS

Lightweight, compact sensors are used in many laboratory research and light industrial applications. Their low mass makes them effective in applications where transducer mass must be minimized and high signal to noise ratio must be maintained.

|                                                                                                 | Sensitivity,<br>mV/g | Acceleration<br>Range, g | Frequency<br>Response, Hz<br>±5%, ±10% | Temperature<br>Range, °C | Weight,<br>grams | Case Material   | Mounting          |
|-------------------------------------------------------------------------------------------------|----------------------|--------------------------|----------------------------------------|--------------------------|------------------|-----------------|-------------------|
| 722/722T<br>   | 10                   | .0002 - 500              | 3 - 10,000<br>2 - 12,000               | -50 to 120               | 34               | stainless steel | 10-32 tapped hole |
| 726/726T<br>   | 100                  | .0001 - 50               | 3 - 10,000<br>2 - 12,000               | -50 to 120               | 34               | stainless steel | 10-32 tapped hole |
| 728/728T*<br>  | 500                  | .00003 - 10              | 3 - 4,000<br>2 - 6,000                 | -50 to 120               | 44               | stainless steel | 10-32 tapped hole |
| 732A/732T<br>  | 10                   | .0005 - 250              | 3 - 15,000<br>1 - 25,000               | -50 to 120               | 15               | stainless steel | 10-32 tapped hole |
| 736/736T<br> | 100                  | .00015 - 50              | 6 - 15,000<br>2 - 25,000               | -50 to 120               | 15               | stainless steel | 10-32 tapped hole |

### TRIAXIAL ACCELEROMETERS

Triaxial units contain three mutually perpendicular accelerometers in a compact housing, allowing the evaluation of three dimensional motion.

|                                                                                              | Sensitivity,<br>mV/g | Acceleration<br>Range, g | Frequency<br>Response, Hz<br>all channels ±10% | Temperature<br>Range, °C | Weight,<br>grams | Case Material          | Mounting                |
|----------------------------------------------------------------------------------------------|----------------------|--------------------------|------------------------------------------------|--------------------------|------------------|------------------------|-------------------------|
| 733<br>   | 10                   | .00015 - 250             | 2 - 4,000                                      | -50 to 120               | 23               | hardcoated<br>aluminum | 10-32<br>captive screw  |
| 755G*<br> | 3                    | .000005 - 1.5            | 6 - 4000                                       | -50 to 80                | 1900             | titanium               | —                       |
| 993A<br>  | 100                  | .0002 - 50               | 2 - 2,000                                      | -50 to 120               | 90               | hardcoated<br>aluminum | ¼ - 28<br>captive screw |

\* Ultra low noise sensor.

## ACCELEROMETERS

WR

### Model HA8 Hand-Held Vibration Probe - Acceleration Output

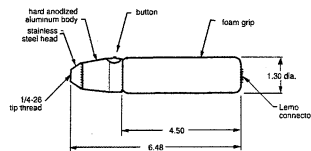


#### Features:

- Rugged
- Easy to use
- Shock protected with intrinsic overload protection
- Built-in electronics
- Miswiring protection

#### Applications:

- Predictive maintenance
- Bearing analysis
- Ship board machinery
- Heavy machinery
- Compressors



Model HA8

#### SPECIFICATIONS

##### DYNAMIC

|                                  |          |        |
|----------------------------------|----------|--------|
| Sensitivity, $\pm 5\%$ , 25 °C   | 100      | mV/g   |
| Acceleration Range               | 50       | g peak |
| Amplitude Nonlinearity           | 1        | %      |
| Frequency Response $\pm 10\%$ :  |          |        |
| screwed down using mounting stud | 2 - 2000 | Hz     |
| through probe tip                | 5 - 500  | Hz     |

##### ELECTRICAL

|                                   |                                    |                                |
|-----------------------------------|------------------------------------|--------------------------------|
| Power Requirement, voltage source | 18 - 30                            | VDC                            |
| current regulating diode          | 2 - 10                             | mA                             |
| Electrical Noise, equiv. g, nom.: |                                    |                                |
| Broadband, 2.5 Hz to 25 kHz       | 100                                | $\mu\text{g}$                  |
| Spectral                          |                                    |                                |
| 10 Hz                             | 10                                 | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 100 Hz                            | 5                                  | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 1000 Hz                           | 3                                  | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| Output Impedance, max.            | 100                                | $\Omega$                       |
| Bias Output Voltage, nom.         | 10                                 | VDC                            |
| Grounding                         | case isolated, internally shielded |                                |

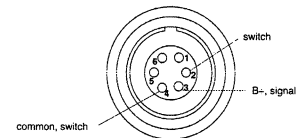
##### ENVIRONMENTAL

|                                        |     |                            |
|----------------------------------------|-----|----------------------------|
| Vibration Limit                        | 500 | g peak                     |
| Shock Limit                            | 100 | g peak                     |
| Electromagnetic Sensitivity, equiv. g. |     | $\mu\text{g}/\text{gauss}$ |

##### PHYSICAL

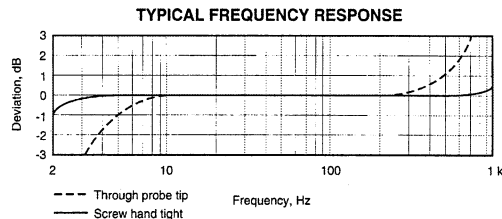
|                           |                                                   |       |
|---------------------------|---------------------------------------------------|-------|
| Weight                    | 150                                               | grams |
| Case Material             | stainless steel head, hard anodized aluminum body |       |
| Grip Material             | black neoprene foam                               |       |
| Output Connector (socket) | Lemo EGG1B306CNL                                  |       |
| Mating Connector (pins)   | Lemo FGG1B306CNADA62 (R6LM)                       |       |

#### Connector Pinout



**OPTIONS:** Customer specified sensitivity, connector.

**ACCESSORIES AVAILABLE:** Cabling, power supplies, amplifiers, signal conditioners, probe tips.



*Due to continued research and product development, Wilcoxon Research reserves the right to amend this specification without notice.*

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## ACCELEROMETERS

### Model 731 Ultra-Quiet Seismic Accelerometer

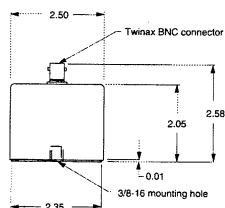


#### Features:

- Ultra high sensitivity
- Ultra low-noise electronics for clear signals at very low vibration levels
- Low frequency capable
- Built-in electronics
- Filtered to eliminate high frequencies
- Capable of measuring to sub micro-g levels
- Miswiring protection

#### Applications:

- Foundations
- Earth tremors
- Structures
- Microscopes
- Optical equipment
- Laboratory research
- Semiconductor manufacturing



Model 731

#### SPECIFICATIONS

##### DYNAMIC

|                                    | 731        | UNITS      |
|------------------------------------|------------|------------|
| Sensitivity, $\pm 5\%$ , 25 °C     | 10         | V/g        |
| Acceleration Range                 | 0.5        | g peak     |
| Amplitude Nonlinearity             | 1          | %          |
| Frequency Response:                |            |            |
| $\pm 5\%$                          | 0.15 - 200 | Hz         |
| $\pm 10\%$                         | 0.10 - 300 | Hz         |
| $\pm 3\text{ dB}$                  | 0.05 - 500 | Hz         |
| Resonance Frequency, mounted, nom. | 950        | Hz         |
| Transverse Sensitivity, max.       | 1          | % of axial |
| Temperature Response               | see graph  |            |

##### ELECTRICAL

|                                                 |          |                                |
|-------------------------------------------------|----------|--------------------------------|
| Power Requirement <sup>1</sup> , voltage source | 18 - 30  | VDC                            |
| current regulating diode                        | 2 - 10   | mA                             |
| Electrical Noise, equiv. g, nom.:               |          |                                |
| Broadband, 2.5 Hz to 25 kHz                     | 0.2      | $\mu\text{g}$                  |
| Spectral                                        |          |                                |
| 2 Hz                                            | 0.03     | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 10 Hz                                           | 0.01     | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 100 Hz                                          | 0.004    | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| Output Impedance, max.                          | 100      | $\Omega$                       |
| Bias Output Voltage, nom.                       | 9        | VDC                            |
| Grounding                                       | isolated |                                |

##### ENVIRONMENTAL

|                                        |           |                             |
|----------------------------------------|-----------|-----------------------------|
| Temperature Range                      | -10 to 65 | °C                          |
| Vibration Limit                        | 10        | g peak                      |
| Shock Limit <sup>2</sup>               | fragile   |                             |
| Electromagnetic Sensitivity, equiv. g. | 20        | $\mu\text{g}/\text{gauss}$  |
| Base Strain Sensitivity                | 0.0001    | $\text{g}/\mu\text{strain}$ |

##### PHYSICAL

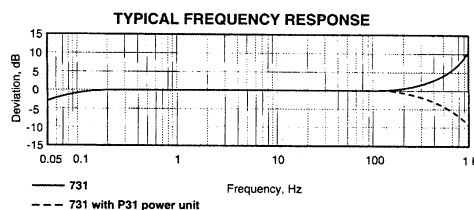
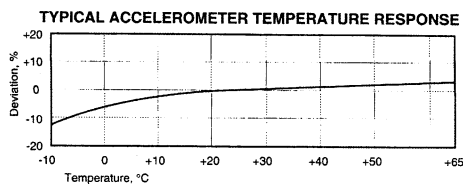
|                           |                                                 |       |
|---------------------------|-------------------------------------------------|-------|
| Weight                    | 540                                             | grams |
| Case Material             | anodized aluminum                               |       |
| Mounting                  | $\frac{1}{8}$ - 16 tapped hole                  |       |
| Output Connector          | Amphenol 31-2225                                |       |
| Cabling: Mating Connector | Amphenol 31-224 (R2T)                           |       |
| Standard Cable            | J9, 2 conductor shielded, PVC jacket, 10 pF/ft. |       |

NOTES: <sup>1</sup> Model P31 powering unit is recommended for single channel battery operation. Model PR710-1 is available for up to 10 channels, rackmount, line powered applications; consult the factory.  
<sup>2</sup> Will withstand a six inch drop to concrete.

OPTIONS: Customer specified sensitivity or frequency response, standard BNC connector.

ACCESSORIES SUPPLIED: SF7 mounting stud, R2T2T-J9-10 cable assembly, panel mount receptacle (when sold without power unit).

ACCESSORIES AVAILABLE: Model P31 and PR710-1 powering units, magnetic mounting bases, cementing studs, triaxial mounting block.



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## ACCELEROMETERS

### Model 732B High Frequency Accelerometer

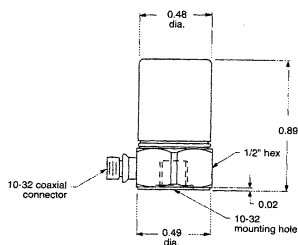


#### Features:

- Rugged
- Versatile
- Wide frequency range
- Standardized sensitivity
- High resonance frequency
- High frequency range
- Hermetically sealed

#### Applications:

- Pumps
- Turbines
- Helicopters
- Ship board machinery
- Laboratory research
- Compressors
- High frequency testing



Model 732B

#### SPECIFICATIONS

##### DYNAMIC

|                                    | 732B       | UNITS      |
|------------------------------------|------------|------------|
| Sensitivity, $\pm 5\%$ , 25 °C     | 10         | mV/g       |
| Acceleration Range                 | 250        | g peak     |
| Amplitude Nonlinearity             | 1          | %          |
| Frequency Response:                |            |            |
| $\pm 5\%$                          | 3 - 15,000 | Hz         |
| $\pm 3\text{ dB}$                  | 1 - 25,000 | Hz         |
| Resonance Frequency, mounted, nom. | 50         | kHz        |
| Transverse Sensitivity, max.       | 5          | % of axial |
| Temperature Response               | see graph  |            |

##### ELECTRICAL

|                                        |               |                                |
|----------------------------------------|---------------|--------------------------------|
| Power Requirement, voltage source      | 18 - 30       | VDC                            |
| current regulating diode <sup>1</sup>  | 0.5 - 10      | mA                             |
| Electrical Noise, equiv. g, nom.:      |               |                                |
| Broadband, 2.5 Hz to 25 kHz            | 2000          | $\mu\text{g}$                  |
| Spectral                               |               |                                |
| 10 Hz                                  | 100           | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 100 Hz                                 | 30            | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 1,000 Hz                               | 10            | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 10,000 Hz                              | 5             | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| Output Impedance, max.                 | 800           | $\Omega$                       |
| Bias Output Voltage, nom. <sup>2</sup> | 6             | VDC                            |
| Grounding                              | case grounded |                                |

##### ENVIRONMENTAL

|                                        |            |                             |
|----------------------------------------|------------|-----------------------------|
| Temperature Range                      | -50 to 120 | °C                          |
| Vibration Limit                        | 500        | g peak                      |
| Shock Limit                            | 5000       | g peak                      |
| Electromagnetic Sensitivity, equiv. g. | 100        | $\mu\text{g}/\text{gauss}$  |
| Base Strain Sensitivity                | 0.005      | $\text{g}/\mu\text{strain}$ |

##### PHYSICAL

|                           |                                       |       |
|---------------------------|---------------------------------------|-------|
| Weight                    | 15                                    | grams |
| Case Material             | stainless steel                       |       |
| Mounting                  | 10-32 tapped hole                     |       |
| Output Connector          | 10-32 coaxial                         |       |
| Cabling: Mating Connector | Microdot 10-32 (R1)                   |       |
| Standard Cable            | J93, coaxial, Teflon jacket, 30 pF/ft |       |

NOTES: <sup>1</sup> A maximum current of 6 mA is recommended for operating temperatures in excess of 100 °C.

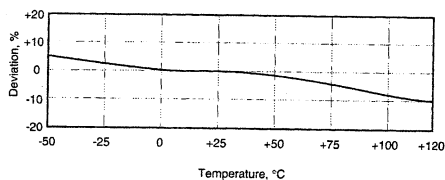
<sup>2</sup> Bias output voltage will vary between 5-10 volts depending upon the value of the constant current diode used.

OPTIONS: Customer specified filtering and sensitivity, top connector (BNC), PVC jacketed cable.

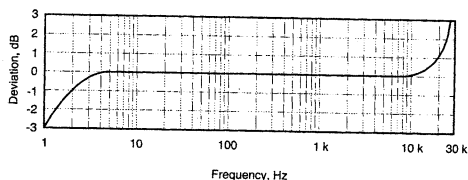
ACCESSORIES SUPPLIED: SF1 mounting stud, R12-J93-10 cable assembly.

ACCESSORIES AVAILABLE: Power supplies, amplifiers, signal conditioners, cementing studs, magnetic mounting bases, isolating studs.

#### TYPICAL TEMPERATURE RESPONSE



#### TYPICAL FREQUENCY RESPONSE



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## ACCELEROMETERS

### Model 793 Industrial Piezoelectric Accelerometer

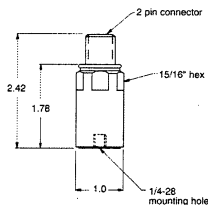


#### Features:

- Intrinsic safety certified option
- Rugged, general purpose
- Corrosion-resistant
- Priced economically for single and OEM quantities
- Hermetic seal
- Standardized sensitivity
- Ground isolated
- ESD protection
- Miswiring protection

#### Applications:

- Engine monitoring
- Machinery monitoring
- Ship board machinery
- Shock & vibration
- Predictive maintenance
- Shock pulse measurements



Model 793

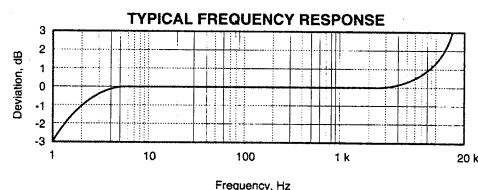
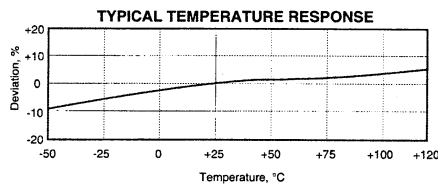
| SPECIFICATIONS                          | 793                                           | UNITS                          |
|-----------------------------------------|-----------------------------------------------|--------------------------------|
| <b>DYNAMIC</b>                          |                                               |                                |
| Sensitivity, $\pm 5\%$ , 25°C           | 100                                           | mV/g                           |
| Acceleration Range <sup>1</sup>         | 50                                            | g peak                         |
| Amplitude Nonlinearity                  | 1                                             | %                              |
| Frequency Response:                     |                                               |                                |
| $\pm 5\%$                               | 3 - 5,000                                     | Hz                             |
| $\pm 10\%$                              | 2 - 7,000                                     | Hz                             |
| $\pm 3\text{ dB}$                       | 1 - 15,000                                    | Hz                             |
| Resonance Frequency, mounted, nom.      | 26                                            | kHz                            |
| Transverse Sensitivity, max.            | 5                                             | % of axial                     |
| Temperature Response                    | see graph                                     |                                |
| <b>ELECTRICAL</b>                       |                                               |                                |
| Power Requirement:                      |                                               |                                |
| voltage source                          | 18 - 30                                       | VDC                            |
| current regulating diode <sup>1,2</sup> | 2 - 10                                        | mA                             |
| Electrical Noise, equiv. g, nom.:       |                                               |                                |
| Broadband, 2.5 Hz to 25 kHz             | 700                                           | $\mu\text{g}$                  |
| Spectral                                |                                               |                                |
| 10 Hz                                   | 10                                            | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 100 Hz                                  | 5                                             | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 1000 Hz                                 | 5                                             | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| Output Impedance, max.                  | 100                                           | $\Omega$                       |
| Bias Output Voltage, nom.               | 12                                            | VDC                            |
| Grounding                               | case isolated,<br>internally shielded         |                                |
| <b>ENVIRONMENTAL</b>                    |                                               |                                |
| Temperature Range                       | -50 to 120                                    | °C                             |
| Vibration Limit                         | 500                                           | g peak                         |
| Shock Limit                             | 5000                                          | g peak                         |
| Electromagnetic Sensitivity, equiv. g.  | 15                                            | $\mu\text{g}/\text{gauss}$     |
| Humidity Limit                          | 100                                           | % relative                     |
| Base Strain Sensitivity                 | 0.004                                         | g/ $\mu\text{strain}$          |
| <b>PHYSICAL</b>                         |                                               |                                |
| Weight                                  | 110                                           | grams                          |
| Case Material                           | stainless steel                               |                                |
| Mounting                                | 1/4 - 28 tapped hole                          |                                |
| Output Connector                        | MIL-C-5015 2-pin                              |                                |
| Cabling: Mating connector               | Amphenol 97-3106A-10SL-4S (R6)                |                                |
| Recommended Cable                       | J5, coaxial, polyurethane jacket,<br>30 pF/ft |                                |

- NOTES:** <sup>1</sup> Signal distortion can occur when measuring high vibration levels, especially with long cables. A24-30VDC powering source is recommended for minimizing distortion. Current regulating diode must be at least 2mA plus 1mA per 1000pF of cable capacitance.
- <sup>2</sup> A maximum current of 6mA is recommended for operating temperatures in excess of 100°C.

**OPTIONS:** Intrinsic safety "E" (793E) certified by Factory Mutual for use in the following environments: Classes I, II, III, Div. I, Groups A, B, C, D, E, F, G, 10 ft. sealed integral cable (see 793S/SG), temperature sensor "T", ultra low noise "U", customer specified filtering, sensitivity, cable length.

**ACCESSORIES SUPPLIED:** SF6 mounting stud.

**ACCESSORIES AVAILABLE:** Magnetic mounting bases, cementing studs, cable assembly R62-J5-XX, power supplies, amplifiers, signal conditioners.

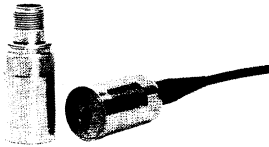


Due to continued research and product development, Wilcoxon Research reserves the right to amend this specification without notice.

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# ACCELEROMETERS

## Model 793L Low Frequency Piezoelectric Accelerometer

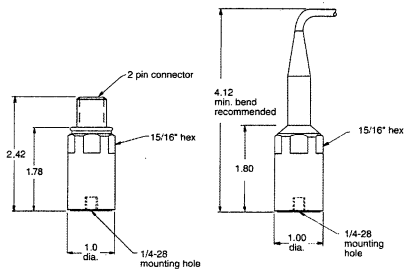


### Features:

- Intrinsic safety certified option
- High sensitivity
- Ultra low-noise electronics for clear signals at very low vibration levels
- Low frequency capable
- Filtered to eliminate high frequencies
- Hermetic seal
- ESD protection
- Miswiring protection

### Applications:

- Machinery monitoring
- Rotating machinery
- Low vibration level measurements
- Laboratory research



Model 793L

Model 793LS

| SPECIFICATIONS                           | 793L                                       | UNITS                          |
|------------------------------------------|--------------------------------------------|--------------------------------|
| <b>DYNAMIC</b>                           |                                            |                                |
| Sensitivity, $\pm 5\%$ , 25 °C           | 500                                        | mV/g                           |
| Acceleration Range <sup>1</sup>          | 10                                         | g peak                         |
| Amplitude Nonlinearity                   | 1                                          | %                              |
| Frequency Response, nom.:                |                                            |                                |
| $\pm 5\%$                                | 0.6 - 700                                  | Hz                             |
| $\pm 10\%$                               | 0.4 - 1000                                 | Hz                             |
| $\pm 3\text{ dB}$                        | 0.2 - 2300                                 | Hz                             |
| Resonance Frequency, mounted, nom.       | 15                                         | kHz                            |
| Transverse Sensitivity, max.             | 7                                          | % of axial                     |
| Temperature Response                     | see graph                                  |                                |
| <b>ELECTRICAL</b>                        |                                            |                                |
| Power Requirement, voltage source        | 18 - 30                                    | VDC                            |
| current regulating diode <sup>1, 2</sup> | 2 - 10                                     | mA                             |
| Electrical Noise, equiv. g, nom.:        |                                            |                                |
| Broadband, 2.5 Hz to 25 kHz              | 10                                         | $\mu\text{g}$                  |
| Spectral 2 Hz                            | 1.5                                        | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 10 Hz                                    | 0.4                                        | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| 100 Hz                                   | 0.2                                        | $\mu\text{g}/\sqrt{\text{Hz}}$ |
| Output Impedance, max.                   | 100                                        | $\Omega$                       |
| Bias Output Voltage, nom.                | 10                                         | VDC                            |
| Grounding                                | case isolated, internally shielded         |                                |
| <b>ENVIRONMENTAL</b>                     |                                            |                                |
| Temperature Range                        | -50 to 120                                 | °C                             |
| Vibration Limit                          | 250                                        | g peak                         |
| Shock Limit                              | 2500                                       | g peak                         |
| Electromagnetic Sensitivity, equiv. g.   | 5                                          | $\mu\text{g}/\text{gauss}$     |
| Humidity Limit                           | 100                                        | % relative                     |
| Base Strain Sensitivity                  | 0.001                                      | g/ $\mu\text{strain}$          |
| <b>PHYSICAL</b>                          |                                            |                                |
| Weight                                   | 135                                        | grams                          |
| Case Material                            | stainless steel                            |                                |
| Mounting                                 | 1/4-28 tapped hole                         |                                |
| Output Connector                         | MIL-C-5015 2-pin                           |                                |
| Cabling <sup>3</sup> : Mating Connector  | Amphenol 97-3106A-10SL-4S (R6)             |                                |
| Recommended Cable                        | J5, coaxial, polyurethane jacket, 30 pF/ft |                                |

- NOTES:** <sup>1</sup> Signal distortion can occur when measuring high vibration levels, especially with long cables. A 24-30 VDC powering source is recommended for minimizing distortion.  
<sup>2</sup> A maximum current of 6 mA is recommended for operating temperatures in excess of 100 °C.  
<sup>3</sup> Cabling for "S" option model is J61, coaxial, polyurethane jacket, 30 pF/ft.

### Connections

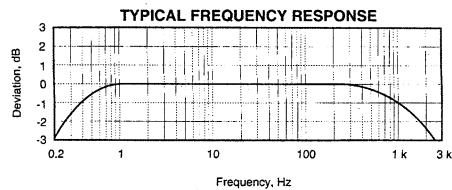
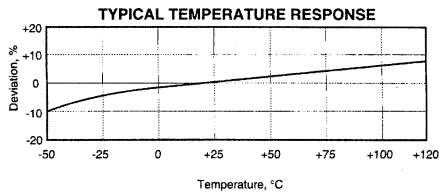
Pin A ..... Signal, power  
Pin B ..... Common

**OPTIONS:** Intrinsic safety "E" (793LE) certified by Factory Mutual for use in the following environments: Classes I, II, III, Div. 1, Groups A, B, C, D, E, F, G, 10 ft. sealed integral cable "S", temperature sensor "T", customer specified sensitivity, filtering, cable length.

**ACCESSORIES SUPPLIED:** SF6 mounting stud.



**ACCESSORIES AVAILABLE:** R62-J5-XX cable assembly, magnetic mounting bases, cementing studs, power supplies, amplifiers, signal conditioners.



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